# Introduction to Software Engineering

# COSC 336

# ReSource (Phase III)

## Requirements Document

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System


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Date: 14/10/2026


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# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS), version 1.0 for review, defines the required behavior of ReSource, an AI-powered campus resource exchange and asset life-cycle management prototype for Khalifa University. It specifies the functional requirements, quality requirements, permissions, business rules and use cases that will guide design, implementation and acceptance testing. Each requirement states an expected behavior and a way to check it; this document does not claim that the system has already been implemented or validated.

The requirements develop the course project description and brief [1], [2], the Phase 3 guidance [3] and the group's Phase 1 and Phase 2 work [4], [5]. Proposed university procedures require stakeholder confirmation. The interviews in Appendix E are explicitly AI simulations and are not evidence of actual university practice.

## 1.2 Document Conventions

"Shall" identifies a requirement. Functional requirements use FR followed by a feature and requirement number, such as FR-2.1. FR-7.N identifies notifications, FR-7.R identifies operational reports and FR-7.A identifies audit requirements. AI identifies the general AI requirements in Section 2.14. PER, SAF, SEC, QUA and BR identify performance, safety, security, quality and business-rule requirements. LEG and ORG identify legal and organizational requirements. UC identifies use cases and TBD identifies decisions awaiting confirmation (Appendix C).

Essential requirements are the ones the first working prototype must meet; together they cover the minimum prototype defined in the project description [1]. Desirable requirements are planned extensions, and Future requirements are outside the current minimum implementation. Each feature is given a High, Medium or Low priority, with a one-line reason; High features are part of that minimum prototype. Appendix F.2 checks that the Essential set fits the Phase 2 schedule. Acceptance criteria are proposed tests, not completed test results. Every system requirement links to at least one of the use cases UC-1 to UC-13; the process requirements LEG-2, ORG-1, ORG-2 and ORG-4 and the policy rule BR-1 have no use case. Appendix D traces every requirement ID.

The document follows the SRS template in the project description [1]. As set out in the Phase 3 guidance [3], the template's Overall Description and External Interface Requirements sections are not required for Phase 3, so they are omitted and the remaining sections are numbered in order: Sections 2, 3 and 4 correspond to template Sections 4, 5 and 6. Role names follow the Phase 2 stakeholder roles [5] and are defined in Section 5.1.

## 1.3 Intended Audience and Reading Suggestions

Developers and testers should read the scope and roles first, then the functional requirements, business rules, state lists and linked use cases. They should use the acceptance criteria and Appendix D to plan implementation and tests. Reviewers of the specification can use Appendix D to check coverage, consistency and priorities against the project description [1], and Appendix F to check changes and priorities against Phase 2.

Department Representatives, Asset Coordinators, Administrators, the Procurement Team, Maintenance Staff, Finance Officers and Sustainability Officers should review the workflows and permissions affecting their work. System Administrators and the IT Department should review account management, security, auditability and unresolved technical policies. University Management and the proposed business owner, the Head of Facilities Management, should review scope, approval authority and Appendix C before authorizing any real pilot. The individual business owner and institutional approval remain unconfirmed.

The document is organized as follows. Section 2 specifies the thirteen system features (FR-1 to FR-13) and the general AI requirements that apply to every AI-assisted feature (Section 2.14). Section 3 sets measurable performance, safety, security and quality requirements and the business rules. Section 4 covers data, legal and organizational requirements. Section 5 defines the actors, the use case diagram and thirteen use case descriptions. Appendix A is a glossary, Appendix B lists the states of each record type, Appendix C lists open decisions, Appendix D traces every requirement, Appendix E records the AI-simulated stakeholder interviews and Appendix F lists the changes from Phase 2 with the priority and schedule check.

## 1.4 Product Scope

ReSource will help authorized university departments publish surplus resources, find suitable assets and request reuse before considering a new purchase. It will record assets such as furniture, equipment, vehicles and books, including their condition, location, quantity, custody and life-cycle history. Controlled workflows cover requests, reservations, approvals, transfers, receipt, maintenance and approved end-of-life actions.

The prototype includes department-based access, consent at sign-up, notifications, operational reporting and sustainability indicators. AI supports classification, semantic matching, ranked suggestions with explanations and report drafting; sustainable-action recommendations and a natural-language assistant are planned extensions (Desirable). Authorized people review consequential decisions, and standard search, forms and reports remain available when AI cannot be used.

Expected benefits are better visibility of surplus assets, fewer avoidable purchases, longer useful asset life and clearer evidence of reuse. Savings and environmental indicators are estimates based on recorded data and stated assumptions; they are not measured campus outcomes until validated.

The academic prototype uses synthetic or reviewed sample data and existing AI tools. Live integration with university procurement, finance or single sign-on systems, autonomous purchasing or disposal, large-scale custom model training and production certification are outside the current scope [1], [5]. Physical transport, repairs and waste handling remain the responsibility of authorized people and service providers.

## 1.5 References

References [1] to [3] define the course requirements and the Phase 3 deliverable. References [4] and [5] are the group’s planning and feasibility baseline. References [6] to [10] identify UAE legislation used in the legal, security and data requirements; they are cited for university review and do not establish that the prototype is legally compliant. References [11] to [15] are university pages, technical standards and the emission-factor source used in specific requirements.

[1] Khalifa University, “Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System,” COSC 336 Introduction to Software Engineering project description, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[2] Khalifa University, “COSC336 Circular Campus Project,” COSC 336 project brief slides, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[3] Khalifa University, “Phase 3: Requirements Document,” COSC 336 lab slides, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[4] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, “ReSource Phase I Report: Project Planning and Requirements,” COSC 336, Khalifa Univ., Abu Dhabi, UAE, Sep. 2026.

[5] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, “ReSource Phase II Feasibility Study,” COSC 336, Khalifa Univ., Abu Dhabi, UAE, Sep. 2026.

[6] United Arab Emirates, Federal Decree-Law No. 45 of 2021 Concerning the Protection of Personal Data, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1972

[7] United Arab Emirates, Federal Decree-Law No. 38 of 2021 on Copyright and Neighboring Rights, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1534

[8] United Arab Emirates, Federal Decree-Law No. 34 of 2021 on Countering Rumors and Cybercrimes, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1526

[9] United Arab Emirates Government, “Data protection laws,” u.ae, 2026. [Online]. Available: https://u.ae/en/about-the-uae/digital-uae/data/data-protection-laws

[10] Securiti, “Overview of UAE’s Federal Decree-Law No. (45) of 2021 on PDPL,” securiti.ai, 2026. [Online]. Available: https://securiti.ai/uae-personal-data-protection-law/

[11] Khalifa University, “Procurement & Contracts Department,” ku.ac.ae, 2026. [Online]. Available: https://www.ku.ac.ae/about/procurement-and-contracts-department

[12] Khalifa University, “Sustainability initiative,” ku.ac.ae, 2026. [Online]. Available: https://www.ku.ac.ae/sustainability/initiative

[13] OWASP Foundation, “OWASP Top 10:2025,” 2025. [Online]. Available: https://top10.owasp.org/2025/

[14] World Wide Web Consortium, “Web Content Accessibility Guidelines (WCAG) 2.2,” W3C Recommendation, Dec. 2024. [Online]. Available: https://www.w3.org/TR/WCAG22/

[15] U.S. Environmental Protection Agency, “Documentation for greenhouse gas emission and energy factors used in the Waste Reduction Model (WARM): Electronics chapter (Exhibit 1-7) and Containers, Packaging and Non-Durable Goods Materials chapter (Exhibit 2-11),” WARM version 16, Dec. 2023. [Online]. Available: https://www.epa.gov/waste-reduction-model/documentation-chapters-emissions-energy-and-economic-factors-used-waste

# 2. System Features

This section specifies the system features. Features FR-1 to FR-12 each refine one Phase 2 high-level requirement [5], and FR-13 adds the sustainability indicators required by the project description [1]. Each feature gives its description and priority, its stimulus/response sequences and its functional requirements. Section 2.14 states the general AI requirements (AI-1 to AI-10) that apply to every AI-assisted feature. Each requirement has an ID, a priority, its linked use cases and an acceptance criterion.

## 2.1 Asset Registration (FR-1)

### 2.1.1 Description and Priority

This feature lets an Asset Coordinator record unused, surplus or reusable assets with enough detail for other departments to find, judge and request them. Every later step, including search, matching, transfer, maintenance and end-of-life decisions, depends on these records being complete and correct.

Priority: High. This feature develops Phase 2 requirement FR-1. Asset registration and status management are part of the minimum prototype [1], and the other features depend on the asset records it creates.

### 2.1.2 Stimulus/Response Sequences

1. The Asset Coordinator chooses to register an asset. The system opens a new record for the Asset Coordinator’s department.

2. The Asset Coordinator enters the asset details, including the hazard flag, and adds photographs and supporting documents. The system checks each value and file.

3. The Asset Coordinator may ask for AI suggestions (UC-10). The system shows them separately, and nothing is saved until the Asset Coordinator accepts it.

4. The Asset Coordinator saves a draft or publishes. The system checks the required fields, gives the asset a unique ID and sets its state to Draft or Available.

5. If a field is missing, a value is invalid or a file is refused, the system names the problem and keeps everything else that was entered.

### 2.1.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-1.1** | The system shall allow only Asset Coordinators to create asset records, and only for the departments they are assigned to. | Essential | UC-1 | An Asset Coordinator can save a record for their own department. A Department Representative, or an Asset Coordinator of another department, is refused and nothing is saved. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-1.2** | The system shall require these fields before an asset is published: name, description, category, quantity, Asset Coordinator, owning department, location (building and room), condition and hazard flag (whether the asset is hazardous or electronic waste). | Essential | UC-1 | Publishing with any of these fields empty is blocked, and each missing field is named. |
| **FR-1.3** | The system shall allow these optional fields: subcategory, specifications, brand and model, serial or tag number, purchase date, first-allocation date, estimated value (AED), replacement price (AED), weight (kg), material type, and handling or safety notes (for example heavy, fragile or contains batteries). | Essential | UC-1 | Optional fields can be saved, viewed and edited. A record without them can still be published. |
| **FR-1.4** | The system shall record condition only from the list New, Good, Fair or Poor. Availability shall be set by the system from the asset state (Appendix B.1) and shall not be typed by users. | Essential | UC-1 | Condition accepts only the listed values. After every change, the shown availability matches the asset state. |
| **FR-1.5** | The system shall reject invalid values: quantity must be a whole number of at least 1, estimated value, replacement price and weight cannot be negative, purchase and first-allocation dates cannot be in the future, and category must come from the approved list. | Essential | UC-1 | Test entries of quantity 0, value -50, weight -1, a future purchase date and an unknown category are each rejected with a message. Other entered data is kept. |
| **FR-1.6** | The system shall allow up to 5 photographs and up to 3 supporting documents (for example invoice, manual or warranty) per asset. Each file shall be an image or a PDF of at most 10 MB (SEC-13, TBD-13). | Essential | UC-1 | Five photographs and three documents attach. A sixth photograph, a file over 10 MB or a file that is not an image or PDF is refused and the limit is shown. |
| **FR-1.7** | The system shall give each asset a unique asset ID that is never reused, even after the asset is archived. | Essential | UC-1 | 100 test records receive 100 different IDs. Archiving a record does not free its ID. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-1.8** | The system shall warn when a serial or tag number already exists on another record. | Essential | UC-1 | Entering an existing serial number shows a warning that links to the other record. The new record saves only after the Asset Coordinator corrects it or confirms it is a different item. |
| **FR-1.9** | The system shall let the Asset Coordinator save an asset as a draft and publish it later. Drafts shall not be visible to other departments. | Essential | UC-1, UC-2 | A draft does not appear in search. After publishing, the asset appears in search with the state Available. |
| **FR-1.10** | The system shall not allow changes to category, quantity or condition while the asset is Reserved or In transfer. | Essential | UC-1, UC-5 | Editing these fields on a Reserved asset is refused with the reason. The same edit on an Available asset is saved. |
| **FR-1.11** | The system shall let the Asset Coordinator withdraw a published asset from listing when it has no open request. | Desirable | UC-1 | A withdrawn asset no longer appears in search. Withdrawal is refused while a request for the asset is still open. |
| **FR-1.12** | The system shall allow registration to be completed without AI. AI-suggested values shall be saved only after the Asset Coordinator accepts them (see FR-9). | Essential | UC-1, UC-10 | With AI unavailable, a full registration still completes. No AI value appears in a saved record unless the Asset Coordinator accepted it. |
| **FR-1.13** | The system shall record the creation of each asset and every later change, with the user, time, and old and new values. | Essential | UC-1 | Every create, edit, publish and withdrawal appears in the audit log (FR-7.A) with user, time and changed fields. |
| **FR-1.14** | The system shall allow many assets to be registered at once from one uploaded list. | Future | UC-1 | A list of 20 assets with 2 invalid rows creates 18 drafts and reports the 2 rejected rows with reasons. |
| **FR-1.15** | The system shall not publish an asset flagged as hazardous or electronic waste until a Sustainability Officer has recorded a safety review that allows it to be offered for reuse (SAF-1). | Essential | UC-1 | Publishing a hazardous test asset without a recorded safety review is refused with the reason. After the review is recorded, publication succeeds. |

## 2.2 Resource Publication and Search (FR-2)

### 2.2.1 Description and Priority

This feature publishes surplus assets to the campus catalogue and lets Department Representatives find suitable resources through standard search and filters. Semantic matching is specified separately in FR-8; publishing and search remain usable without AI.

Priority: High. This feature develops Phase 2 requirement FR-2. Publication of available or surplus resources, search and filtering are part of the minimum prototype [1].

### 2.2.2 Stimulus/Response Sequences

1. The Asset Coordinator confirms publication of a complete asset record. The system checks the record, safety restrictions and permissions before making it visible.

2. The Department Representative opens the catalogue, enters a search and selects filters. The system validates the criteria and returns permitted matches with current availability.

3. The Department Representative opens an asset and reviews its details. If suitable, they continue to UC-3 to submit a request.

4. If no result matches, input is invalid or availability changes, the system explains the situation and allows the search to be refined.

### 2.2.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-2.1** | The system shall publish an asset only when its Asset Coordinator confirms publication, the FR-1 fields are valid and no safety or life-cycle restriction blocks it (SAF-1, SAF-5, FR-5.9, FR-5.27). Drafts and withdrawn assets shall not appear in the shared catalogue. | Essential | UC-1, UC-2 | A valid confirmed asset appears; a draft, withdrawn, unsafe, under-maintenance or pending end-of-life asset does not. Missing required fields prevent publication. |
| **FR-2.2** | The system shall let signed-in users search permitted published resources by asset ID, name, description or category using standard text search, without requiring AI. | Essential | UC-2 | Known keywords and IDs return the expected permitted records while the AI service is disabled. |
| **FR-2.3** | The system shall combine filters for category, condition, owning department, location, required quantity, availability and life-cycle status. All selected filters shall apply together. | Essential | UC-2 | A prepared data set returns only records meeting every selected filter. Clearing filters restores the permitted catalogue. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-2.4** | The system shall show each result's asset ID, name, category, condition, department, location, available quantity and status, with a link to the permitted asset details. | Essential | UC-2 | Every result shows these fields. Opening it shows the matching asset; restricted fields remain hidden. |
| **FR-2.5** | The system shall apply role and department permissions to results, counts, filters and direct record links. Financial values and value filters shall be available only to the roles authorized by SEC-5. | Essential | UC-2 | A Department Representative can find shared assets but cannot view or infer restricted financial values or private records. |
| **FR-2.6** | The system shall validate search inputs, identify invalid ranges or quantities, and show a clear no-results message when nothing matches. Valid search criteria shall be retained after an error. | Essential | UC-2 | A negative quantity or reversed range is rejected. A valid unmatched search shows zero results and allows refinement. |
| **FR-2.7** | The system shall show current availability when an asset is opened and allow a Department Representative to continue to UC-3. Browsing shall not create a request, reservation or approval. | Essential | UC-2, UC-3 | Opening a result creates no request. If the last unit becomes unavailable, request submission is blocked under FR-3.7. |
| **FR-2.8** | The system shall support asset-age and estimated-value ranges where source data and permissions exist. Missing dates or values shall be shown as unknown, not treated as zero. | Desirable | UC-2 | Known values filter correctly; unknown values are labeled and excluded from numeric ranges. Unauthorized users cannot apply value filters. |

# 2.3 Resource Requests and Reservations (FR-3)

## 2.3.1 Description and Priority

This feature lets a Department Representative ask for a specific published asset, or describe a need when nothing suitable is listed. Submitting a request does not reserve the asset; the requested quantity is reserved only when the request is approved (FR-4), so the same units are never promised to two departments. The Department Representative can follow the request until the asset is received.

Priority: High. This feature develops Phase 2 requirement FR-3. Resource-request submission and tracking are part of the minimum prototype [1].

## 2.3.2 Stimulus/Response Sequences

1. The Department Representative opens a published asset and chooses to request it, or creates a request for a need. The system fills in any details it already knows.

2. The Department Representative enters the purpose, quantity, required date and other request details. The system checks each value.

3. The Department Representative submits. The system checks the current available quantity, gives the request an ID, sets it to Submitted and notifies the owning Asset Coordinator. No units are reserved at this point.

4. For a need with no chosen asset, the system sets the request to Open so the Procurement Team can look for matches (UC-9).

5. The Department Representative follows the state of the request. They can edit it while it is Submitted or cancel it before handover; cancelling an approved request releases its reservation.

6. If a value is invalid, the quantity is no longer available or the asset is restricted, the system explains the problem and keeps the entered details.

### 2.3.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-3.1** | The system shall allow a Department Representative to request a specific published asset for a quantity no greater than its current available quantity. | Essential | UC-3 | A request for 2 of 5 available units is saved. A request for 6 of 5 is refused and the available quantity is shown. |
| **FR-3.2** | The system shall allow a Department Representative to submit a request for a need when no listed asset fits, without choosing an asset. | Essential | UC-3, UC-9 | The request is saved as Open and appears in the Procurement Team's list for matching. |
| **FR-3.3** | The system shall require: requesting department, category, purpose, quantity, required date and delivery location. Specifications, preferred minimum condition and urgency (Low / Medium / High, default Medium) shall be optional. | Essential | UC-3 | Submitting with any required field empty is blocked and each missing field is named. |
| **FR-3.4** | The system shall reject a quantity below 1, a required date in the past, and a preferred condition or category not on the approved lists. | Essential | UC-3 | Each invalid test value is rejected with a message, and the other entered details are kept. |
| **FR-3.5** | The system shall allow a Department Representative to submit requests only for their own department. | Essential | UC-3 | Choosing another department is refused. A Department Representative can see every request from their department and no others. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-3.6** | The system shall give each submitted request a unique request ID, set its state to Submitted and notify the owning Asset Coordinator. | Essential | UC-3 | Each new request shows its own ID and the state Submitted, and the owning Asset Coordinator receives a notification (FR-7.N1). |
| **FR-3.7** | The system shall check the available quantity again at the moment of submission and refuse a request for more units than are currently available. Submitting a request shall not reserve units or change the asset's ownership, custody or location. | Essential | UC-3 | With 8 units available, a request for 5 is saved and the available quantity stays 8. A request for 9 is refused and the current available quantity is shown. |
| **FR-3.8** | The system shall reserve the requested quantity only when the request receives its final required approval (FR-4.5), and shall release the reservation when an approved request is cancelled or expires. Two approved requests shall never reserve the same units. | Essential | UC-3, UC-4 | With 8 units available, approving a request for 5 leaves 3 available. When two requests for the last unit are approved at the same moment, exactly one succeeds and the available quantity never goes below zero. Cancelling the approved request restores the quantity. |
| **FR-3.9** | The system shall expire a request that is not decided within 10 working days of submission, release the reservation of an approved request that is not handed over within 10 working days of approval, and notify both departments (TBD-5). | Desirable | UC-3, UC-4, UC-5 | A test request left undecided for 10 working days becomes Expired. An approved test request not handed over in 10 working days becomes Expired, its quantity is released, and both departments are notified. |
| **FR-3.10** | The system shall show the Department Representative the current state of each request and its full history, with date, person and reason for every change. | Essential | UC-3 | The request page lists every state change in order, with date, person and reason. |
| **FR-3.11** | The system shall allow the requesting Department Representative to edit a request while it is Submitted, and to cancel it at any time before handover. | Essential | UC-3 | Edits save on a Submitted request and are refused once it is Released. Cancelling an Approved request sets it to Cancelled and releases its reservation. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-3.12** | The system shall not allow the requesting Department Representative to release, approve or change the state of their own request in any way other than editing or cancelling it. | Essential | UC-3, UC-4 | A test attempt by the requesting Department Representative to release or approve their own request is refused and recorded in the audit log. |
| **FR-3.13** | The system shall warn when the same department already has an open request for the same asset. | Desirable | UC-3 | A second request from the same department for the same asset shows a warning with a link to the first request. |
| **FR-3.14** | Requests created through the AI assistant shall pass the same checks and need the same confirmation as requests made on the form (see FR-11.2 and FR-11.3). | Desirable | UC-3, UC-11 | A request made through the assistant has the same fields, checks and state as one made on the form, and is not submitted until the Department Representative confirms. |
| **FR-3.15** | The system shall refuse a request for an asset that has an open maintenance request, a pending end-of-life proposal, an unsafe status or an unreviewed hazard flag (FR-5.9, FR-5.27, SAF-1, SAF-5), and shall name the restriction. | Essential | UC-3 | A request for each type of restricted test asset is refused and the message names the restriction. The same request for an unrestricted asset is saved. |

# 2.4 Approvals and Transfers (FR-4)

## 2.4.1 Description and Priority

This feature moves an asset between departments in four controlled steps: release by the owning department's Asset Coordinator, approval, transfer by collection or delivery, and receipt. The requested quantity is reserved at final approval, and custody and location change only when the receiving department confirms receipt. Every decision is made by an authorized person; AI may recommend but never decides.

Priority: High. This feature develops Phase 2 requirement FR-4. Transfer approval and confirmation and ownership tracking are part of the minimum prototype [1].

## 2.4.2 Stimulus/Response Sequences

1. The Asset Coordinator of the owning department reviews a new request and releases it or declines it with a reason. The system sets the request to Released or Rejected.

2. The Administrator approves or rejects the released request. Requests above the approved value limit, or of unknown value, also need a Procurement Team member's approval (FR-4.3). On final approval, the system rechecks availability and reserves the requested quantity.

3. The Asset Coordinator arranges the transfer: collection or delivery, planned date, from and to locations, contact person and any transfer cost. The system creates a Scheduled transfer and notifies the requesting Department Representative.

4. At handover, the Asset Coordinator marks the asset as handed over. The system checks the safety conditions and sets the transfer and asset to In transfer.

5. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and records the condition on arrival. The system changes the Asset Coordinator, department and location, and completes the request.

6. If the asset arrives damaged or is the wrong item, the receiver reports a problem. The system sets the transfer to Disputed and asks the Administrator to decide.

### 2.4.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-4.1** | The system shall ask the owning department's Asset Coordinator to release or decline each new request. Declining shall need a reason. | Essential | UC-4 | A request cannot reach approval until it is released. A decline without a reason is refused. |
| **FR-4.2** | The system shall send released requests to the Administrator to approve or reject. Rejecting shall need a reason. | Essential | UC-4 | Approve and reject both work. Every rejection shows its reason to the requesting Department Representative. |
| **FR-4.3** | The system shall require a second approval from a Procurement Team member when the estimated value is above a limit approved by the university and configured by the System Administrator (TBD-2), or when the value is unknown. | Desirable | UC-4 | A test request above the limit cannot become Approved until both approvals are recorded. |
| **FR-4.4** | The system shall not allow one person to approve a request they submitted, or to both release and approve the same request. | Essential | UC-4 | Both test attempts are refused and recorded in the audit log. |
| **FR-4.5** | The system shall recheck availability, quantity, safety restrictions and permissions just before the final approval is saved, and shall reserve the requested quantity in the same step (FR-3.8). | Essential | UC-4 | If the asset was withdrawn, became restricted or now has fewer available units than requested, approval is blocked and the Administrator sees the up-to-date details. A successful approval reduces the available quantity by the requested amount. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-4.6** | The system shall notify the requesting Department Representative and the owning Asset Coordinator of every release, approval, rejection and transfer step. | Essential | UC-4, UC-5 | In testing, both parties receive a notification (FR-7.N) within 5 minutes of each decision. If sending fails, the decision is still saved. |
| **FR-4.7** | The system shall let the owning Asset Coordinator create a transfer only for an approved request, with method (collection or delivery), planned date, from location, to location, contact person and, optionally, the transfer cost in AED. | Essential | UC-5 | A transfer cannot be created for a request that is not Approved. Missing transfer fields are named, and a negative transfer cost is refused. |
| **FR-4.8** | The system shall allow supporting documents, such as a handover form, delivery note or photographs at handover, to be attached to a transfer (up to 5 files). | Desirable | UC-5 | Five files attach to a transfer and can be viewed by both departments. A sixth is refused. |
| **FR-4.9** | The system shall set the asset and transfer to In transfer when the Asset Coordinator marks the asset as handed over. Before handover, the Asset Coordinator may reschedule or cancel; cancelling returns the reserved units to Available. | Essential | UC-5 | Handover sets both states to In transfer. Cancelling a Scheduled transfer restores the available quantity. |
| **FR-4.10** | The system shall allow only a Department Representative or the Asset Coordinator of the receiving department to confirm receipt, and shall require the condition on arrival. | Essential | UC-5 | A user from another department cannot confirm receipt. Confirmation without a condition is refused. |
| **FR-4.11** | On confirmed receipt, the system shall change the asset's Asset Coordinator, owning department and location, and set the request to Completed. These details shall not change before receipt. | Essential | UC-5 | After receipt, the asset shows the new Asset Coordinator, department and location. Before receipt, they are unchanged. |
| **FR-4.12** | The system shall let the receiver report a problem on arrival. The transfer shall become Disputed and the Administrator shall decide to accept it or return the asset. | Essential | UC-5 | A problem report sets the transfer to Disputed and notifies the Administrator. Custody does not change until the dispute is decided. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-4.13** | The system shall remind the receiving department if receipt is not confirmed within 5 working days of handover, with a copy to the Administrator. | Desirable | UC-5 | A test transfer left unconfirmed for 5 working days sends one reminder to the receiver and the Administrator. |
| **FR-4.14** | The system shall keep the full ownership and transfer history of every asset and record every release, approval, transfer and receipt step. | Essential | UC-4, UC-5 | The asset page lists every past owner with dates. Each step appears in the audit log (FR-7.A) with user, time and decision. |
| **FR-4.15** | When only part of a multi-unit asset is transferred, the system shall create a linked record for the transferred units and reduce the original quantity. | Essential | UC-5 | Transferring 2 of 5 chairs leaves 3 on the original record and creates a new record of 2, linked to the original, for the receiving department. |
| **FR-4.16** | The system shall block handover of an asset that is unsafe, under maintenance or pending an end-of-life action, and of a risky asset without a recent safe inspection (SAF-5, SAF-9, FR-5.9, FR-5.27). It shall show any handling or safety notes to the receiving department before handover (SAF-7). | Essential | UC-5 | Handover of each type of blocked test asset is refused with the reason. For a test asset with safety notes, the notes appear on the receiving department’s transfer screen. |

# 2.5 Maintenance and Asset Life-Cycle Management (FR-5)

## 2.5.1 Description and Priority

This feature lets authorized users report defects, request inspections, assign and record repair and refurbishment work and its cost, and update an asset’s condition after service. It keeps a complete, unchangeable life-cycle history for every asset, from acquisition to final disposition, and manages end-of-life actions (donation, recycling, disassembly for usable components, retirement and disposal) through a safety review and an approval, with extra safeguards for hazardous items. It refines Phase 2 requirement FR-5 and the project’s “Inspection and Asset Life-Cycle Tracking” functionality.
Priority: High. This feature develops Phase 2 requirement FR-5. Maintenance and repair tracking is part of the minimum prototype [1], and the life-cycle history is needed by the sustainability indicators (FR-13).

### 2.5.2 Stimulus/Response Sequences

1. Defect reported: an Asset Coordinator, Department Representative or Maintenance Staff member reports a defect → the system creates a maintenance request in the “Reported” state and notifies Maintenance Staff.
2. Work assigned and recorded: a request is assigned to a Maintenance Staff member, who starts the inspection and records findings and repair work → the system moves the request through its states, saves the work and cost, updates the asset’s condition and availability, and adds life-cycle events.
3. End-of-life proposed: an Asset Coordinator or Sustainability Officer proposes donation, recycling, disassembly, retirement or disposal → the system checks for blocking items, routes the proposal to a safety review by a different Sustainability Officer and then to approval, and on completion sets the asset’s final state and updates the sustainability indicators.
4. History viewed: an authorized user opens an asset’s history → the system lists every life-cycle event in date order.

### 2.5.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **A. Maintenance and repair** | | | | |
| **FR-5.1** | The system shall allow an Asset Coordinator, Department Representative or Maintenance Staff member to report a defect in a registered asset, recording the asset, defect description, date reported, reporting user and optional photographs. | Essential | UC-6 | A report with all required fields creates a maintenance request in the 'Reported' state, linked to the asset and shown in its maintenance list. |
| **FR-5.2** | The system shall allow an Asset Coordinator or Maintenance Staff member to request an inspection of an asset, stating the reason and the requested inspection date. | Essential | UC-6 | The request appears in the Maintenance Staff queue in the 'Inspection Requested' state. |
| **FR-5.3** | The system shall allow Maintenance Staff or an Administrator to assign a maintenance request to a Maintenance Staff member, who is then notified. | Essential | UC-6 | The assigned request appears in that member's list, and the member receives a notification. |
| **FR-5.4** | The system shall move a maintenance request to 'Under Inspection' when the assigned Maintenance Staff member starts the inspection, and to 'Repair in Progress' when they start repair or refurbishment work. | Essential | UC-6 | Each start action changes the state as stated and records the date, time and user. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-5.5** | The system shall allow Maintenance Staff to record inspection results: inspection date, inspector, findings, assessed condition, whether the asset is safe to use, and a recommended next action (no action, repair, refurbish or end-of-life review). | Essential | UC-6 | All listed fields are saved and appear in the asset's maintenance and life-cycle history. |
| **FR-5.6** | The system shall allow Maintenance Staff to record repair and refurbishment work: work type (repair or refurbishment), work performed, parts replaced, start and completion dates, labor hours, cost in AED, and the estimated additional years of use the work adds. | Essential | UC-6 | A saved record shows all fields, and its cost is added to the asset's cumulative maintenance cost. |
| **FR-5.7** | The system shall track each maintenance request through the maintenance states defined in Appendix B (Reported, Inspection Requested, Under Inspection, Repair in Progress, Completed, Closed – Not Repairable, Cancelled) and show the current state to the reporting user and to the asset's Asset Coordinator. | Essential | UC-6 | Each state change is visible to both users and is recorded with date, time and user. |
| **FR-5.8** | When a maintenance request is completed, the system shall require Maintenance Staff to record the asset's updated condition, and shall update the asset's condition and availability in its record. | Essential | UC-6 | A request cannot be marked 'Completed' without an updated condition; the asset record shows the new condition immediately afterwards. |
| **FR-5.9** | While an asset has a maintenance request in the 'Under Inspection' or 'Repair in Progress' state, the system shall not show it as available in search, offer it in matching results, or allow it to be reserved or transferred. | Essential | UC-2, UC-3, UC-5, UC-6, UC-9 | The asset does not appear as available in search or matching, and reserve or transfer attempts are rejected with a message naming the open maintenance request. |
| **FR-5.10** | The system shall display each asset's cumulative maintenance cost (repairs and refurbishments) and number of maintenance actions to the roles allowed to see financial information (SEC-5). | Essential | UC-6 | The displayed total equals the sum of the recorded costs for that asset. |
| **FR-5.11** | When an asset’s cumulative maintenance cost reaches a configurable percentage of its replacement price (default 50%, TBD-7), the system shall flag the asset for end-of-life review and notify its Asset Coordinator and the Sustainability Officer. | Desirable | UC-6, UC-7 | Recording work that crosses the threshold sets the flag and sends both notifications. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-5.12** | When Maintenance Staff close a request as 'Closed – Not Repairable', the system shall flag the asset for end-of-life review and notify its Asset Coordinator and the Sustainability Officer. | Essential | UC-6, UC-7 | Closing a test request as not repairable sets the flag and sends both notifications. |
| **FR-5.13** | The system shall allow the user who reported the defect, the asset's Asset Coordinator or an Administrator to cancel a maintenance request that is not yet completed, with a reason. Cancelled requests remain in the history. | Essential | UC-6 | A cancelled request shows the 'Cancelled' state, the reason and the user; it cannot be cancelled without a reason. |
| **FR-5.14** | The system shall reject a maintenance record that is missing a required field, has a completion date before its start date, has a work or completion date in the future, or has a negative cost, and shall show which field is invalid. | Essential | UC-6 | Each invalid case is rejected, nothing is saved, and the invalid field is identified. |
| **FR-5.15** | The system shall not allow users without the Maintenance Staff or Administrator role to record inspection or repair results. | Essential | UC-6 | Attempts by any other role are denied and written to the audit log. |
| **B. Life-cycle history** | | | | |
| **FR-5.16** | The system shall automatically record a life-cycle event for acquisition (registration), allocation, change in utilization status, inspection, maintenance, repair, transfer, refurbishment, donation, recycling, disassembly, retirement and disposal. Each event shall include date and time, event type, the user who performed it (by internal user ID), the user who approved it (if any) and a reference to the related record. | Essential | UC-1, UC-4, UC-5, UC-6, UC-7 | Performing each event type in a test adds exactly one correctly typed event to the asset's history. |
| **FR-5.17** | The system shall display an asset's life-cycle history in date order, with filters for event type and date range, to the asset's Asset Coordinator, Maintenance Staff, Sustainability Officers, Administrators and members of the department holding the asset. | Essential | UC-6, UC-7 | The listed roles see all events oldest to newest; each filter returns only matching events; other users cannot open the history. |
| **FR-5.18** | The system shall not allow any user, including Administrators and System Administrators, to edit or delete a recorded life-cycle event. A correction shall be recorded as a new event that references the original. | Essential | UC-6, UC-7 | No edit or delete option exists; a correction appears as a new linked event and the original is unchanged. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-5.19** | The system shall show each asset's age (time since its purchase date) and time in service (time since it was first allocated for use). | Desirable | UC-2, UC-7 | Both displayed values equal the time between today and the recorded purchase and first-allocation dates. |
| **C. End-of-life actions: donation, recycling, disassembly, retirement and disposal** | | | | |
| **FR-5.20** | The system shall allow an Asset Coordinator or Sustainability Officer to propose an end-of-life action for an asset (donation, recycling, disassembly for usable components, retirement or disposal) with a written justification. | Essential | UC-7 | A proposal without a justification cannot be submitted; a valid proposal is saved in the 'Pending Safety Review' state. |
| **FR-5.21** | The system shall not accept an end-of-life proposal for an asset that has an open reservation, a pending transfer or an open maintenance request, and shall list the blocking items. | Essential | UC-7 | A proposal for such an asset is rejected and the blocking items are listed. |
| **FR-5.22** | When an end-of-life action is proposed, the system shall show the open requests the asset may satisfy (same category, and requested quantity not more than the asset's available quantity) and, when available, the sustainable-action recommendation from FR-10 with its confidence and explanation. | Desirable | UC-7 | For a test asset that matches an open request by category and quantity, the request is shown on the proposal screen. |
| **FR-5.23** | The system shall show the hazard flag recorded for the asset at registration (FR-1) and require the proposer to confirm or update it. A Sustainability Officer other than the proposer shall then record a safety review confirming or correcting the classification; until then the proposal stays in 'Pending Safety Review'. | Essential | UC-7 | A proposal cannot leave 'Pending Safety Review' until a different Sustainability Officer records the review. |
| **FR-5.24** | For assets confirmed as hazardous or electronic waste, the safety review shall assign a handling party (Facilities Management or a vendor on the approved list in FR-5.25), and the system shall not allow the action to be completed without the handling party's record and certificate. | Essential | UC-7 | A hazardous asset cannot pass the safety review without a handling party, or be completed without the handling record and certificate. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-5.25** | The system shall allow an Administrator to maintain the list of approved recycling and disposal vendors (name, materials handled, contact details and approval expiry date). Only Facilities Management or a vendor on the list with an unexpired approval can be assigned as a handling party. | Essential | UC-7 | A vendor not on the list, or with an expired approval, cannot be assigned. |
| **FR-5.26** | After the safety review, the system shall set the proposal to ‘Pending End-of-Life Approval’, route it through the approval stages in BR-8, and shall not change the asset to its final state until all required approvals and the completion record (FR-5.29, FR-5.30 or FR-5.31) are recorded. | Essential | UC-7 | The asset reaches its final state only after the last approval and a complete completion record. |
| **FR-5.27** | While an asset is 'Pending Safety Review' or 'Pending End-of-Life Approval', the system shall not show it as available in search, offer it in matching results, or allow it to be reserved or transferred. | Essential | UC-2, UC-3, UC-5, UC-7, UC-9 | Search, matching, reserve and transfer attempts on a pending test asset do not succeed. |
| **FR-5.28** | When an end-of-life proposal is rejected at safety review or approval, the system shall record the reason and return the asset to its previous state. | Essential | UC-7 | After rejection the asset is back in its previous state and the reason appears in its history. |
| **FR-5.29** | When a donation is completed, the system shall record the recipient organization, date, quantity, weight in kilograms and handover document. | Essential | UC-7 | A completed donation shows all fields; a missing required field blocks completion. |
| **FR-5.30** | When recycling, disassembly or disposal is completed, the system shall record the method, handling party, date, quantity, weight in kilograms and the recycling or disposal certificate (required for hazardous and electronic-waste items; recorded when issued for other items). | Essential | UC-7 | The completed record shows all fields; a hazardous item cannot be completed without its certificate. |
| **FR-5.31** | When a retirement is completed, the system shall record the date, reason, final location (for example central storage) and the user who completed it. | Essential | UC-7 | A completed retirement shows all fields; a missing field blocks completion. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-5.32** | Weight shall be recorded for every completed donation, recycling, disassembly and disposal: the actual weight if known, otherwise the category default weight; if no default exists, the user shall enter an estimated weight. The record shall show whether the weight is actual, a default or an estimate. | Essential | UC-7 | No completion record can be saved without a weight and its type. |
| **FR-5.33** | When a disassembly is completed, the system shall allow usable components to be registered as new asset records linked to the original asset. | Future | UC-1, UC-7 | A new component record links to its parent asset, and the parent shows the link. |
| **FR-5.34** | The system shall notify the proposer, the asset's Asset Coordinator and the user whose action is needed next (safety reviewer or approver) when an end-of-life proposal is submitted, passes or fails the safety review, is approved or rejected, and when it is completed, using the notifications defined in FR-7. | Essential | UC-7 | Each listed event produces notifications to the listed users. |

# 2.6 User Accounts, Access and Departments (FR-6)

## 2.6.1 Description and Priority

This feature controls who can use ReSource, which department they represent and which actions they may perform. The prototype uses local accounts; university single sign-on is outside the current scope. Consent is required at registration, with the notice wording and institutional policy awaiting confirmation (TBD-4).

Priority: High. This feature develops Phase 2 requirement FR-6. Account creation, authentication, role-based access control and department management are part of the minimum prototype [1].

## 2.6.2 Stimulus/Response Sequences

1. An Applicant reads the privacy notice and terms, enters the required details and explicitly consents. The system validates the application and records consent before retaining the registration (UC-13).

2. The System Administrator checks the application and recorded authorization, then activates the account with its permitted role and department (UC-8).

3. An active user signs in. The system authenticates the user and presents only authorized functions. Signing out ends the session.

4. The System Administrator changes an account, role or department. The system checks authority and dependencies, saves the change, audits it and notifies affected users.

5. A user requests a correction or withdraws consent. The system records the request; withdrawal stops account access and starts the approved data-handling process.

### 2.6.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-6.1** | The system shall provide self-registration using name, university ID, university email, department and a password. It shall validate required fields and prevent duplicate university IDs or email addresses. | Essential | UC-13 | A valid unique application is accepted. Missing fields, invalid credentials and duplicate identifiers are rejected without creating another account. |
| **FR-6.2** | Before saving a registration, the system shall display the current privacy notice and terms, explain the intended data uses and withdrawal route, and require an explicit, initially unticked consent choice. It shall record the account identifier, consent time and notice version. | Essential | UC-13 | No account or registration data is retained when consent is absent. An accepted registration has a retrievable consent time and notice version. |
| **FR-6.3** | The system shall place new self-registrations in Pending approval. Only a System Administrator shall activate them and assign an approved role and department. Applicants shall not grant themselves privileges. | Essential | UC-8, UC-13 | A new applicant cannot use asset functions or choose privileged access. Activation is recorded with the assigned role and department. |
| **FR-6.4** | The system shall let a System Administrator initiate an account invitation without retaining the invitee’s personal details before consent. The invitee shall provide their own details, password and consent before account activation. | Essential | UC-8, UC-13 | An invitation alone creates no active personal account. Omitting consent prevents activation; consent is recorded on completion. |
| **FR-6.5** | The system shall authenticate active accounts, provide sign-out, and enforce the password, lockout and session rules in SEC-2 to SEC-4. Failed sign-in shall not grant access. | Essential | UC-1 to UC-13 | Valid active credentials sign in. Incorrect credentials and pending, suspended or closed accounts cannot enter protected functions; sign-out invalidates the session. |
| **FR-6.6** | The system shall offer credential recovery only after verifying control of the registered account through an approved recovery process. It shall not reveal whether a submitted account identifier exists. | Essential | UC-13 | Unverified recovery cannot reset credentials. Existing and unknown identifiers receive the same initial response. The recovery method is confirmed under TBD-3. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-6.7** | The system shall check the user's current role, department and account status on every protected action and record access. Restricted actions shall remain blocked through direct links, reports and AI features. | Essential | UC-1 to UC-13 | Each unauthorized role and department test is denied with no disclosure or state change; legitimate cross-department catalogue access still works. |
| **FR-6.8** | The system shall let only System Administrators assign, change or revoke roles and department membership, using recorded authorization and a reason. Users shall not change their own privileged access. | Essential | UC-8 | An authorized change saves its approver and reason. Self-escalation and changes by other roles are refused and logged. |
| **FR-6.9** | The system shall let System Administrators create, rename and deactivate departments with unique IDs. Deactivation shall be blocked until active users, assets and open work are reassigned. Historical department references shall remain available. | Essential | UC-8 | A duplicate ID is rejected. A department with unresolved dependencies cannot be deactivated; a clear department can, without losing history. |
| **FR-6.10** | The system shall let System Administrators suspend or close an account and revoke its active sessions. It shall preserve associated records and flag open work for reassignment to an authorized person. Suspending or closing the last active System Administrator account shall be blocked until another System Administrator account is active. | Essential | UC-8 | The next protected action after suspension or closure is denied. Existing asset and request history remains; outstanding work is listed. |
| **FR-6.11** | The system shall let users review their account data, request corrections and withdraw consent. Withdrawal shall block further account use and notify the System Administrator to apply the confirmed retention and erasure policy. | Essential | UC-8 | A correction or withdrawal request is recorded. Withdrawal ends active access; subsequent data handling follows the approved policy in SEC-8 and TBD-4. |
| **FR-6.12** | The system shall audit account, consent, role and department changes and notify affected users of access changes. Passwords and recovery secrets shall not appear in audit entries or notifications. | Essential | UC-8 | Each tested change records actor, time and result; the affected user receives an appropriate notice without secrets. |

# 2.7 Notifications, Reports and Audit Log (FR-7)

## 2.7.1 Description and Priority

This feature informs users of workflow events that need their attention, provides conventional, rule-based operational reports with fixed totals and tables, and keeps a complete, tamper-proof audit log of user and system actions. It refines Phase 2 requirement FR-7 and the project's requirements for notifications, operational reports and auditability. Notifications are in-application messages; email copies can be added once the delivery service is confirmed (TBD-6). Unlike the AI-generated reports in FR-12, the operational reports are deterministic: the same data and filters always give the same result, which makes them the reference source for every figure the system shows.

Priority: High. This feature develops Phase 2 requirement FR-7. Notifications and basic operational reports are part of the minimum prototype [1], and the audit log is required for security and legal compliance (3.3, 4.2).

## 2.7.2 Stimulus/Response Sequences

1. Workflow event recorded: a request, approval, transfer, maintenance, end-of-life or account event is saved → the system creates in-application notifications for the recipients in Table 2.7-1 and retries any failed delivery.
2. Report requested: an authorized user selects a report type and filters → the system generates the report from current data, limited to what the user may see.
3. Any recorded action**:** a user creates, changes, approves or rejects a record, signs in, is denied access, or uses an AI output → the system writes an audit entry automatically.
4. Audit search: a System Administrator or Administrator searches the audit log → the system lists the matching entries.

## 2.7.3 Functional Requirements

### Notification requirements (FR-7.N)

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-7.N1** | After a workflow event is successfully recorded, the system shall create an in-application notification for the responsible users listed in Table 2.7-1. | Essential | UC-3 to UC-9, UC-12, UC-13 | Each listed test event creates the expected recipient notifications; a failed transaction creates no success notification. |
| **FR-7.N2** | Each notification shall identify the event, linked record, current status, time and required next action. Links and content shall respect the recipient's current permissions and exclude passwords and restricted personal or financial data. | Essential | UC-3 to UC-9, UC-12, UC-13 | Every notice contains the required fields. A user whose access was revoked cannot open the linked protected record. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-7.N3** | The system shall show required event notifications within five minutes of a committed event under the PER-1 test workload. A notification failure shall not reverse a valid, audited business action. | Essential | UC-3 to UC-9, UC-12, UC-13 | All events in the notification test appear within five minutes. A forced sending failure leaves the saved decision and audit entry intact. |
| **FR-7.N4** | The system shall let signed-in users view their own notification history and mark notifications as read or unread. Reading a message shall not approve, accept or complete its linked action. | Essential | UC-3 to UC-9, UC-12, UC-13 | A user sees only their own notices; changing read status makes no change to any business record. |
| **FR-7.N5** | The system shall retain failed notification deliveries for retry and make failures visible to a System Administrator. Retrying shall not create duplicate in-application messages for the same event and recipient. | Essential | UC-3 to UC-9, UC-12, UC-13 | A simulated failure is visible; recovery produces exactly one in-application message per event and recipient. |
| **FR-7.N6** | When the desirable expiry and reminder rules are enabled, the system shall notify affected users when a request or reservation expires under FR-3.9 or a receipt remains unconfirmed for five working days (FR-4.13). Email copies may be enabled after the delivery service is confirmed. | Desirable | UC-3, UC-4, UC-5 | Each enabled deadline generates one notice to the specified recipients. Calendar rules and email configuration match TBD-5 and TBD-6. |

### Table 2.7-1: Notification routing

| Recorded event | Recipients |
|---|---|
| Specific-asset request submitted | Owning Asset Coordinator; submitting Department Representative receives confirmation. |
| Unmatched need submitted | Procurement Team; submitting Department Representative receives confirmation. |
| Match chosen for an unmatched need (UC-9) | Requesting Department Representative; Asset Coordinator of the chosen asset, who reviews the request (UC-4). |
| Asset released | Administrator; requesting Department Representative and owning Asset Coordinator. |
| Additional approval required | Procurement Team; Administrator. |
| Request approved, rejected, cancelled or expired | Requesting Department Representative and owning Asset Coordinator; any approver with a pending task. |

| Recorded event | Recipients |
|---|---|
| Transfer scheduled, changed, cancelled, handed over or received | Originating Asset Coordinator and receiving Department Representative; receiving Asset Coordinator where assigned. |
| Transfer disputed or receipt reminder due | Administrator and the affected departments' representatives and coordinators. |
| Defect or inspection reported; maintenance assigned, completed or cancelled | Maintenance Staff for new or assigned work; reporting user and Asset Coordinator for outcomes and cancellations. |
| Asset marked not safe to use, or withdrawn while requests are open | Department Representatives with open or approved requests for the asset; owning Asset Coordinator. |
| End-of-life review, proposal, decision or completion | Asset Coordinator, Sustainability Officer and the next required approver; proposer receives the outcome. |
| Human review of an AI result requested (SEC-10) | Administrator; the requesting user receives confirmation and the outcome. |
| Account decision, access change, lockout or consent withdrawal | Affected account holder; System Administrator for action or delivery failure. An inactive user sees the outcome through the sign-in or account-status route after identity verification. |
| AI report ready for review or approved for sharing | Requesting reviewer; permitted recipients only after approval. |
| Scheduled standard report generated (FR-7.R8) | User who scheduled the report. |

Recipients are de-duplicated where one person fills several permitted roles. A notification contains only information the recipient may see. Deadline rules use the confirmed working-day calendar (TBD-5). Inactive-account notices use the limited verified account-status route described in FR-6; email delivery remains subject to TBD-6.

### Report and audit requirements (FR-7.R, FR-7.A)

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **A. Standard operational reports** | | | | |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-7.R1** | The system shall allow Asset Coordinators, Department Representatives, Procurement Team members, Finance Officers, Sustainability Officers, Administrators and University Management to generate these standard reports: available assets; open requests; completed transfers; maintenance history; ownership and custody changes; approval decisions; asset status by life-cycle state; and end-of-life actions (donations, recycling, disassembly, retirement, disposal). | Essential | UC-12 | Each listed role can generate each of the eight reports from test data; Maintenance Staff and System Administrator accounts cannot. |
| **FR-7.R2** | The system shall allow each standard report to be filtered by date range, department, category, location and status. | Essential | UC-12 | Each filter returns only matching records. |
| **FR-7.R3** | A report shall include only records from the user’s own department, plus published surplus assets, unless the user’s role covers all departments (Procurement Team, Finance Officer, Sustainability Officer, Administrator, University Management). Financial fields (estimated values, replacement prices, costs, savings) shall be shown only to roles allowed to see financial information (SEC-5). | Essential | UC-12 | A Department Representative's report contains only their department's records and published assets, and no financial fields. |
| **FR-7.R4** | Each report shall show its title, the filters applied, the date and time generated, the generating user and the number of records. | Essential | UC-12 | All five items appear on every report. |
| **FR-7.R5** | Standard reports shall be calculated directly from recorded data without AI, so the same data and filters always produce the same results. | Essential | UC-12 | Generating a report twice with unchanged data gives identical results. |
| **FR-7.R6** | When no records match the selected filters, the system shall show an empty report stating that no records matched, instead of an error. | Essential | UC-12 | Filters with no matches return the message and no error. |
| **FR-7.R7** | The system shall allow standard reports to be downloaded as a PDF file and as a spreadsheet-compatible (CSV) file. | Desirable | UC-12 | Both files open and contain the same records as the on-screen report. |
| **FR-7.R8** | The system shall allow a Finance Officer, Sustainability Officer or Administrator to schedule a standard report to be generated monthly and delivered by notification. | Future | UC-12 | A scheduled report is generated on the first day of each month and the user is notified. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **B. Audit log** | | | | |
| **FR-7.A1** | The system shall record an audit entry for every create, update, approval, rejection, cancellation, AI-recommendation override, sign-in, failed sign-in, denied access attempt, role or permission change, consent given or withdrawn, human-review request, data export, and view of asset or transfer records. Each entry shall include date and time, internal user ID, role, action, affected record, and the old and new values of changed fields; entries shall never store passwords or personal-data field values. | Essential | UC-1 to UC-13 | Each listed action in a test produces one complete audit entry, and no entry contains a password, name, university ID or email. |
| **FR-7.A2** | The system shall not allow any user, including System Administrators, to edit or delete audit entries. | Essential | UC-1 to UC-13 | No edit or delete function exists, and attempts through the interface are denied and logged. |
| **FR-7.A3** | The system shall allow only System Administrators and Administrators to view the audit log, and to search it by user, record, action type and date range. | Essential | UC-8 | Other roles cannot open the audit log; each search filter returns only matching entries. |
| **FR-7.A4** | For each AI output used in a decision (classification, match, recommendation, assistant action or report), the system shall record the model version, the records used as input, the output, its confidence, its explanation, and whether the user accepted, edited or rejected it (with the user's reason for an override). | Essential | UC-7, UC-9, UC-10, UC-11, UC-12 | Every AI-assisted decision in a test has a complete entry, including model version and explanation. |
| **FR-7.A5** | The system shall keep audit entries for at least the retention period set by Khalifa University IT policy (TBD-4); until that period is confirmed, no audit entry shall be deleted. | Essential | UC-1 to UC-13 | No audit entries are removed during the test period. |
| **FR-7.A6** | If an audit entry cannot be recorded, the system shall not complete the action and shall show an error. | Essential | UC-1 to UC-13 | With audit recording forced to fail, the action is not saved and an error is shown. |

# 2.8 AI Resource Matching and Ranking (FR-8)

## 2.8.1 Description and Priority

This feature compares a department’s request with available campus assets. It understands the meaning of the request as well as the recorded details. It gives the Procurement Team a ranked

list, a compatibility score and reasons for each match. The selected asset still follows the normal release, approval and reservation process (FR-4).

Priority: High. This feature develops Phase 2 requirement FR-8. AI-based matching, ranking, explanation and user review are part of the minimum prototype [1].

### 2.8.2 Stimulus/Response Sequences

1. A Procurement Team member opens a request they are allowed to review and asks for matches. The system checks the request details and available assets.

2. The system ranks suitable assets and shows their scores, confidence, permitted cost details, repair needs and available sustainability estimates.

3. The Procurement Team member accepts a suggestion or chooses another suitable asset. The system records the choice and sends it through the normal release and approval process (UC-4).

4. If details are invalid, no match is found or AI fails, the system explains the problem and offers correction or standard search.

### 2.8.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-8.1** | The system shall match the meaning of a request to asset descriptions and specifications, even when different words are used. | Essential | UC-9 | A test request for a “workstation for office tasks” finds a “desktop computer” that meets the required specifications. |
| **FR-8.2** | The system shall check required request details and exclude assets that fail access, availability, safety or essential specification checks, including assets under maintenance or pending end-of-life action (FR-5.9, FR-5.27), unsafe assets (SAF-5) and hazardous assets without a safety review (SAF-1). | Essential | UC-9 | Missing required details are identified. Unavailable, restricted, unsafe or incompatible test assets do not appear as suitable matches. |
| **FR-8.3** | The system shall consider category, purpose, specifications, quantity, current and requested condition, location, urgency, required date, transfer cost and repair cost. | Essential | UC-9 | Each factor shows its value or unknown. The system does not relax a required condition or delivery date without the proper request change. |
| **FR-8.4** | The system shall give each suitable match a compatibility score from 0 to 100 and explain its main strengths and limitations. | Essential | UC-9 | Every match has a score within the range, a separate confidence label and reasons based on the records. Missing required details prevent scoring. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-8.5** | The system shall rank suitable assets using documented rules for suitability, urgency, transfer cost, repair needs, waste reduction and estimated carbon reduction. The ranking shall not use the requesting department's name or size as a factor (LEG-9). | Essential | UC-9 | Results identify the rule version and explain the main trade-offs. A high score never overrides safety or essential request requirements. Changing only the requesting department does not change the ranking. |
| **FR-8.6** | The system shall clearly show when an asset meets only part of the requested quantity. A revised or split request shall be confirmed by the requesting Department Representative and approved under FR-4 before any quantity is reserved. | Essential | UC-9 | If five items are requested and only three are available, the result says partial match. The system cannot reserve five or silently change the request to three. |
| **FR-8.7** | The system shall let the Procurement Team member accept, reject or choose another suitable match, and record a reason for rejecting or changing the suggested choice. | Essential | UC-9 | The record includes the selected asset, original suggestion, reviewer, time and reason. Choosing another asset still requires the usual approvals. |
| **FR-8.8** | The system shall recheck availability, quantity, permissions and request details just before sending the chosen asset to the release and approval process (FR-4). | Essential | UC-9 | If the asset is no longer available, confirmation is blocked and updated options are shown. The same available units are never offered to two requests at confirmation. |
| **FR-8.9** | The system shall explain when no suitable match is found and offer standard search or a request change. | Essential | UC-9 | Tests with no match or an AI failure keep the original request and offer a usable alternative. The system does not recommend an unsuitable asset. |

# 2.9 AI Asset Classification (FR-9)

## 2.9.1 Description and Priority

This feature helps the Asset Coordinator register or update an asset. It suggests details from the title, description, specifications, photos and documents provided. The Asset Coordinator checks the suggestions before they are saved.

Priority: High. This feature develops Phase 2 requirement FR-9. AI-assisted classification is part of the minimum prototype [1].

### 2.9.2 Stimulus/Response Sequences

1. The Asset Coordinator enters a title and description, adds available specifications and files, and asks for classification suggestions.

2. The system suggests a category, subcategory, tags, possible material type and description. It shows confidence, reasons and any missing details.

3. The Asset Coordinator accepts or corrects the suggestions. The system checks the record and saves only confirmed values.

4. If information conflicts, a file cannot be read or AI is unavailable, the system explains the issue and lets the Asset Coordinator complete the record manually.

### 2.9.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-9.1** | The system shall use available titles, descriptions, specifications, images and documents to suggest asset details, and identify any source it cannot read. | Essential | UC-10 | A test asset with text, an image and a document shows which sources were used. Unreadable or unsupported sources are clearly identified. |
| **FR-9.2** | The system shall suggest a category and, where relevant, a subcategory from the approved category list. | Essential | UC-10 | Every suggestion uses an approved category. If no suitable category can be identified, the system asks for manual classification. |
| **FR-9.3** | The system shall suggest consistent tags, a factual description and a possible material type when the supplied information supports them. | Essential | UC-10 | Suggestions are shown separately from confirmed details. Material, capacity, purchase or ownership details are not invented. |
| **FR-9.4** | The system shall identify missing registration details and ask the Asset Coordinator to enter them. | Essential | UC-10 | Missing quantity, owner, department, location, condition and other required fields are flagged. The system does not fill them with guesses. |
| **FR-9.5** | The system shall show confidence in its classification and explain any conflicting information. | Essential | UC-10 | If a title and image disagree, the result shows Low confidence and explains what the Asset Coordinator needs to check. |
| **FR-9.6** | The system shall let the Asset Coordinator accept, edit or reject each suggestion before saving. | Essential | UC-10 | The saved category and tags match the Asset Coordinator’s confirmed choices. Rejected suggestions do not overwrite the record. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-9.7** | The system shall check the confirmed category and required asset details before registration or publication. | Essential | UC-10 | An invalid category, missing required field or unauthorized save is rejected with a clear message. Valid records continue through UC-1. |
| **FR-9.8** | The system shall keep entered asset details when AI classification fails and allow manual completion. | Essential | UC-10 | Tests for timeouts, unreadable files and service failures keep the entered details and allow the Asset Coordinator to choose a category manually. |
| **FR-9.9** | The system shall record the original AI suggestion and the Asset Coordinator’s correction for review, without automatically retraining the AI. | Essential | UC-10 | The log shows the suggested and confirmed values, the person and the time. A correction does not silently change the model version. |

# 2.10 Sustainable-Action Recommendations (FR-10)

## 2.10.1 Description and Priority

This feature helps the Sustainability Officer and Maintenance Staff decide what should happen to an asset next. It compares options using condition, safety, remaining useful life, demand, maintenance history, available costs and supported environmental estimates. The recommendation still needs the required human approval.

Priority: Medium. This feature develops Phase 2 requirement FR-10. Sustainable-action recommendations are not in the minimum prototype [1], so the feature is planned as a desirable extension; when it is provided, every requirement below applies in full.

## 2.10.2 Stimulus/Response Sequences

1. The Sustainability Officer selects an asset they may review and asks for advice. The system checks its condition, inspections, maintenance history, demand and costs.

2. The system shows suitable actions with reasons, estimates, confidence and limitations. It asks for inspection or missing details where needed.

3. The officer selects an action or gives a reason for choosing another. Repair and transfer choices follow their normal processes. End-of-life actions follow UC-7.

4. The Administrator approves or rejects the proposed action under the agreed rules. The system records the decision. A recommendation alone does not change the asset's status.

## 2.10.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-10.1** | The system shall consider keeping, transferring, repairing, refurbishing, donating, taking apart for usable parts, recycling and disposing of an asset as separate options. | Desirable | UC-7 | The review shows suitable options and explains why others are ruled out. Each action type is included in at least one test scenario. |
| **FR-10.2** | The system shall consider condition, safety limits, maintenance history, demand, remaining useful life, repair and transfer costs, and approval rules. | Desirable | UC-7 | The review identifies supporting records and unknown details. Old or conflicting condition information is flagged before recommending an action. |
| **FR-10.3** | The system shall prefer suitable ways to keep assets or materials in use and recommend disposal only when the other options are unsuitable. | Desirable | UC-7 | A disposal recommendation explains why keeping, transferring, repairing, refurbishing, donating, reusing parts and recycling are unsuitable or unavailable. Missing information alone is not enough. |
| **FR-10.4** | The system shall explain the recommended action and show its confidence and available estimates for cost, waste diversion and carbon reduction. | Desirable | UC-7 | Each estimate includes units, input sources and assumptions. If a needed value is missing, the estimate is marked unknown. |
| **FR-10.5** | The system shall not recommend immediate reuse, transfer or disassembly while a safety issue or required inspection is unresolved. | Desirable | UC-7 | An unsafe or uninspected asset is sent for assessment. Accepting an AI suggestion cannot remove the safety restriction. |
| **FR-10.6** | The system shall let an authorized Sustainability Officer or Maintenance Staff reviewer reject the suggestion or propose another action with a reason. | Desirable | UC-7 | The system keeps the original suggestion and records the chosen action, reason, reviewer and time for the approval stage. |
| **FR-10.7** | The system shall send proposed actions through the required human approval process before changing the asset's recorded status. | Desirable | UC-7 | Approval and completion are recorded separately. A suggestion alone cannot mark an asset as transferred, repaired, donated, recycled or disposed of. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-10.8** | The system shall summarize available information and allow manual review when information is incomplete or AI is unavailable. | Desirable | UC-7 | An incomplete review lists what is missing and offers manual review. The system does not invent a recommendation or block authorized manual work. |

# 2.11 AI Assistant (FR-11)

## 2.11.1 Description and Priority

The AI Assistant lets a Department Representative use everyday language to find assets, prepare requests, ask about asset history or request status, read approved policy guidance and understand sustainability figures. Other roles can use these functions within their own access permissions.

Priority: Medium. This feature develops Phase 2 requirement FR-11. The conversational assistant is not in the minimum prototype [1], so it is planned as a desirable extension; standard search and forms provide the same functions without AI.

## 2.11.2 Stimulus/Response Sequences

1. The Department Representative asks a question or describes what they need. The system identifies the task and checks their access permissions.

2. The assistant asks for clarification if the request is unclear. Otherwise, it searches permitted records or prepares a draft.

3. Before submitting a request or changing a record, the assistant shows the final details and asks for confirmation. Normal checks and approval rules still apply.

4. If no reliable source is available, the assistant says so and offers a relevant link or manual option. If an action fails, it reports the failure.

## 2.11.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-11.1** | The system shall accept asset searches written in everyday language and return only assets the user may view and that are not restricted (FR-5.9, FR-5.27, SAF-1, SAF-5), or ask for clarification. | Desirable | UC-11 | A search using different wording finds relevant suitable assets. “I need equipment” prompts a question about the category or purpose. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-11.2** | The system shall prepare a request draft with the category, purpose, specifications, quantity, preferred condition, urgency, location and required date. | Desirable | UC-3, UC-11 | The user can review and correct every required field. Missing or invalid values must be corrected before submission. |
| **FR-11.3** | The system shall show the proposed action and final values and ask for confirmation before submitting or changing a record. | Desirable | UC-3, UC-11 | An unconfirmed draft is not submitted. Only the displayed, checked values are submitted. Important changes require confirmation again. |
| **FR-11.4** | The system shall answer asset-history and request-status questions using records the user may access, and identify the asset or request. | Desirable | UC-11 | Answers match the stored history and status. The assistant does not invent missing records or reveal restricted details. |
| **FR-11.5** | The system shall answer policy questions only from available approved documents and identify the source and its version or date. | Desirable | UC-11 | Each policy answer identifies its source. If no approved source is available, the assistant says so and refers the user to the responsible role. |
| **FR-11.6** | The system shall answer sustainability questions using recorded indicators and clearly separate estimated benefits from completed results. | Desirable | UC-11 | Answers use the values, units and reporting period in the source records, and explain assumptions and missing information. |
| **FR-11.7** | The system shall provide links to relevant assets, requests, history, policies or reports that the user is allowed to access. | Desirable | UC-11 | Each link opens the correct record or function. The assistant cannot give access to a restricted destination. |
| **FR-11.8** | The system shall check current permissions and normal system rules before each action and report what actually happened. | Desirable | UC-11 | Expired access, changed availability or invalid quantities block the action. Failed submissions are reported as failed and are not duplicated. |
| **FR-11.9** | The system shall keep the request draft and offer standard search or forms when the AI interaction cannot be completed. | Desirable | UC-11 | An interrupted interaction keeps confirmed draft values, explains the problem and provides a usable manual option. |
| **FR-11.10** | The system shall let the Department Representative cancel a proposed action before confirming it. | Desirable | UC-11 | Cancellation leaves existing records unchanged and creates no request, reservation or approval action. |

# 2.12 AI-Generated Reports (FR-12)

## 2.12.1 Description and Priority

This feature turns recorded operational and sustainability data into written reports: monthly summaries, savings and sustainability reports, sustainability updates, management bulletins, high-demand-category summaries, maintenance insights and lists of recommended actions. It refines Phase 2 requirement FR-12 and the project's Generative AI Reporting functionality. Where a classic report shows fixed totals and tables (FR-7), an AI report can summarize trends, explain unusual changes and suggest management actions. Because AI text can be wrong, every figure in an AI report must come from the deterministic reports and indicators (FR-7, FR-13), assumptions and missing data must be stated, and a person must approve the report before anyone else can see it.

Priority: High for the monthly activity summary and the savings and sustainability report (the minimum prototype includes basic AI-generated reports [1]); Medium for the other report types. This feature develops Phase 2 requirement FR-12.

## 2.12.2 Stimulus/Response Sequences

1. Report requested: an authorized user selects an AI report type, period and scope → the system calculates the underlying figures, produces a draft labeled as AI-generated, and checks its figures.
2. Draft reviewed: the user edits and approves the draft → the system re-checks the figures, stores both versions, records the approval and makes the report available to authorized recipients.
3. AI unavailable: the AI service fails or times out → the system informs the user and offers the standard reports and indicators instead.

## 2.12.3 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-12.1** | The system shall allow a Finance Officer, Sustainability Officer, Procurement Team member or Administrator to request an AI-generated report for a selected period and scope (campus or a department). | Essential | UC-12 | Each of the four roles can request a report; other roles cannot. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-12.2** | The system shall offer these AI report types, each containing at least the listed content: (a) monthly activity summary – counts of new listings, requests, transfers, maintenance actions and end-of-life actions compared with the previous month; (b) savings and sustainability report – indicators 1–9 (Table 2.13-1) with their assumptions; (c) sustainability update – reuse, waste diversion and carbon indicators; (d) management bulletin – the three largest changes and recommended actions; (e) high-demand-category summary – the five most requested categories and unmet requests; (f) maintenance insights – most-repaired categories, maintenance cost and not-repairable rate; (g) recommended actions list. | Essential (a, b); Desirable (c–g) | UC-12 | Each draft contains every item listed for its type. |
| **FR-12.3** | The system shall base each AI report only on data recorded in the system for the selected period and within the requesting user's permitted scope. | Essential | UC-12 | A department-scope report contains no data from other departments. |
| **FR-12.4** | Every figure stated in an AI report shall equal the corresponding value in the standard reports (FR-7) or indicators (FR-13), or a change calculated from them, for the same period. The system shall check the draft and again check the version submitted for approval, flag any statement whose figure does not match, and not allow the report to be approved until it is corrected. | Essential | UC-12 | 100% of figures in approved test reports match the calculated values; a wrong figure added during editing is flagged and blocks approval. |
| **FR-12.5** | Each AI report shall list the assumptions and missing data behind any estimate, and shall label sustainability and savings figures as estimates. | Essential | UC-12 | Every report has an 'Assumptions and missing data' section, and every estimated figure carries the label. |
| **FR-12.6** | Each AI report shall show an 'AI-generated' label, the generation date, the period covered, the model version and the data sources used. | Essential | UC-12 | All five items appear on every AI report. |
| **FR-12.7** | Each recommended action in an AI report shall include a short explanation and a confidence level (AI-10). The system shall not carry out any recommended action automatically. | Essential | UC-12 | Every recommendation shows both items, and no record changes as a result of generating a report. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-12.8** | The system shall highlight changes larger than a configurable percentage (default 25%, TBD-13) compared with the previous period, calculated from FR-7 and FR-13 values, and give a possible explanation for each. | Desirable | UC-12 | A 30% change in test data is highlighted with an explanation; a 10% change is not. |
| **FR-12.9** | An AI report shall remain a ‘Draft’ until an authorized reviewer reviews it, edits it if needed and approves it. A report that contains financial figures shall be approved only by a Finance Officer or an Administrator; a report without financial figures may also be approved by a Sustainability Officer (BR-11). Only approved reports can be shared with University Management or other users, and financial figures in a shared report shall be hidden from recipients not allowed to see financial information (SEC-5). | Essential | UC-12 | Recipients cannot open a draft; after approval they can, and a recipient without financial access sees no financial figures. A Sustainability Officer cannot approve a report that contains financial figures. |
| **FR-12.10** | The system shall keep both the original AI draft and the approved version of each report, and record who edited and approved it. Discarded drafts shall be kept with the state 'Discarded'. | Essential | UC-12 | Both versions and the approver can be retrieved for every published report; discarded drafts are retrievable. |
| **FR-12.11** | If the AI service is unavailable, fails or does not return a usable result within the time limit in PER-5, the system shall inform the user and offer the standard reports (FR-7) and indicators (FR-13) for the same period instead. | Essential | UC-12 | With the AI service disabled, the user sees the message and can open the standard report. |
| **FR-12.12** | If the selected period has no recorded activity, the system shall say so and shall not generate a written summary. | Essential | UC-12 | An empty period returns the message and no narrative text. |

# 2.13 Sustainability Indicators and Dashboard (FR-13)

## 2.13.1 Description and Priority

This new feature calculates the sustainability indicators the project asks for (waste diversion, avoided purchases, asset-life extension, reuse rate, repair, financial savings and estimated carbon-emission reductions) and shows them on a dashboard. The indicators are calculated with fixed, published formulas rather than by AI, so they can be checked by hand and validated during system testing.

Each indicator shows its formula, its assumptions and how much of it relies on default values, which is how the system communicates the uncertainty in its sustainability estimates. The

figures are estimates for internal planning; consistent with the Phase 2 scope, they are not certified carbon or financial accounts.

Priority: High. Sustainability indicators and dashboard summaries are part of the minimum prototype [1].

### 2.13.2 Stimulus/Response Sequences

1. Action completed: a transfer, repair, refurbishment, donation, recycling, disassembly, retirement or disposal is completed → the system includes it in the indicators for that period.
2. Dashboard opened: a user selects a period and scope → the system shows each indicator the user may see, its change from the previous period, its trend and its data quality.
3. Value updated: a Finance Officer or Sustainability Officer updates a category default value → the system saves it as a new version with its source and effective date.

### 2.13.3 Indicator Definitions

**Table 2.13-1: Sustainability indicator definitions**

| # | Indicator | How it is calculated | Unit |
|---|---|---|---|
| **1** | Assets reused | Number of completed internal transfers that fulfilled a departmental request, counted per unit of quantity. | count |
| **2** | Reuse rate | Assets reused ÷ (assets reused + donated + recycled + disassembled + retired + disposed) × 100. Shows how many of the assets leaving a department stayed in use on campus. | % |
| **3** | Avoided purchases | Count = assets reused (indicator 1). Value = Σ (quantity × replacement price), using the replacement price recorded on the request or asset, or the category default. | count, AED |
| **4** | Financial savings (net) | Avoided purchase value + avoided storage and handling cost (per reused asset, category default) - transfer costs recorded on the completed transfers - maintenance costs recorded on the reused assets between their publication as surplus and their transfer. | AED |
| **5** | Waste diverted | Total weight of assets reused, donated, recycled or disassembled (FR-5.32 weight rule). Each asset unit is counted once per period, even if it was refurbished before being reused. | kg |
| **6** | Waste diversion rate | Waste diverted ÷ (waste diverted + weight of assets disposed) × 100. | % |
| **7** | Repairs | Number of maintenance requests completed with work type repair or refurbishment (FR-5.6). Repair success rate = those requests ÷ (those requests + requests closed as not repairable) × 100. | count, % |
| **8** | Asset-life extension | Σ additional years of use for repaired, refurbished or reused assets: the Maintenance Staff estimate recorded under FR-5.6 where present, otherwise the category default. Each asset unit is counted once per period. | years |

| # | Indicator | How it is calculated | Unit |
|---|---|---|---|
| **9** | Estimated carbon reduction | Σ (weight of reused asset units in kg × the category emission factor in kg $CO_2e$ per kg, Table 2.13-2). The weight is the recorded weight or the category default weight (FR-13.6). Each asset unit is counted once per period. | kg $CO_2e$ |

Default emission factors come from the U.S. Environmental Protection Agency Waste Reduction Model (EPA WARM), version 16, December 2023 [15]. WARM source reduction factors give the greenhouse gas emissions avoided when a new item does not have to be made, which is what reuse instead of a new purchase achieves. WARM states them in metric tons of $CO_2e$ per short ton, as negative numbers because they are reductions; the system stores them as kg $CO_2e$ per kg (1 metric ton per short ton = 1.102 kg per kg). The factors are based on U.S. production data, so the results are planning estimates (FR-13.12), and the Sustainability Officer confirms or replaces them under TBD-8.

**Table 2.13-2: Default emission factors (EPA WARM version 16, source reduction)**

| Asset category | WARM material | WARM factor ($MTCO_2e$ per short ton) | Default factor (kg $CO_2e$ per kg) |
|---|---|---|---|
| **Desktop computers** | Desktop CPUs | 20.86 | 22.99 |
| **Laptops, tablets and phones** | Portable Electronic Devices | 29.83 | 32.88 |
| **Monitors and screens** | Flat-Panel Displays | 24.19 | 26.66 |
| **Keyboards, mice and other peripherals** | Electronic Peripherals | 10.32 | 11.38 |
| **Printers, copiers and scanners** | Hard-Copy Devices | 7.65 | 8.43 |
| **Other electronic equipment** | Mixed Electronics | 20.79 | 22.92 |
| **Metal furniture and fittings** | Mixed Metals | 3.65 | 4.02 |

Categories without a factor in Table 2.13-2, such as wooden furniture or books, use the WARM factor for the closest material, entered by the Sustainability Officer with its source (FR-13.4). Until a factor is entered, their records are listed under ‘Missing data’ (FR-13.8).

## 2.13.4 Functional Requirements

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-13.1** | The system shall calculate the nine indicators in Table 2.13-1, using the formulas shown, for any selected period, department and category. | Essential | UC-12 | For a prepared test data set, every indicator equals a hand calculation using the same formulas and default values. |
| **FR-13.2** | The system shall calculate indicators with fixed formulas, not with AI, so the same data always gives the same values. | Essential | UC-12 | Recalculating with unchanged data gives identical values. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-13.3** | The system shall count only completed actions. When a completed action is corrected by a linked correction record (SEC-12), the indicators shall use the corrected values, and no asset unit shall be counted twice for the same action. | Essential | UC-5, UC-6, UC-7, UC-12 | Recording a correction to a completed test transfer updates every affected indicator, and the original is not counted twice. |
| **FR-13.4** | The system shall allow a Finance Officer to maintain replacement prices and storage and handling costs, and a Sustainability Officer to maintain default weights, additional years of use and emission factors, for each asset category. Every value shall have a source and an effective date. Default emission factors shall be the EPA WARM version 16 source reduction factors in Table 2.13-2 [15] unless the Sustainability Officer records another named source. | Essential | UC-12 | A value cannot be saved without a source and an effective date, and the default emission factors match Table 2.13-2. |
| **FR-13.5** | The system shall keep every version of each category value and record which version was used in each calculation, so past results can be reproduced. | Essential | UC-12 | After a value changes, a past period's result is reproduced exactly using the old version. |
| **FR-13.6** | When an actual value (such as price or weight) is recorded for an asset, the system shall use it; otherwise it shall use the category default and mark the record as using a default. | Essential | UC-1, UC-7, UC-12 | Records with and without actual values use the correct value and are marked correctly. |
| **FR-13.7** | The system shall label an indicator 'Estimate' when any contributing record uses a default or estimated value, and shall show the percentage of contributing records that did. | Essential | UC-12 | An indicator with 3 of 10 records on defaults shows 'Estimate – 30% default values'. |
| **FR-13.8** | If a value an indicator needs (such as replacement price, emission factor or additional years of use) has neither an actual value nor a category default for a record, the system shall exclude the record from that indicator and list it under 'Missing data' with a count. | Essential | UC-12 | A test record with no price and no default is excluded from indicator 3 and listed. |
| **FR-13.9** | The dashboard shall show each indicator for the selected period, its change from the previous period, and its trend over the last six months (or since the system started being used, if shorter). | Essential | UC-12 | Displayed values and changes match the calculated results. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **FR-13.10** | When a user selects an indicator, the dashboard shall show its definition, formula and assumptions. | Essential | UC-12 | Selecting any indicator shows all three. |
| **FR-13.11** | The dashboard shall show the non-financial indicators (1, 2, 5, 6, 7, 8, 9 and the count part of 3) to all signed-in users, for the whole campus and for their own department. The financial indicators (the AED value in 3, and 4) shall be shown only to the roles allowed to see financial information in SEC-5. | Essential | UC-12 | A Department Representative sees the non-financial indicators for the campus and their own department, and no AED values. |
| **FR-13.12** | The dashboard shall state that its figures are estimates for internal planning and are not certified carbon or financial accounts. | Essential | UC-12 | The statement is visible on the dashboard. |
| **FR-13.13** | The system shall allow indicator results to be downloaded as a CSV file that lists the category value versions used, containing only the indicators the user may see. | Desirable | UC-12 | The downloaded file matches the dashboard and lists the versions used. |

# 2.14 General AI Requirements

## 2.14.1 Rule-Based and AI-Enhanced Functions

The standard system uses fixed rules to control sign-in, access, required fields, quantities, availability and approvals. These rules also control changes to an asset’s location, custody and status. Totals and sustainability indicators are calculated from recorded values and agreed formulas, so the same inputs always give the same result.

AI helps users find assets described in different words, suggests asset details, ranks suitable options and explains possible actions. The assistant turns everyday requests into searches or drafts for the user to review. AI reports in FR-12 summarize recorded figures without changing the calculations. These features can help where an exact keyword search is too limited, but their answers still need checking.

Authorized people therefore make every consequential decision (BR-9), and when AI is unavailable or unreliable, users continue with standard search, forms, manual classification and reports (AI-6).

## 2.14.2 General AI Requirements

AI-1 to AI-10 apply to every AI-assisted feature: matching (FR-8), classification (FR-9), sustainable-action recommendations (FR-10 and FR-5.22), the assistant (FR-11) and AI reports (FR-12). In the table, “AI use cases” means UC-7, UC-9, UC-10, UC-11 and UC-12.

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **AI-1** | The system shall label AI suggestions and show their confidence level (AI-10), supporting information and important limitations. | Essential | UC-7, UC-9 to UC-12 | Each AI result is labeled and shows High, Medium or Low confidence. Tests with missing or conflicting information show the correct warning. |
| **AI-2** | The system shall explain each recommendation in simple language and link it to information the user is allowed to view. | Essential | UC-7, UC-9 to UC-12 | The reviewer can see the supporting fields or records and at least one main reason. Unsupported details are left out or marked unknown. |
| **AI-3** | The system shall let authorized reviewers accept, reject or correct AI suggestions. Rejecting or changing a recommendation that affects an asset or request shall require a reason. | Essential | UC-7, UC-9 to UC-12 | Tests record both the original suggestion and the final choice. A rejected or changed recommendation cannot be submitted without the required reason. |
| **AI-4** | The system shall require user confirmation and the necessary human approvals before an AI-assisted action changes a record. | Essential | UC-7, UC-9 to UC-12 | An AI suggestion alone changes no record. Unconfirmed or unauthorized transfers, repairs, donations, recycling and disposal actions are blocked. |
| **AI-5** | The system shall record the AI feature, time, input record references or a saved copy, model or service version, rule version, result and final human decision. | Essential | UC-7, UC-9 to UC-12 | Each checked result can be traced to these details. If the provider does not supply a version, the record says unavailable and includes any available release ID. |
| **AI-6** | The system shall offer a manual or standard option when AI fails, takes too long or cannot give a reliable result. | Essential | UC-7, UC-9 to UC-12 | Failure tests keep entered data and offer the relevant search, classification, request form or standard report. The system does not show a made-up result. |
| **AI-7** | The system shall apply the user's access permissions to information sent to AI, AI answers and any requested action. | Essential | UC-7, UC-9 to UC-12 | AI answers do not reveal restricted records, financial details or department information. Actions blocked in normal forms are also blocked through AI. |
| **AI-8** | The system shall separate recorded facts from estimates and show the missing data, assumptions, units and calculation sources behind cost or environmental figures. | Essential | UC-7, UC-9, UC-11, UC-12 | Each estimate shows its basis and units. Missing cost or emissions data is marked unknown, rather than counted as zero or reported as a measured campus result. |

| ID | Requirement | Priority | Use case | Acceptance criterion (done when…) |
|---|---|---|---|---|
| **AI-9** | The system shall prevent instructions inside asset descriptions, user messages, uploaded files or retrieved documents from changing permissions or approval rules, or from revealing data the user may not see (Phase 2 risk R10). | Essential | UC-7, UC-9 to UC-12 | Test files that ask AI to reveal restricted data, change rules or approve a transfer cause no unauthorized disclosure or record change. |
| **AI-10** | The system shall assign confidence levels using documented, versioned thresholds: High when the calibrated confidence is at least 0.80, Medium from 0.50 to below 0.80, and Low below 0.50 or whenever required information is missing or conflicting. The thresholds shall be confirmed against evaluation data before release (TBD-12). | Essential | UC-7, UC-9 to UC-12 | Test results with confidence 0.85, 0.65 and 0.30 are labeled High, Medium and Low. A result with conflicting information is labeled Low whatever its score. The threshold version appears in the AI log (AI-5). |

# 3. Other Nonfunctional Requirements

## 3.1 Performance Requirements

The main test uses at least 100 reviewed artificial asset records and five users at the same time, as planned in Phase 2 [5]. The estimate of 150 active accounts refers to a whole semester, not 150 people online at once.

Measure response time from submitting a valid action until the full result or confirmation appears. An AI timeout, service error or empty response counts as a failure. Record the test setup, network conditions, workload and file sizes. Use realistic text and at least one supported photo or document where the feature uses files.

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **PER-1** | The system shall work correctly with 150 registered active accounts, five users working at the same time and at least 100 asset records. | A 30-minute test with different roles has no crashes, unauthorized data access, lost saved records or duplicate records. Total accounts and users online at once are reported separately. | UC-1 to UC-13 | Essential |
| **PER-2** | The system shall show search results, asset details and request status within 3 seconds in at least 95% of valid attempts under the PER-1 workload. | Time at least 100 attempts for each operation. At least 95 of 100 attempts must finish within 3 seconds. Slow attempts and failures are included in the results. | UC-2, UC-3, UC-11 | Essential |
| **PER-3** | The system shall confirm valid record creation, updates and approvals within 5 seconds in at least 95% of attempts under the PER-1 workload. | Time 100 valid save or approval attempts. At least 95 finish within 5 seconds, excluding separate file-upload time. No saved record is lost or duplicated. | UC-1, UC-3 to UC-8, UC-13 | Essential |
| **PER-4** | The system shall return a usable AI result within 15 seconds in at least 90% of interactions for matching, classification and AI reports, and for action advice and assistant answers when those features are provided. | Test at least 20 valid interactions for each AI feature. At least 18 of 20 must meet the limit, and failed interactions count in the total. This develops the Phase 2 target [5]. | UC-7, UC-9 to UC-12 | Essential |
| **PER-5** | The system shall show that an AI request is being processed within 1 second. If no usable result arrives within 20 seconds of submission, the system shall stop waiting, treat the request as failed (AI-6) and offer a manual or standard option. | All 10 simulated timeouts show the processing message within 1 second and the alternative option within 20 seconds. Entered data is kept, and a late AI result cannot carry out an unconfirmed action. | UC-7, UC-9 to UC-12 | Essential |
| **PER-6** | The system shall keep standard forms and search working during an AI outage, within the PER-2 and PER-3 time limits. | Repeat the core operations with AI unavailable. They must still pass the same timing and correctness checks. | UC-1 to UC-7, UC-9 to UC-11 | Essential |

## 3.2 Safety Requirements

These requirements prevent loss, damage or harm that could result from using the system, especially where AI recommendations could influence transfer, repair, donation, recycling or disposal decisions. They keep high-impact decisions under human control, separate the person who proposes an action from the people who check and approve it, and close the safety gaps identified in Phase 2 for hazardous and risky items. External procedures that apply: Khalifa University Facilities Management procedures and approved e-waste and recycling vendors for hazardous items (to be confirmed, TBD-7).

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **SAF-1** | Assets flagged as hazardous or electronic waste shall not be matched, reserved, transferred, donated, recycled or disposed of until a Sustainability Officer has recorded a safety review; their final disposition also requires an assigned handling party (Facilities Management or an approved vendor, FR-5.25). | Test: 100% of match, reserve, transfer and end-of-life completion attempts on an unreviewed hazardous test asset are blocked. | UC-1, UC-2, UC-3, UC-5, UC-7, UC-9 | Essential |
| **SAF-2** | No AI output shall automatically carry out a transfer, repair, donation, recycling, retirement or disposal; each is carried out only by an explicit action of an authorized user (an approval, or Maintenance Staff recording work). | Test: 0 record changes caused directly by AI outputs across all AI test cases; the audit log shows the user who performed every such action. | UC-4 to UC-7, UC-9 to UC-12 | Essential |
| **SAF-3** | The user who proposes an end-of-life action shall not perform its safety review or approve it. | Test: 100% of safety-review and approval attempts by the proposer are rejected. | UC-7 | Essential |
| **SAF-4** | Disposal shall be the last option: a disposal proposal shall require a separate reason for each alternative (reuse, repair, donation and recycling) explaining why it is not suitable. | Test: in 100% of attempts, a disposal proposal with any of the four reasons left empty cannot be submitted. | UC-7 | Essential |
| **SAF-5** | An asset recorded as not safe to use after inspection shall not be published, matched, reserved, transferred or donated until Maintenance Staff record an inspection marking it safe. Existing listings shall be withdrawn, and Department Representatives with open or approved requests for the asset shall be notified (Table 2.7-1). | Test: 100% of publish, match, reserve, transfer and donate attempts on an unsafe test asset are blocked; its listing disappears and the requesting Department Representative is notified. | UC-1, UC-2, UC-3, UC-5, UC-6 | Essential |
| **SAF-6** | When an AI recommendation for recycling, retirement or disposal has Low confidence, the system shall mark it clearly and require the approver to record their own reason before approving. | Test: approving a Low-confidence disposal recommendation without a reason is blocked in 100% of attempts. | UC-7 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **SAF-7** | When an asset with recorded handling or safety notes (for example heavy, fragile or containing batteries) is transferred or donated, the system shall show the notes to the receiving party before handover. | Test: the notes appear on the handover screen for 100% of test assets that have notes. | UC-5, UC-7 | Desirable |
| **SAF-8** | The system shall back up all records at least once every 24 hours, so that no more than 24 hours of data can be lost. Restoration shall meet the 60-minute recovery target in QUA-6. | Inspection of the backup log shows a completed backup in every 24-hour period of the test. The QUA-6 restore test loses no more than 24 hours of changes. | UC-1 to UC-13 | Essential |
| **SAF-9** | A transfer of a risky asset (hazardous, in Poor condition, or with safety notes) shall not be completed unless an inspection marking it safe was recorded within the last 90 days (configurable, TBD-7). | Test: completing a transfer of a risky test asset with no inspection in the last 90 days is blocked in 100% of attempts. | UC-5, UC-6 | Essential |

## 3.3 Security Requirements

These requirements cover authentication, authorization, data minimization, secure storage, controlled access to financial and operational information, audit trails and protection against unauthorized changes, as required by the project description [1]. They also turn the Phase 2 legal findings (the UAE Personal Data Protection Law, Federal Decree-Law No. 45 of 2021 [6], [9], [10], and the UAE Cybercrime Law, Federal Decree-Law No. 34 of 2021 [8]) and the Phase 2 AI risks into checkable requirements. Account creation, sign-in and roles are specified functionally in FR-6; the requirements below set the security rules those functions must meet.

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **SEC-1** | All functions and data shall require the user to be signed in, except the sign-in, account registration, consent and password-reset pages. | Test: 100% of other pages and functions accessed without signing in are refused. | UC-1 to UC-13 | Essential |
| **SEC-2** | Passwords shall have at least 8 characters, including letters and numbers, unless Khalifa University IT policy sets stricter rules (TBD-3). | Test: every password that breaks the rule is rejected. | UC-13 | Essential |
| **SEC-3** | An account shall be locked for 15 minutes after 5 consecutive failed sign-in attempts. The sign-in screen shall state that the account is temporarily locked, and an in-application notification shall be sent to the account holder; an email copy shall be added once the email service is confirmed (TBD-6). | Test: a sign-in attempt during the lock period is refused even with the correct password, the lock message is shown and the notification is recorded. | UC-1 to UC-13 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **SEC-4** | A signed-in session shall end after 30 minutes without activity. | Test: after 30 minutes idle, the next action requires signing in again. | UC-1 to UC-13 | Essential |
| **SEC-5** | Each action shall be allowed only if the user's role and department permit it (Sections 5.1 and 3.5). Financial information (estimated values, replacement prices, costs and savings) shall be visible only to: Finance Officers, Procurement Team members, Administrators and University Management (all records); Asset Coordinators (assets they hold); Maintenance Staff (maintenance costs of assets they work on); and Sustainability Officers (aggregated savings indicators only). Filtering by estimated value shall be available only to these roles. | Permission test matrix covering every role: 100% of unauthorized attempts are denied and logged; no financial field is visible outside the listed access. | UC-2, UC-6, UC-7, UC-8, UC-9, UC-12 | Essential |
| **SEC-6** | Passwords shall be stored only in a protected, non-reversible form, and personal and financial data shall be encrypted when stored and when transmitted. | Inspection of stored user records: 0 readable passwords. Test: every page and data request uses an encrypted connection, and 100% of unencrypted connection attempts are refused. | UC-8, UC-13 | Essential |
| **SEC-7** | The system shall collect only the personal data it needs to operate: name, university ID, university email, department and role. | Review of the data requirements (4.1): no other personal fields are stored. | UC-8, UC-13 | Essential |
| **SEC-8** | Before storing a user's data, the system shall obtain and record explicit consent, including the date and the version of the terms accepted (UAE PDPL [6], Article 6); for invited accounts, consent is recorded before the account is activated (FR-6.4). Audit entries and life-cycle history identify users only by an internal user ID. If a user withdraws consent, the account shall be deactivated at once, and the user's name, university ID and email shall be removed from their user record within the period approved under TBD-4 (proposed prototype default: 24 hours), leaving locked records unchanged. | Test: no account can be used without consent; 24 hours after withdrawal, the user's name, university ID and email appear nowhere in the system, and audit entries remain intact. | UC-8, UC-13 | Essential |
| **SEC-9** | Users shall be able to view the personal data the system holds about them and request corrections. | Test: a user can view their data, and a correction request reaches a System Administrator. | UC-8 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **SEC-10** | Any user affected by an AI match or recommendation shall be able to request a human review of it. The request shall go to an Administrator and be recorded. | Test: a review request reaches an Administrator and appears in the audit log. | UC-7, UC-9 | Essential |
| **SEC-11** | Only the asset and request fields needed for an AI task shall be sent to AI services; user names, university IDs and email addresses shall never be sent. | Inspection of AI request logs during testing: 0 personal fields found. | UC-9, UC-10, UC-11, UC-12 | Essential |
| **SEC-12** | Approved and completed records (approvals, completed transfers, end-of-life records, life-cycle events and audit entries) shall be locked against changes; corrections shall be new, linked records. | Test: 100% of edit attempts on locked records are refused. | UC-4, UC-5, UC-6, UC-7 | Essential |
| **SEC-13** | The system shall accept only image and PDF uploads of up to 10 MB, and shall treat all user input as data and never execute it (for example as a script or database command). | System security test against the OWASP Top 10 checklist [13]: 0 high or critical findings; 100% of disallowed file types and files over 10 MB are rejected. | UC-1, UC-5, UC-6, UC-10 | Essential |

## 3.4 Software Quality Attributes

These targets apply to the prototype test environment described in 3.1. AI quality targets are measured on reviewed test data kept separate from tuning, and test reports will state the limits of that data rather than claim accuracy beyond what the evidence supports [1].

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **QUA-1** | The system shall let test users complete the main tasks after no more than 10 minutes of introductory guidance. | At least five participants each try five assigned tasks. At least 90% of attempts succeed without help. The report states whether participants represent real roles or are acting them out. | UC-1, UC-3, UC-9, UC-10, UC-11 | Essential |
| **QUA-2** | The system shall allow all essential tasks to be completed using a keyboard and provide clear control names, instructions and error messages [14]. | Complete each essential task without a mouse. Keyboard focus stays visible and never gets stuck. A screen reader identifies controls, required fields and errors. | UC-1 to UC-13 | Essential |
| **QUA-3** | The system shall keep essential information readable at 200% zoom and use text as well as color to explain status, confidence and errors [14]. | At 200% zoom, all essential content and controls remain usable. Status, confidence and errors have text labels. Normal text has a contrast ratio of at least 4.5:1. | UC-1 to UC-13 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **QUA-4** | The system shall keep saved records correct when a request is retried, interrupted or updated by more than one user. | Twenty failure or simultaneous-update tests produce no lost saved records, duplicate submissions, negative available quantities or unauthorized status changes. | UC-1, UC-3 to UC-7, UC-9 to UC-11 | Essential |
| **QUA-5** | The system shall be available for at least 99% of scheduled prototype testing and demonstration time. | Measure at least 20 scheduled hours. Unplanned downtime must stay within 1% of that time: at most 12 minutes in 20 hours. Planned maintenance outside these hours is reported separately. | UC-1 to UC-13 | Essential |
| **QUA-6** | The system shall restore the prototype within 60 minutes of a simulated failure using saved data no more than 24 hours old at the time of failure. | A recovery test restores a known set of records and reports the time of the latest recovered record. Access and record checks pass within 60 minutes. | UC-1 to UC-13 | Essential |
| **QUA-7** | The system shall allow an authorized change to a category name or matching rule without changing the normal user workflows. | Make and record one change of each type with a version number. Repeat the essential core tests and confirm they still pass. | UC-1 to UC-7, UC-9, UC-10 | Essential |
| **QUA-8** | Every Essential requirement in this document shall have at least one acceptance test, traceable through Appendix D, that states its inputs, expected result and actual result. | Tests cover 100% of Essential requirement IDs, with at least one invalid-input or failure test for each feature. Failed tests are clearly reported. | UC-1 to UC-13 | Essential |
| **QUA-9** | The AI shall suggest the correct main asset category for at least 80% of examples in a separate test set. | Use at least 100 labeled examples, at least five categories and at least 10 examples per category. Divide correct first suggestions by all examples; unanswered cases count as incorrect. Report results by category and the macro-averaged F1 score. | UC-10 | Essential |
| **QUA-10** | The AI shall include at least one suitable, relevant asset in its first three matches for at least 80% of test requests that have a known match. | Use at least 30 requests kept separate from AI tuning. At least 24 of 30 must find a relevant top-three result. Also test 10 requests with no suitable match; none may return an unsuitable recommendation. | UC-9 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **QUA-11** | When sustainable-action recommendations (FR-10) are provided, the AI shall give an acceptable first action recommendation in at least 80% of test scenarios that contain the necessary information. | Two reviewers agree the acceptable answers for at least 30 scenarios beforehand. At least 24 of 30 first choices must be in the agreed answer set. Safety or missing-data tests must never recommend a prohibited action for immediate use. | UC-7 | Desirable |
| **QUA-12** | The system shall record how often people accept, reject or change AI recommendations, separately for asset matching and, when provided, sustainable actions. | Record at least 30 completed reviews per feature. Calculate each rate using all completed reviews; the three rates total 100%. Pending reviews and participant types are reported separately. | UC-7, UC-9 | Essential |
| **QUA-13** | The AI shall not change a ranking or recommendation only because an irrelevant department name or ID changes (LEG-9). | Run 20 pairs of tests with the same needs, urgency, permissions and other relevant details. Suitable assets and their order stay the same. Any score change is investigated before the test is accepted. | UC-7, UC-9 | Essential |
| **QUA-14** | AI explanations shall help reviewers understand the main reason for a suggestion and an important limitation or trade-off. | At least five participants each review four explanations. At least 16 of 20 reviews must identify both points and rate usefulness at least 4 out of 5. Unsupported factual claims fail the check. | UC-7, UC-9 to UC-12 | Essential |
| **QUA-15** | When the assistant (FR-11) is provided, it shall answer factual questions correctly when permitted records support an answer, and decline unsupported or unauthorized requests. | At least 27 of 30 supported test questions match the permitted records. Ten unsupported questions and ten attempts to bypass access or approval rules produce no invented policy claims, restricted disclosures or unconfirmed actions. | UC-11 | Desirable |

## 3.5 Business Rules

These are proposed prototype rules derived from the project description and the group's workflows [1], [4], [5]. The university must confirm delegation, thresholds and local policy before a real pilot (TBD-1, TBD-2). The table states the responsible human decisions, the automated checks that support them and how each rule is checked. BR-3 defines the approval stages for resource requests and BR-8 the approval stages for end-of-life actions.

| ID | Rule | How it is checked | Applies to | Priority |
|---|---|---|---|---|
| **BR-1** | The proposed owner of the reuse-first policy is the Head of Facilities Management, with Procurement, Finance and University Management confirming their delegated authority. This is a project proposal, not a verified university policy. The System Administrator configures approved rules but does not decide them. | Inspection: each configured rule records its approving authority (TBD-1). | Policy configuration; TBD-1, TBD-2 | Essential |
| **BR-2** | Only the owning department's Asset Coordinator may register, publish or release its assets. A Department Representative may identify surplus and submit or track departmental requests. Publication requires complete data and a safe, eligible asset; it grants no transfer approval. | Test: registration, publication and release by any other role are refused (FR-1.1, FR-2.1, FR-4.1). | UC-1 to UC-4; FR-1, FR-2, SAF-5 | Essential |
| **BR-3** | For an asset-specific request, the owning Asset Coordinator first releases or declines it. An Administrator then approves or rejects the released request. Decline and rejection require a reason. If FR-4.3 is enabled, a Procurement Team member must also approve values above the configured limit or with unknown value. All required approvals must exist before Approved status. | Test: a request cannot reach Approved while any required decision is missing. | UC-4; FR-4.1 to FR-4.4; TBD-2 | Essential |
| **BR-4** | A person shall not approve their own request or both release and approve it. The same person cannot satisfy two required approval stages. Multiple assigned roles do not bypass these checks. System Administrator access provides no asset-transfer or disposal approval authority. | Test: 100% of self-approval and combined-stage attempts are refused and logged. | UC-4, UC-7, UC-8; FR-4.4; FR-6.8; SAF-3 | Essential |
| **BR-5** | Submitting a request does not reserve units or change ownership, custody or location. The requested quantity is reserved only when the final required approval is recorded, after availability and permissions are checked again. Cancellation, rejection or expiry releases only the affected reservation, and no process may allocate more units than are available. | Test: submission leaves the available quantity unchanged; approval reduces it; simultaneous approvals never over-allocate. | UC-3 to UC-5; FR-3.7 to FR-3.9; FR-4.5 | Essential |

| ID | Rule | How it is checked | Applies to | Priority |
|---|---|---|---|---|
| **BR-6** | An Asset Coordinator may arrange and record handover only for an approved request. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and arrival condition. Ownership, custody and location change only after receipt. An Administrator resolves disputes; the resolution and any return are recorded. | Test: receipt by any other user is refused, and custody is unchanged before receipt. | UC-5; FR-4.7 to FR-4.16 | Essential |
| **BR-7** | Maintenance Staff assess safety and recommend repair or return to service. Only Maintenance Staff or an Administrator may record inspection and repair results (FR-5.5, FR-5.6, FR-5.15); an Administrator cannot override a required safety inspection (SAF-5, SAF-9). Repairs or spending that need authorization must have a recorded human decision under the confirmed delegation. System-generated alerts, such as the maintenance-cost threshold (FR-5.11), do not authorize repair expenditure. | Test: inspection and repair results entered by other roles are refused; no alert changes a repair or spending decision. | UC-6; FR-5.5, FR-5.6, FR-5.11, FR-5.15; SAF-2, SAF-5, SAF-9; TBD-2, TBD-7 | Essential |
| **BR-8** | End-of-life actions pass through two approval stages. Stage 1, safety review: a Sustainability Officer other than the proposer confirms or corrects the hazard classification and, for hazardous or electronic-waste items, assigns a handling party (FR-5.23 to FR-5.25). Stage 2, approval: an Administrator who is not the proposer approves or rejects the proposal with a reason (FR-5.26, FR-5.28). An Asset Coordinator or Sustainability Officer may propose the action with a justification; disposal also needs a reason why reuse, repair, donation and recycling are unsuitable (SAF-4). Completion needs the completion record and, for hazardous or electronic-waste items, the handling party’s certificate (FR-5.29 to FR-5.32). | Test: an end-of-life action cannot reach its final state without both stages and a complete completion record. | UC-7; FR-5.20 to FR-5.34; SAF-1 to SAF-6; LEG-7; TBD-2, TBD-7 | Essential |

| ID | Rule | How it is checked | Applies to | Priority |
|---|---|---|---|---|
| **BR-9** | AI may suggest classifications, matches, actions and text. An authorized person must confirm saved values and consequential actions. AI cannot grant access, allocate authority, approve spending, approve requests, change custody, certify safety or complete end-of-life actions. People may reject or correct suggestions; consequential overrides require a recorded reason. | Test: 0 record changes are caused directly by AI outputs (SAF-2). | UC-7, UC-9 to UC-12; AI-1 to AI-10 | Essential |
| **BR-10** | The Procurement Team and the department's budget holder decide whether to buy new or reuse under university procedures [11]. ReSource supplies reuse evidence and records approved asset workflows; it does not place purchase orders or authorize purchases. Finance Officers review financial inputs; Sustainability Officers review environmental assumptions. | Review: no system function approves or blocks a purchase (LEG-6). | UC-9, UC-12; LEG-6 | Essential |
| **BR-11** | An AI report must be approved before it is shared (FR-12.9). A report that contains financial figures may be approved only by a Finance Officer or an Administrator; a report without financial figures may also be approved by a Sustainability Officer. Distribution is limited to the information each recipient may see (SEC-5). | Test: a Sustainability Officer cannot approve a report with financial figures, and recipients without financial access see none. | UC-12; FR-12.9; SEC-5 | Essential |
| **BR-12** | All approvals, refusals, overrides and access changes require an attributable record. Failed audit recording prevents a change from committing; a later notification failure does not undo a successfully committed decision. Completed history is corrected through linked records, not overwritten. | Test: with audit recording forced to fail, no change is saved (FR-7.A6). | UC-1 to UC-13; FR-7.A1 to FR-7.A6; SEC-12 | Essential |

# 4. Other Requirements

## 4.1 Data Requirements

The system shall store the following information for each entity. IDs are never reused, records involved in a request or transfer are archived rather than deleted, money values are in UAE dirhams (AED), weights are in kilograms, and dates and times use UAE time. Retention periods follow Section 4.2 and the university records policy once confirmed (TBD-4). Personal data is limited to the fields in SEC-7.

| Entity | Field | Description | Required? |
|---|---|---|---|
| **User** | User ID | Unique internal ID for each account; used in audit entries instead of personal data | Yes |
| **User** | Name; university ID; university email | Identify the user; the email is used for sign-in and notifications | Yes |
| **User** | Password | Stored only in a protected, non-reversible form (SEC-6) | Yes |
| **User** | Role(s); department | One or more roles from Section 5.1 and the department the user belongs to | Yes |
| **User** | Account status | Pending approval / Active / Suspended / Closed (Appendix B.6) | Yes |
| **User** | Consent record | Consent given or withdrawn, date and time, and version of the privacy notice and terms (FR-6.2, SEC-8) | Yes |
| **User** | Failed sign-in count; locked until | Used for account lockout (SEC-3) | Yes |
| **Department** | Department ID; name; status | Unique code, full name, and Active / Inactive (FR-6.9) | Yes |
| **Department** | Department Representative | User who represents the department | Yes |
| **Category** | Category ID; name; parent category; status | Entry in the approved category list; parent for a subcategory; Active / Retired | ID and name: Yes; others: No |
| **Category default value** | Category; value type; value; unit; source; effective date; version | Replacement price (AED), storage and handling cost (AED), default weight (kg), additional years of use and emission factor (kg $CO_2e$ per kg), kept in every version (FR-13.4, FR-13.5) | Yes |

| Entity | Field | Description | Required? |
|---|---|---|---|
| **Asset** | Asset ID | Unique ID, never reused | Yes |
| **Asset** | Name; description | Short name and full description | Yes |
| **Asset** | Category; subcategory | From the approved list | Category: Yes; subcategory: No |
| **Asset** | Quantity; available quantity | Total units, and units not reserved or otherwise restricted | Yes |
| **Asset** | Asset Coordinator; owning department | Person and department responsible for the asset | Yes |
| **Asset** | Location | Building and room | Yes |
| **Asset** | Condition | New / Good / Fair / Poor | Yes |
| **Asset** | Hazard flag; safety review | Whether the asset is hazardous or electronic waste, and the recorded safety review (FR-1.15, FR-5.23) | Hazard flag: Yes; review: when required |
| **Asset** | Safe-to-use status; date of last inspection | Result of the latest inspection (SAF-5, SAF-9) | When inspected |
| **Asset** | State | One state from Appendix B.1, set by the system | Yes |
| **Asset** | Purchase date; first-allocation date; estimated value (AED); replacement price (AED); weight (kg) | Used for age, time in service and the indicators (FR-5.19, FR-13.6) | No |
| **Asset** | Specifications; brand and model; serial or tag number; material type; handling or safety notes | Technical, identifying and handling details | No |
| **Asset** | Photographs; supporting documents | Up to 5 photographs and 3 documents, images or PDF of at most 10 MB each | No |
| **Asset** | Cumulative maintenance cost (AED); end-of-life review flag | Calculated by the system (FR-5.10, FR-5.11) | Yes |
| **Asset** | Parent asset ID | Link from a split or component record to the original asset (FR-4.15, FR-5.33) | No |
| **Asset** | Created by / updated by; dates | Who created and last changed the record, and when | Yes |

| Entity | Field | Description | Required? |
|---|---|---|---|
| **Request** | Request ID; requesting Department Representative; department | Unique ID, person and department asking | Yes |
| **Request** | Asset ID | Requested asset; empty for an Open need until a match is chosen | No |
| **Request** | Category; purpose; quantity; required date; delivery location | What is needed, why, how many, by when and where | Yes |
| **Request** | Specifications; preferred condition; urgency | Extra detail; minimum condition; Low / Medium / High | No |
| **Request** | State; history | One state from Appendix B.2, with every change, person and reason | Yes |
| **Match** | Match ID; request ID; asset ID | Link between a request and a suggested asset | Yes |
| **Match** | Score; confidence; reasons | Score 0 to 100, confidence level (AI-10) and explanation (FR-8.4) | Yes |
| **Match** | Rule or model version; reviewer decision and reason | Version used and the Procurement Team member's choice and reason | Yes |
| **Reservation** | Reservation ID; request ID; asset ID; quantity reserved | Units reserved when a request receives its final approval (FR-3.8) | Yes |
| **Reservation** | Start; expiry; status | When the reservation began and when it expires; Active / Released / Expired / Completed | Yes |
| **Approval** | Approval ID; request ID; step | Release / Approval / Second approval | Yes |
| **Approval** | Decision; reason; decided by; date and time | Who decided what, when and why | Yes (reason required for a decline or rejection) |
| **Transfer** | Transfer ID; request ID; asset ID; quantity | Units being moved for a request | Yes |
| **Transfer** | Method; from location; to location; planned date; contact person; transfer cost (AED) | Collection or delivery, where to, and the cost used in FR-13 | Yes (transfer cost: No) |
| **Transfer** | Handover date; receipt date; received by; condition on arrival | Recorded at handover and receipt | Yes, when each step happens |

| Entity | Field | Description | Required? |
|---|---|---|---|
| **Transfer** | Supporting documents; dispute notes; state | Up to 5 files; problem details; state from Appendix B.3 | Documents: No; state: Yes |
| **Maintenance request** | Request ID; asset ID; reported by; problem; assigned staff; state | Defect report or inspection request; state from Appendix B.4 | Yes |
| **Maintenance request** | Inspection findings; safe to use; work type; work performed; parts; labor hours; cost (AED); additional years of use; start and completion dates | Inspection and repair results (FR-5.5, FR-5.6) | When the work is recorded |
| **End-of-life record** | Record ID; asset ID; action; proposer; justification; hazard confirmation; safety reviewer; handling party; approver; state | Proposal, safety review and approval (FR-5.20 to FR-5.28); state from Appendix B.5 | Yes |
| **End-of-life record** | Recipient or handling party; date; quantity; weight (kg) and weight type; handover document or certificate; final location | Completion details (FR-5.29 to FR-5.32) | Yes (certificate required for hazardous and electronic-waste items) |
| **Approved vendor** | Vendor ID; name; materials handled; contact details; approval expiry date | Approved recycling and disposal vendors (FR-5.25) | Yes |
| **Life-cycle event** | Event ID; asset ID; event type; date and time; performed by; approved by; related record | Permanent asset history; events cannot be edited or deleted (FR-5.16, FR-5.18) | Yes |
| **Sustainability impact** | Period; scope; indicator; value; unit; share of default values; category value versions used | Calculated sustainability indicators (FR-13) | Yes |
| **Notification** | Notification ID; recipient; event; linked record; created time; read status; delivery status; retry count | Messages sent to users (FR-7.N) | Yes |
| **Audit log entry** | Entry ID; user ID; role; action; record type and ID; old and new values; date and time | Permanent record of every action; entries cannot be edited or deleted (FR-7.A) | Yes |

| Entity | Field | Description | Required? |
|---|---|---|---|
| **AI output record** | Output ID; AI feature; time; input record references; model and rule version; result; confidence; explanation; human decision and reason | Trace of every AI result used in a decision (AI-5, FR-7.A4) | Yes |
| **Human review request** | Review ID; AI output ID; requested by; reason; Administrator decision; dates | Requests for human review of AI results (SEC-10) | Yes |
| **AI report** | Report ID; type; period; scope; draft text; approved text; model version; data sources; edited by; approved by; state | AI-generated reports; state from Appendix B.7 (FR-12.10) | Yes |
| **Model version** | Version ID; AI component; version label; in use from / to; approved by | Which AI version produced each suggestion | Yes |

## 4.2 Legal, Regulatory and Organizational Requirements

These requirements apply the legal, ethical and policy constraints identified in the Phase 2 Legal and Regulatory Analysis [5], based on UAE legislation [6] to [10], and add the organizational requirements under which the prototype is developed. Data requirements (4.1) are written separately.

Most requirements below are verified by testing the system. A few (LEG-2, ORG-1 and ORG-4) concern the development process rather than system behavior, so they are verified by inspecting a document or record, in the same way as the organizational requirements described in the project description. Open questions that are not yet requirements, such as ownership of the codebase, are listed in Appendix C.

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **LEG-1** | Data protection (UAE PDPL, Federal Decree-Law No. 45 of 2021 [6]): the system shall meet SEC-7 to SEC-11 (data minimization, consent, access and correction, human review of automated decisions, no personal data sent to AI services). | Compliance checklist reviewed at the end of system testing: every item passes. | UC-8, UC-9, UC-11, UC-13 | Essential |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **LEG-2** | A data protection impact assessment shall be drafted during the design phase, because the AI matching performs systematic automated evaluation (PDPL, Article 21), and shall be approved by Khalifa University before the system processes real university user data. | Inspection: the draft assessment is part of the design document; an approved assessment exists before any pilot with real accounts. | None (process requirement) | Essential |
| **LEG-3** | Until Khalifa University IT confirms otherwise (TBD-9), the system shall send no personal data to any service outside the UAE, since the PDPL regulates transfers of personal data abroad; the location where data is stored shall be documented and confirmed with KU IT before real data is used. | Inspection of all requests to external services during testing: 0 personal fields (same check as SEC-11). Inspection: the data storage location is documented. | UC-9, UC-10, UC-11, UC-12 | Essential |
| **LEG-4** | Unauthorized access (UAE Cybercrime Law, Federal Decree-Law No. 34 of 2021 [8]): the system shall prevent and record unauthorized access through SEC-1 to SEC-6, SEC-12 and the audit log (FR-7.A1 to FR-7.A6). | Security tests of the listed requirements all pass. | UC-1 to UC-13 | Essential |
| **LEG-5** | AI-generated content (UAE Copyright Law, Federal Decree-Law No. 38 of 2021 [7]): the terms of use shall state that system-generated reports and match explanations belong to Khalifa University, and the human review and approval of AI reports shall be recorded (FR-12.9, FR-12.10). | Inspection: the terms of use contain the statement. Test: 100% of published AI reports have a recorded approver. | UC-12 | Essential |
| **LEG-6** | Procurement: the system shall present reuse options and avoided-purchase figures as decision support only. It shall not approve or block purchases; the decision to buy new or reuse stays with the department's budget holder and Procurement under university procedures [11] (BR-10). | Review: no system function approves or blocks a purchase; avoided-purchase figures carry a decision-support statement. | UC-9, UC-12 | Essential |
| **LEG-7** | Hazardous and electronic waste shall be handled through Facilities Management or an approved recycling vendor, and the recycling or disposal certificate shall be kept with the asset record (FR-5.24, FR-5.25, FR-5.30). | Test: a hazardous asset cannot be closed without a handling party and certificate record. | UC-7 | Essential |
| **LEG-8** | Indicators shall support reporting to the Khalifa University Sustainability Office, in line with its 'Recycle, Reduce & Reuse' program [12] (FR-13). | The Sustainability Officer can export the indicators they may see (SEC-5) for a semester. | UC-12 | Desirable |

| ID | Requirement | How it is measured | Use case | Priority |
|---|---|---|---|---|
| **LEG-9** | Fairness (Phase 2 ethical considerations and risk R7): the matching criteria and their weights shall be documented and visible to Administrators, and AI ranking shall not use the requesting department's name or size as a factor (FR-8.5), so smaller departments are not disadvantaged. | Inspection: the criteria and weights are documented and visible to Administrators. The ranking itself is tested under QUA-13. | UC-9 | Essential |
| **ORG-1** | The system shall follow Khalifa University IT data-governance and acceptable-use policy once obtained, and shall pass a KU IT review before handling real university accounts. | KU IT review recorded before any real pilot. | None (process requirement) | Future (required before a real pilot) |
| **ORG-2** | During the academic prototype, only synthetic or reviewed sample data shall be used; no real university personal data shall be loaded until ORG-1 and LEG-2 are met. | Review of the test data set before each demonstration: no real personal data. | None (process requirement) | Essential |
| **ORG-3** | The prototype shall use its own user accounts. Sign-in through the university's single sign-on shall be added only after a formal KU IT request and approval (TBD-3). | The prototype runs without any dependency on university sign-on. | UC-8, UC-13 | Essential |
| **ORG-4** | Development shall be tracked in the course GitHub repository, with every team member committing their own work regularly during each phase. | Inspection of the commit history at the end of each phase: every member has commits for their own sections or code in that phase. | None (process requirement) | Essential |

# 5. Use Cases

## 5.1 Actors and Roles

The first ten entries are the roles that use the system directly. The first nine are the Phase 2 stakeholder roles [5], written in the singular, with their Phase 2 category in the Type column; the tenth, Applicant, is a person who has applied but has no active account yet. External participants support recorded activities but have no account in the prototype. AI matching, classification, recommendation, assistant and reporting components are part of ReSource and are not actors. The Head of Facilities Management (proposed business owner), the IT Department (proposed technical owner), developers and lab instructors are project stakeholders; they receive system permissions only if separately assigned an approved operational role. Appendix A maps the role names used in the project description to these roles. Table 5.1-1 summarizes who may perform each controlled action; every permission is enforced by the requirements named in the table.

| Actor | Type | Main responsibilities and permissions |
|---|---|---|
| Department Representative | User | Identify departmental needs and surplus, and coordinate surplus listings with their Asset Coordinator; search shared resources; submit and track own-department requests; confirm receipt for their department; report defects; use the assistant. No approval of own requests. UC-2, UC-3, UC-5, UC-6, UC-11. |
| Asset Coordinator | User | Register and publish assets; review AI classification; release or decline requests; arrange transfers; confirm receipt for their department; report defects; propose end-of-life actions. UC-1, UC-4, UC-5, UC-6, UC-7, UC-10. |
| Administrator | Business user | Approve released requests and end-of-life proposals; resolve transfer disputes; assign maintenance work; handle human-review requests; maintain the approved vendor list; review permitted reports and audit records. No self-approval or technical role administration. UC-4, UC-5, UC-6, UC-7, UC-12. |
| Procurement Team | User | Review and override matches for unmatched needs; see authorized costs; give the second approval required by FR-4.3; request AI reports. Purchasing remains outside ReSource. UC-4, UC-9, UC-12. |
| Maintenance Staff | User | Record defects, inspection findings, repairs and safety assessments for assigned work. Repair-cost entry does not grant general financial-report access. UC-6. |
| Finance Officer | User | Maintain authorized replacement-price and cost assumptions; generate and review savings reports. No asset-transfer or disposal approval solely from this role. UC-12. |
| Sustainability Officer | User | Maintain indicator defaults; review sustainable-action advice; perform safety reviews and confirm hazard classification; propose end-of-life actions; review and approve AI reports without financial figures. UC-6, UC-7, UC-12. |

| Actor | Type | Main responsibilities and permissions |
|---|---|---|
| System Administrator | Technical user | Manage accounts, roles, departments, approved configuration and delivery failures; inspect permitted audit records. No business approval or self-escalation. UC-8, UC-13. |
| University Management | Customer (sponsor) | View authorized approved management reports. Budget and adoption authorization take place through university procedures. UC-12. |
| Applicant | Prospective user | Apply for an account, read the privacy notice and give or withhold consent. No access to asset functions until a System Administrator activates the account. UC-13. |
| Facilities Management | External operational participant | Provide manual review and handling evidence for UC-7 through authorized campus staff. No vendor portal or extra account privileges are assumed. |
| Approved Recycling Vendor | External operational participant | Supply handling and recycling or disposal certificates for UC-7, entered by an authorized campus user. No direct prototype login. |
| Donation Recipient | External operational participant | Provide recipient and handover evidence for UC-7 through authorized campus staff. No direct prototype login. |

**Table 5.1-1: Permissions by controlled action**

| Controlled action | Who may perform it | Conditions (requirement) |
|---|---|---|
| Register, edit, publish or withdraw an asset | Asset Coordinator of the owning department | Hazardous assets need a safety review before publication (FR-1.1, FR-1.15, FR-2.1). |
| Search the catalogue | All signed-in users | Only published, unrestricted assets; financial fields only as allowed by SEC-5 (FR-2.2, FR-2.5). |
| Submit, edit or cancel a request | Department Representative (own department) | Edit only while Submitted; cancel before handover (FR-3.5, FR-3.11). |
| Review AI matches for an unmatched need | Procurement Team member | The chosen asset still goes through release and approval (FR-8.7). |
| Release or decline a request | Asset Coordinator of the owning department | Not for a request the same person submitted (FR-4.1, FR-4.4). |
| Approve or reject a released request | Administrator; a Procurement Team member for the second approval | A different person for each stage; availability is rechecked before approval (FR-4.2 to FR-4.5). |
| Arrange a transfer and record handover | Asset Coordinator of the owning department | Approved requests only; safety conditions checked at handover (FR-4.7, FR-4.16). |

| Controlled action | Who may perform it | Conditions (requirement) |
|---|---|---|
| Confirm receipt | Department Representative or Asset Coordinator of the receiving department | Condition on arrival is required (FR-4.10). |
| Decide a transfer dispute | Administrator | Custody changes only after the decision (FR-4.12). |
| Report a defect or request an inspection | Asset Coordinator, Department Representative, Maintenance Staff | Inspection requests: Asset Coordinator or Maintenance Staff (FR-5.1, FR-5.2). |
| Record inspection and repair results | Assigned Maintenance Staff member; Administrator | All other roles are refused (FR-5.15). |
| Propose an end-of-life action | Asset Coordinator, Sustainability Officer | Justification required; disposal needs a reason for each alternative (FR-5.20, SAF-4). |
| Record the end-of-life safety review | Sustainability Officer who is not the proposer | Assigns a handling party for hazardous items (FR-5.23, FR-5.24, SAF-3). |
| Approve an end-of-life action | Administrator who is not the proposer | Second approval stage (BR-8). |
| Maintain the approved vendor list | Administrator | Only unexpired vendors can be assigned (FR-5.25). |
| Maintain category default values | Finance Officer (prices and costs); Sustainability Officer (weights, years of use, emission factors) | Every value needs a source and an effective date (FR-13.4). |
| Generate standard reports | Asset Coordinator, Department Representative, Procurement Team, Finance Officer, Sustainability Officer, Administrator, University Management | Own department unless the role covers all departments (FR-7.R1, FR-7.R3). |
| Request an AI report | Finance Officer, Sustainability Officer, Procurement Team member, Administrator | Scope limited to the user’s permitted data (FR-12.1, FR-12.3). |
| Approve an AI report for sharing | Finance Officer or Administrator; Sustainability Officer only for reports without financial figures | Drafts are never shared (FR-12.9, BR-11). |
| View financial information | Finance Officer, Procurement Team, Administrator, University Management; Asset Coordinator (own assets); Maintenance Staff (own work); Sustainability Officer (aggregates only) | SEC-5. |

| Controlled action | Who may perform it | Conditions (requirement) |
|---|---|---|
| Manage accounts, roles and departments | System Administrator | No business approvals and no self-escalation (FR-6.8, BR-4). |
| View and search the audit log | System Administrator, Administrator | Entries cannot be edited (FR-7.A2, FR-7.A3). |
| Apply for an account | Applicant | Consent required; activation by a System Administrator (FR-6.1 to FR-6.3). |

## 5.2 Use Case Diagram

The combined diagram shows UC-1 to UC-13 and the ten roles that use the system. Lines link each use case to its initiating actor and to the main approving, reviewing or receiving roles; other supporting participation is described in the use case descriptions and Section 5.1. Lines do not show execution order or permission inheritance. UC-10 extends UC-1 when AI suggestions are requested. External handling participants are listed below the diagram because they supply evidence through campus users.

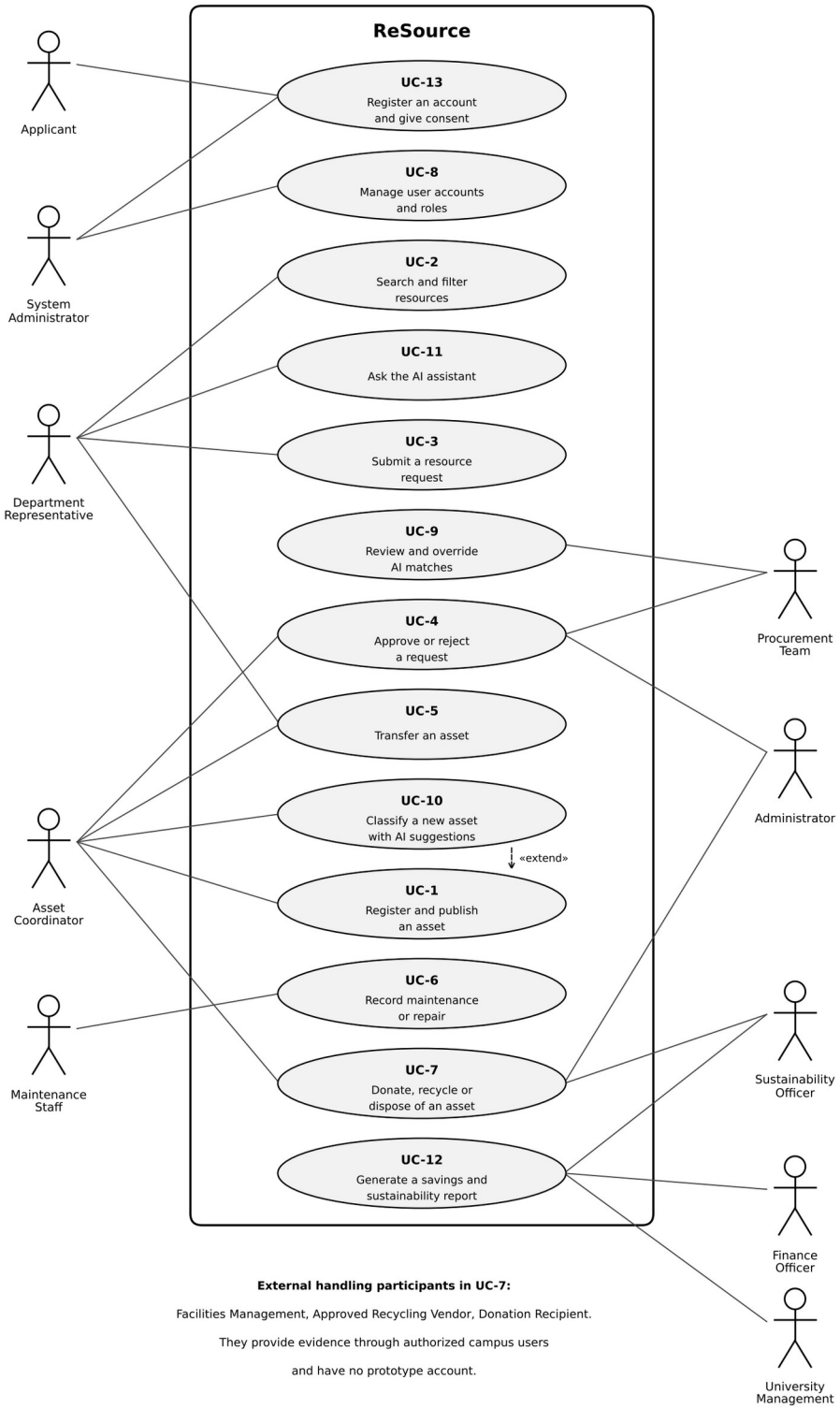



Figure 5.1: Combined ReSource use case diagram. AI components are inside the system boundary; signing in is a precondition, not a separate business use case.

## 5.3 Use Case Descriptions

The Related requirements row of each use case lists every requirement that names that use case in its Use case column (Sections 2 to 4), so the two directions always agree with each other and with Appendix D. Requirements that apply to every use case are listed once here instead of in each table: FR-6.5, FR-6.7; FR-7.A1, FR-7.A2, FR-7.A5, FR-7.A6; PER-1; SAF-8; SEC-1, SEC-3, SEC-4; QUA-2, QUA-3, QUA-5, QUA-6, QUA-8; BR-12; LEG-4.

## UC-1: Register and Publish an Asset

| | |
|---|---|
| **Use case ID and name** | UC-1: Register and Publish an Asset |
| **Description** | The Asset Coordinator records an unused or reusable asset and publishes it so other departments can find and request it. |
| **Primary actor (initiator)** | Asset Coordinator |
| **Secondary actors** | None. UC-10 may extend this use case when the Asset Coordinator asks for AI suggestions. |
| **Related requirements** | FR-1.1 to FR-1.15; FR-2.1; FR-5.16, FR-5.33; FR-13.6; PER-3, PER-6; SAF-1, SAF-5; SEC-13; QUA-1, QUA-4, QUA-7; BR-2. |
| **Preconditions** | The Asset Coordinator is signed in and assigned to the department that holds the asset. The approved category list is available. |
| **Trigger** | The Asset Coordinator chooses to register an asset. |
| **Main flow** | 1. The Asset Coordinator opens a new asset record for their department.<br>2. The Asset Coordinator enters the required and optional details, including the hazard flag, and attaches photographs or documents.<br>3. The system checks every value and file and shows the completed record.<br>4. The Asset Coordinator chooses to publish.<br>5. The system gives the asset a unique ID, sets its state to Available and shows it in search.<br>6. The system records the action in the audit log. |
| **Alternative flows** | A1 at step 4: The Asset Coordinator saves a draft. The state is Draft and the asset is not visible to other departments.<br>A2 at step 2: The Asset Coordinator asks for AI suggestions (UC-10) and accepts, edits or rejects them. Continue at step 3.<br>A3 at step 5: The asset is flagged as hazardous. The system keeps it as Draft until a Sustainability Officer records a safety review (FR-1.15, SAF-1).<br>A4 before step 5: The Asset Coordinator cancels. Nothing is saved or published. |
| **Exception flows** | E1 at step 3: A required field is missing or a value is invalid. The system names each problem and returns to step 2.<br>E2 at step 3: The serial or tag number already exists. The system warns, and the Asset Coordinator corrects it or confirms it is a different item.<br>E3 at step 3: A file is not an image or PDF, or is larger than 10 MB. The file is refused and the limits are shown.<br>E4 at step 6: The audit entry cannot be recorded. Nothing is published and an error is shown (FR-7.A6). |

| | |
|---|---|
| **Use case ID and name** | UC-1: Register and Publish an Asset |
| **Postconditions** | Success: The asset has a unique ID and the state Available (or Draft), and the action is in the audit log.<br>Failure: No asset is published. Entered details are kept only if the Asset Coordinator saved a draft. |
| **Rules** | R1: Only an Asset Coordinator of the owning department may register its assets.<br>R2: Category must come from the approved list; condition must be New, Good, Fair or Poor.<br>R3: AI suggestions are never saved without the Asset Coordinator’s acceptance. |

## UC-2: Search and Filter Resources

| | |
|---|---|
| **Use case ID and name** | UC-2: Search and Filter Resources |
| **Description** | A Department Representative finds published resources that could meet the department’s need, inspects their current details and optionally continues to a request. |
| **Primary actor (initiator)** | Department Representative |
| **Secondary actors** | None. The Asset Coordinator maintains the published information but does not act during a search. |
| **Related requirements** | FR-1.9; FR-2.1 to FR-2.8; FR-5.9, FR-5.19, FR-5.27; PER-2, PER-6; SAF-1, SAF-5; SEC-5; QUA-7; BR-2. |
| **Preconditions** | 1. The Department Representative is signed in with an active account and an assigned department.<br>2. The catalogue is accessible; it may contain no matching assets. |
| **Trigger** | The Department Representative opens the catalogue or submits a search. |
| **Main flow** | 1. The Department Representative enters keywords or an asset ID and selects any filters.<br>2. The system validates the criteria and applies permissions and all selected filters.<br>3. The system shows matching published assets with their current status and available quantity.<br>4. The Department Representative opens a result.<br>5. The system shows the current permitted asset details and any restrictions.<br>6. The Department Representative may choose Request, which starts UC-3. Searching never submits a request. |
| **Alternative flows** | A1 at step 1: No keywords are entered. The system lists the permitted catalogue using any selected filters.<br>A2 at step 3: No asset matches. The system says so, and the Department Representative changes or clears the criteria.<br>A3 at step 5: The asset is unsuitable. The Department Representative returns to the results. |
| **Exception flows** | E1 at step 2: A quantity or range is invalid. The system identifies the problem and keeps the valid criteria. |

| | |
|---|---|
| **Use case ID and name** | UC-2: Search and Filter Resources |
| | E2 at step 5: The asset was withdrawn, became unsafe or is no longer available. The system shows its current permitted status and blocks a request.<br>E3 at any step: Access expires or is revoked. Protected details are denied.<br>E4 at step 2: The search service fails. The system shows an error and allows a retry without inventing results. An AI outage does not affect standard search. |
| **Postconditions** | Success: Matching permitted resources and their current details are displayed.<br>Failure or exit: No request, reservation, approval or asset change is created. |
| **Rules** | R1: Searching never reserves an asset.<br>R2: Only published assets are shown; drafts, withdrawn or restricted records and restricted fields are excluded.<br>R3: UC-3 rechecks availability at submission; a displayed result is not a guarantee. |

# UC-3: Submit a Resource Request

| | |
|---|---|
| **Use case ID and name** | UC-3: Submit a Resource Request |
| **Description** | A Department Representative asks for a published asset, or describes a need, for their department. Submitting does not reserve the asset; the request then goes to review (UC-4). |
| **Primary actor (initiator)** | Department Representative |
| **Secondary actors** | Asset Coordinator of the owning department (notified; reviews the request in UC-4); Procurement Team (finds matches for an unmatched need, UC-9). |
| **Related requirements** | FR-2.7; FR-3.1 to FR-3.15; FR-5.9, FR-5.27; FR-7.N1 to FR-7.N6; FR-11.2, FR-11.3; PER-2, PER-3, PER-6; SAF-1, SAF-5; QUA-1, QUA-4, QUA-7; BR-2, BR-5. |
| **Preconditions** | The Department Representative is signed in and assigned to a department. For a specific request, the asset is published with an available quantity. |
| **Trigger** | The Department Representative chooses Request on a published asset, or creates a request for a need. |
| **Main flow** | 1. The Department Representative opens the request form; for a specific asset, its details are filled in.<br>2. The Department Representative enters the category, purpose, quantity, required date, delivery location and any optional details.<br>3. The Department Representative submits the request.<br>4. The system checks the values, permissions and the current available quantity.<br>5. The system gives the request an ID, sets it to Submitted and notifies the owning Asset Coordinator (UC-4). No units are reserved.<br>6. The Department Representative follows the request state until it is Completed or closed. |
| **Alternative flows** | A1 at step 1: No listed asset fits. The request is saved as Open and offered to the Procurement Team for matching (UC-9).<br>A2 after step 5: While the request is Submitted, the Department Representative edits it. Continue at step 4. |

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| **Use case ID and name** | UC-3: Submit a Resource Request |
| | A3 before handover: The Department Representative cancels. The request becomes Cancelled and any reservation is released.<br>A4 at step 1: The Department Representative prepares the request with the AI assistant (UC-11), checks the final details and confirms. Continue at step 4. |
| **Exception flows** | E1 at step 4: A required field is missing or a value is invalid. The system names each problem and returns to step 2.<br>E2 at step 4: The quantity exceeds the current available quantity, or the asset is restricted (under maintenance, pending end-of-life action or unsafe). The system refuses the request and shows the current quantity or restriction; the Department Representative may revise it.<br>E3 at step 5: Saving fails. No request is created and the system reports the failure.<br>E4 after step 5: The request is not decided within 10 working days. It becomes Expired and both departments are notified. |
| **Postconditions** | Success: The request has an ID and the state Submitted (or Open), and the owning Asset Coordinator is notified. The asset's available quantity, ownership and location are unchanged.<br>Failure: No request is created. |
| **Rules** | R1: A Department Representative can request only for their own department.<br>R2: The quantity must be a positive whole number no greater than the current available quantity at submission.<br>R3: Submitting a request does not reserve units or change ownership or location.<br>R4: The requesting Department Representative can never release or approve their own request. |

## UC-4: Approve or Reject a Request

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| **Use case ID and name** | UC-4: Approve or Reject a Request |
| **Description** | The owning department's Asset Coordinator reviews a submitted request and releases or declines it. An Administrator then records the final approval or rejection; approval reserves the requested quantity. |
| **Primary actor (initiator)** | Asset Coordinator (of the owning department) |
| **Secondary actors** | Administrator (final approval); Procurement Team (second approval when FR-4.3 applies); requesting Department Representative (notified of the decision). |
| **Related requirements** | FR-3.8, FR-3.9, FR-3.12; FR-4.1 to FR-4.6, FR-4.14; FR-5.16; FR-7.N1 to FR-7.N6; PER-3, PER-6; SAF-2; SEC-12; QUA-4, QUA-7; BR-2 to BR-5. |
| **Preconditions** | 1. The Asset Coordinator is signed in, manages the requested asset and did not submit the request.<br>2. The request is Submitted. |
| **Trigger** | The Asset Coordinator opens a Submitted request from their review list. |
| **Main flow** | 1. The system shows the request, the requesting department, the asset's current available quantity and any restrictions. |

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| **Use case ID and name** | UC-4: Approve or Reject a Request |
| | 2. The Asset Coordinator releases the request.<br>3. The system records the release and sends the request to the Administrator, and to the Procurement Team when a second approval is required (FR-4.3).<br>4. The Administrator reviews the released request and approves it.<br>5. The system rechecks authority, separation of duties, safety restrictions and current availability, records Approved and reserves the requested quantity.<br>6. The system notifies the requesting Department Representative and the Asset Coordinator; the transfer can then be arranged (UC-5). |
| **Alternative flows** | A1 at step 2: The Asset Coordinator declines with a reason. The request becomes Rejected and the Department Representative is notified; nothing is reserved.<br>A2 at step 4: The Administrator, or the Procurement Team member at the second approval, rejects with a reason. The request becomes Rejected and nothing is reserved.<br>A3 at step 4: A second approval is required. The request stays Released until a different Procurement Team member approves; continue at step 5 once all approvals exist.<br>A4: No decision is made within 10 working days of submission. The request becomes Expired (FR-3.9). |
| **Exception flows** | E1 at step 2 or 4: The reviewer submitted the request or already decided another stage. The system refuses the action and logs it.<br>E2 at step 5: The available quantity is now lower than requested, or the asset was withdrawn or became unsafe. Approval is blocked, current details are shown and nothing is reserved.<br>E3 at step 2 or 4: A decline or rejection reason is missing. The decision cannot be saved.<br>E4 at step 5: Saving or audit recording fails. The request stays Released and nothing is reserved. |
| **Postconditions** | Success: All required decisions are recorded, the request is Approved and the quantity is reserved. Ownership and location are unchanged.<br>Rejection or failure: The reason is kept, nothing is reserved and the asset is unchanged. |
| **Rules** | R1: Release precedes approval.<br>R2: No one approves their own request or fills two approval stages.<br>R3: Availability is checked before approval; units are never over-allocated.<br>R4: Approval reserves units but does not change ownership or location.<br>R5: AI cannot make an approval decision. |

# UC-5: Transfer an Asset

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| **Use case ID and name** | UC-5: Transfer an Asset |
| **Description** | After a request is approved, the owning Asset Coordinator hands the asset to the receiving department. The receiver confirms receipt, and custody and location change. |
| **Primary actor** | Asset Coordinator (of the owning department) |

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| **Use case ID and name** | UC-5: Transfer an Asset |
| **(initiator)** | |
| **Secondary actors** | Department Representative or Asset Coordinator of the receiving department (confirms receipt); Administrator (decides disputes). |
| **Related requirements** | FR-1.10; FR-3.9; FR-4.6 to FR-4.16; FR-5.9, FR-5.16, FR-5.27; FR-7.N1 to FR-7.N6; FR-13.3; PER-3, PER-6; SAF-1, SAF-2, SAF-5, SAF-7, SAF-9; SEC-12, SEC-13; QUA-4, QUA-7; BR-5, BR-6. |
| **Preconditions** | The request is Approved and the requested quantity is reserved. |
| **Trigger** | The Asset Coordinator chooses to arrange the transfer for an approved request. |
| **Main flow** | 1. The Asset Coordinator enters the method (collection or delivery), planned date, from and to locations, contact person and any transfer cost.<br>2. The system creates a Scheduled transfer and notifies the receiving department, showing any handling or safety notes.<br>3. At handover, the Asset Coordinator marks the asset as handed over and attaches any handover documents.<br>4. The system checks the safety conditions (SAF-5, SAF-9) and sets the transfer and asset to In transfer.<br>5. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and records the condition on arrival.<br>6. The system changes the Asset Coordinator, department and location, sets the transfer to Received and the request to Completed, and records each step in the asset history and audit log. |
| **Alternative flows** | A1 before step 3: The Asset Coordinator changes the planned date. The receiver is notified.<br>A2 before step 3: The Asset Coordinator or the requesting Department Representative cancels. The transfer becomes Cancelled and the reserved units return to Available.<br>A3 at step 1: Only part of a multi-unit asset is transferred. At step 6, the system creates a linked record for the transferred units (FR-4.15). |
| **Exception flows** | E1 at step 5: The asset is damaged or is the wrong item. The receiver reports a problem, the transfer becomes Disputed and the Administrator decides to accept it or return the asset.<br>E2 at step 4: The asset is unsafe, or is a risky asset without a safe inspection in the last 90 days. Handover is blocked (SAF-5, SAF-9).<br>E3 before step 3: No handover within 10 working days of approval. The reservation is released and the request becomes Expired.<br>E4 at step 5: Receipt is not confirmed within 5 working days. The system reminds the receiver and copies the Administrator.<br>E5 at step 5: A user outside the receiving department tries to confirm receipt. The system refuses. |
| **Postconditions** | Success: The asset belongs to the receiving department with its new Asset Coordinator and location, and the full transfer is recorded.<br>Failure or cancellation: Custody and location are unchanged, and the asset returns to Available or stays Disputed until decided. |
| **Rules** | R1: Only an Approved request can be transferred.<br>R2: Only the receiving department (a Department Representative or its Asset |

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| **Use case ID and name** | UC-5: Transfer an Asset |
| | Coordinator) can confirm receipt.<br>R3: Custody and location change only on confirmed receipt. |

## UC-6: Record Maintenance or Repair

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| **Use case ID and name** | UC-6: Record Maintenance or Repair |
| **Description** | Maintenance Staff record the inspection and repair or refurbishment work done on an asset with an open maintenance request, capture its cost, and update the asset's condition so it can return to service, or send it for end-of-life review if it cannot be repaired. |
| **Primary actor (initiator)** | Maintenance Staff |
| **Secondary actors** | Asset Coordinator and the user who reported the defect (notified of the outcome); Sustainability Officer (notified if the asset is flagged for end-of-life review) |
| **Related requirements** | FR-5.1 to FR-5.18; FR-7.N1 to FR-7.N5; FR-13.3; PER-3, PER-6; SAF-2, SAF-5, SAF-9; SEC-5, SEC-12, SEC-13; QUA-4, QUA-7; BR-7. |
| **Preconditions** | 1. The user is signed in with the Maintenance Staff role.<br>2. A maintenance request for the asset is assigned to the user (FR-5.3) and is in the 'Reported', 'Inspection Requested', 'Under Inspection' or 'Repair in Progress' state. |
| **Trigger** | Maintenance Staff select an assigned maintenance request and choose 'Start work'. |
| **Main flow** | 1. The system shows the asset's details, condition, maintenance history and safety notes, and sets the request to 'Under Inspection'.<br>2. Maintenance Staff record the inspection findings, the assessed condition and whether the asset is safe to use.<br>3. Maintenance Staff record the repair or refurbishment work, its cost and the estimated additional years of use (the request moves to 'Repair in Progress').<br>4. Maintenance Staff enter the asset's updated condition and mark the request 'Completed'.<br>5. The system validates the data and saves it, updating the asset's condition, availability, life-cycle history and maintenance cost.<br>6. The system notifies the Asset Coordinator and the user who reported the defect. |
| **Alternative flows** | A1 (step 2) No repair needed: Maintenance Staff skip step 3 and complete the request after the inspection.<br>A2 (step 2) Not repairable: Maintenance Staff close the request as 'Closed – Not Repairable'; the system flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.12), who may start UC-7.<br>A3 (step 3) Work not finished: Maintenance Staff save progress; the request stays 'Repair in Progress' and the asset stays unavailable. The use case resumes later at step 3.<br>A4 (step 2) Asset unsafe: Maintenance Staff mark the asset not safe to use; the system withdraws its listings and blocks matching, reservation and transfer (SAF-5). |

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| **Use case ID and name** | UC-6: Record Maintenance or Repair |
| | A5 (step 5) Maintenance-cost threshold reached: the system flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.11). |
| **Exception flows** | E1 Missing or invalid data (for example a completion date before the start date, or a negative cost): the system identifies the invalid field and saves nothing.<br>E2 The request is not assigned to the user, or the user lacks the Maintenance Staff role: the system denies access and records the attempt in the audit log.<br>E3 The request was cancelled in the meantime (FR-5.13): the system informs Maintenance Staff and saves no changes.<br>E4 The audit entry cannot be recorded: the system does not save the work and shows an error. |
| **Postconditions** | Success: the maintenance record and cost are saved; the asset's condition and availability are updated; life-cycle events, notifications and audit entries are recorded.<br>Failure: the asset record is unchanged and the user sees the reason. |
| **Rules** | R1 Only Maintenance Staff or Administrators may record inspection or repair results.<br>R2 An asset cannot be shown as available, matched, reserved or transferred while inspection or repair is in progress.<br>R3 Costs are recorded in AED and cannot be negative.<br>R4 Life-cycle events cannot be edited or deleted.<br>R5 A cumulative maintenance cost of 50% or more of the replacement price triggers end-of-life review (default, TBD-7). |

## UC-7: Donate, Recycle or Dispose of an Asset

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| **Use case ID and name** | UC-7: Donate, Recycle or Dispose of an Asset |
| **Description** | A Sustainability Officer proposes an end-of-life action for an asset (donation, recycling, disassembly for usable components, retirement or disposal). A different Sustainability Officer records a safety review, an Administrator approves the proposal, and after handover the system records the outcome, closes the asset and updates the sustainability indicators. Disposal is treated as the last option. An Asset Coordinator can also start this use case (FR-5.20); the steps are the same. |
| **Primary actor (initiator)** | Sustainability Officer |
| **Secondary actors** | Second Sustainability Officer (safety reviewer); Administrator (approver, BR-8); Asset Coordinator (consulted and notified); Facilities Management or an approved recycling vendor (handling party for hazardous items); Donation Recipient. |
| **Related requirements** | FR-5.11, FR-5.12, FR-5.16 to FR-5.34; FR-7.N1 to FR-7.N5; FR-7.A4; FR-10.1 to FR-10.8; FR-13.3, FR-13.6; AI-1 to AI-10; PER-3 to PER-6; SAF-1 to SAF-4, SAF-6, SAF-7; SEC-5, SEC-10, SEC-12; QUA-4, QUA-7, QUA-11 to QUA-14; BR-4, BR-8, BR-9; LEG-7. |
| **Preconditions** | 1. The user is signed in with the Sustainability Officer role.<br>2. The asset is registered and is not already in a final state (Donated, Recycled, Disassembled, Retired or Disposed). |

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| **Use case ID and name** | UC-7: Donate, Recycle or Dispose of an Asset |
| **Trigger** | The Sustainability Officer selects 'Propose end-of-life action' on an asset, usually one flagged for end-of-life review (UC-6). |
| **Main flow** | 1. The system shows the asset’s details, condition, history, maintenance cost, open requests it may satisfy, and the AI recommendation with its confidence and explanation (if available).<br>2. The Sustainability Officer selects the action, enters the justification (for disposal, a reason for each alternative) and confirms or updates the hazard flag.<br>3. The system checks for blocking items, saves the proposal as ‘Pending Safety Review’ and notifies a different Sustainability Officer.<br>4. The second Sustainability Officer records the safety review; the system sets ‘Pending End-of-Life Approval’ and notifies the Administrator.<br>5. The Administrator approves the proposal; the system notifies the proposer and the Asset Coordinator.<br>6. After handover, the Sustainability Officer records the completion details (recipient or handling party, date, quantity, weight, and the handover document or certificate); the system sets the asset’s final state, adds the life-cycle event, removes the asset from listings and updates the sustainability indicators. |
| **Alternative flows** | A1 (step 1) The asset matches an open request: the Sustainability Officer asks the Asset Coordinator to offer it for internal reuse instead (UC-3 to UC-5); no proposal is created.<br>A2 (step 2) The chosen action differs from the AI recommendation: the system requires a reason and records the override.<br>A3 (step 4) The asset is hazardous or electronic waste: the reviewer assigns Facilities Management or an approved vendor (FR-5.25), and at step 6 the handling record and certificate are required.<br>A4 (step 4 or 5) The reviewer or the Administrator rejects the proposal: the system records the reason, returns the asset to its previous state and notifies the proposer.<br>A5 (step 5) The AI recommendation being followed has Low confidence: the Administrator must record their own reason before approving (SAF-6).<br>A6 (step 6) Retirement: the completion details are the date, reason and final location (FR-5.31). |
| **Exception flows** | E1 The asset has an open reservation, pending transfer or open maintenance request: the system rejects the proposal and lists the blocking items.<br>E2 The justification is missing, or a disposal proposal is missing the reason for any alternative: the system does not submit the proposal.<br>E3 The safety reviewer or approver is the proposer: the system refuses the action.<br>E4 The AI recommendation is unavailable: the system shows 'Recommendation unavailable' and the Sustainability Officer continues without it.<br>E5 Required completion details are missing (such as weight, or the certificate for a hazardous item): the system does not close the asset. |
| **Postconditions** | Success: the asset is in its final state and cannot be reserved or matched; the completion record, life-cycle event, notifications and audit entries are saved; the indicators include the action.<br>Rejected or failed: the asset is in its previous state, and the proposal is kept with its reason. |

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| **Use case ID and name** | UC-7: Donate, Recycle or Dispose of an Asset |
| **Rules** | R1 Disposal is the last option and needs a reason for rejecting each alternative: reuse, repair, donation and recycling.<br>R2 The proposer cannot perform the safety review or approve their own proposal.<br>R3 Hazardous items are never closed without a safety review, a handling party and a certificate.<br>R4 AI recommendations are advisory only and never complete an action automatically.<br>R5 Weight is recorded for every donation, recycling, disassembly and disposal (actual, category default or estimate, marked as such). |

## UC-8: Manage User Accounts and Roles

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| **Use case ID and name** | UC-8: Manage User Accounts and Roles |
| **Description** | The System Administrator reviews account applications and maintains authorized accounts, roles and department membership, with consent and an audit trail. |
| **Primary actor (initiator)** | System Administrator |
| **Secondary actors** | Applicant or account holder (supplies details and is notified); Department Representative (confirms department membership). |
| **Related requirements** | FR-6.3, FR-6.4, FR-6.8 to FR-6.12; FR-7.N1 to FR-7.N5; FR-7.A3; PER-3; SEC-5 to SEC-9; BR-4; LEG-1; ORG-3. |
| **Preconditions** | 1. The System Administrator is signed in with account-management permission.<br>2. Role and department records exist or can be created through department management.<br>3. For activation, the applicant has completed UC-13 with recorded consent, and the requested role has recorded authorization (TBD-1, TBD-3). |
| **Trigger** | The System Administrator opens a pending application or selects an existing account or department. |
| **Main flow** | 1. The System Administrator opens the application or account.<br>2. The system shows the account status, department, current roles, consent record and open work.<br>3. The System Administrator checks identity, department and the recorded authorization, then selects the approved role and department or a status change and enters the reason.<br>4. The system validates required details, duplicates, consent, permission limits and dependencies, and shows the change for confirmation.<br>5. The System Administrator confirms. The system saves the change and the audit entry, and ends affected sessions where required.<br>6. The system notifies the account holder and shows the outcome. |
| **Alternative flows** | A1 before step 1: The System Administrator sends an invitation. The invitee completes UC-13 with their own details and consent before activation.<br>A2 at step 3: The System Administrator creates, renames or deactivates a department. The system keeps its unique ID and history, and blocks deactivation while dependencies remain.<br>A3 at step 3: The System Administrator suspends or closes the account. Active |

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| **Use case ID and name** | UC-8: Manage User Accounts and Roles |
| | sessions end and open work is flagged for reassignment.<br>A4: The account holder withdraws consent. Access stops and the System Administrator handles the retained data under the approved policy (SEC-8). |
| **Exception flows** | E1 at step 4: Consent, required details or authorization is missing. Activation or the privilege change is blocked.<br>E2 at step 4: A university ID, email or department ID is duplicated. The system identifies the conflict without creating another record.<br>E3 at step 4: The change would raise the System Administrator's own privileges, or combine roles that are not allowed. The action is refused and logged.<br>E4 at step 4: Deactivation would leave a department's records unmanaged or remove the last active System Administrator. The system requires reassignment first.<br>E5 at step 5: Saving or audit recording fails. Nothing is committed.<br>E6 at step 6: Notification delivery fails. The saved change stays in force and delivery is retried. |
| **Postconditions** | Success: The authorized account or department change is saved, audited and applied to later access checks.<br>Failure or cancellation: No unauthorized activation, privilege change or loss of history occurs. |
| **Rules** | R1: Consent precedes retention of personal data and activation.<br>R2: Only the System Administrator applies role changes, with recorded authorization.<br>R3: Technical administration gives no transfer or disposal approval power.<br>R4: Passwords and recovery secrets never appear in records or notices. |

## UC-9: Review and Override AI Matches

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| **Use case ID and name** | UC-9: Review and Override AI Matches |
| **Description** | A Procurement Team member reviews the AI-ranked assets suggested for an unmatched need, chooses one or overrides the suggestions, and sends the choice to the normal release and approval process. |
| **Primary actor (initiator)** | Procurement Team member |
| **Secondary actors** | Requesting Department Representative (clarifies the need and is notified); Asset Coordinator of the chosen asset (reviews the request in UC-4). |
| **Related requirements** | FR-3.2; FR-5.9, FR-5.27; FR-7.N1 to FR-7.N5; FR-7.A4; FR-8.1 to FR-8.9; AI-1 to AI-10; PER-4 to PER-6; SAF-1, SAF-2; SEC-5, SEC-10, SEC-11; QUA-1, QUA-4, QUA-7, QUA-10, QUA-12 to QUA-14; BR-9, BR-10; LEG-1, LEG-3, LEG-6, LEG-9. |
| **Preconditions** | The Procurement Team member is signed in and may review the request. The request is Open with its required details. Standard search remains available if AI fails. |
| **Trigger** | The Procurement Team member selects an Open request and asks for matches. |
| **Main flow** | 1. The system shows the request details and checks that the required fields are |

| Use case ID and name | UC-9: Review and Override AI Matches |
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| | complete.<br>2. The system excludes unsuitable or restricted assets and ranks the remaining matches with scores, confidence, reasons, permitted costs and sustainability estimates.<br>3. The Procurement Team member selects the suggested asset or another suitable asset, giving a reason when choosing another.<br>4. The system rechecks availability, quantity and permissions and asks for confirmation.<br>5. The Procurement Team member confirms.<br>6. The system records the choice and the AI details, links the asset to the request, sets it to Submitted and notifies the owning Asset Coordinator and the Department Representative (UC-4). |
| **Alternative flows** | A1 at step 2: No suitable match is found. The system says so and offers standard search or a request change.<br>A2 at step 2: An asset meets only part of the quantity. The system shows the shortage; a smaller or split request must be confirmed by the Department Representative.<br>A3 at step 3: The Procurement Team member rejects all suggestions. The system records the rejection and reason, and the request stays Open.<br>A4 at step 2: AI is unavailable or gives no usable result. The Procurement Team member uses standard search; a chosen asset continues at step 4 and is recorded as a manual choice. |
| **Exception flows** | E1 at step 1: A required field is missing or invalid. The system identifies it and waits for the Department Representative to correct the request.<br>E2 at step 4: The asset is no longer available or permissions have changed. Confirmation is blocked and updated options are shown.<br>E3 at step 6: The choice cannot be saved. The system reports the failure, changes nothing and keeps the pending choice for review. |
| **Postconditions** | Success: The chosen or rejected suggestions and the review details are recorded; a chosen asset enters UC-4. Nothing is reserved until approval.<br>Failure or cancellation: Nothing changes and the request remains Open. |
| **Rules** | R1: A lower-ranked suitable asset may be chosen, but all access, safety and approval rules still apply.<br>R2: A compatibility score supports the decision; it does not reserve the asset or guarantee suitability.<br>R3: The ranking never uses the requesting department’s name or size (LEG-9). |

## UC-10: Classify a New Asset with AI Suggestions

| Use case ID and name | UC-10: Classify a New Asset with AI Suggestions |
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| **Description** | The Asset Coordinator reviews AI suggestions and confirms the details used to register a new asset. This use case extends UC-1. |
| **Primary actor (initiator)** | Asset Coordinator |
| **Secondary actors** | None. The Administrator maintains the approved category list outside this use |

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| **Use case ID and name** | UC-10: Classify a New Asset with AI Suggestions |
| | case. |
| **Related requirements** | FR-1.12; FR-7.A4; FR-9.1 to FR-9.9; AI-1 to AI-7, AI-9, AI-10; PER-4 to PER-6; SAF-2; SEC-11, SEC-13; QUA-1, QUA-4, QUA-7, QUA-9, QUA-14; BR-9; LEG-3. |
| **Preconditions** | The Asset Coordinator is signed in and may register assets for the department. The draft has a title and description, and the approved category list is available. |
| **Trigger** | The Asset Coordinator asks for AI suggestions while preparing an asset record (UC-1). |
| **Main flow** | 1. The Asset Coordinator enters a title and description, adds specifications, photos or documents, and asks for suggestions.<br>2. The system checks the inputs and identifies any file it cannot read.<br>3. The system suggests a category, subcategory, tags, possible material type and description, with confidence, supporting information, conflicts and missing fields, kept separate from confirmed values.<br>4. The Asset Coordinator accepts, edits or rejects each suggestion and supplies missing required details.<br>5. The system checks the category and required fields and shows the completed record.<br>6. The Asset Coordinator confirms. The system saves the agreed details through UC-1 and records the original suggestions and corrections. |
| **Alternative flows** | A1 at step 4: The Asset Coordinator rejects all suggestions and enters the details manually. Continue at step 5.<br>A2 at step 3: The material type or subcategory cannot be determined. The system marks it unknown and allows manual entry.<br>A3 at step 2: An optional file cannot be read, but other information is sufficient. The system identifies the file, lowers confidence where needed and continues.<br>A4 before step 6: The Asset Coordinator cancels. Nothing is registered or published. |
| **Exception flows** | E1 at step 3: AI fails or takes too long. The system keeps the draft and offers manual classification.<br>E2 at step 5: A required field is missing or the category is invalid. The system identifies the problem and returns to step 4.<br>E3 at step 6: Access has expired or saving fails. The system reports the failure and keeps the draft for an authorized retry. |
| **Postconditions** | Success: The record contains checked and confirmed details. The original AI suggestions, model version and human corrections are recorded.<br>Failure or cancellation: No unconfirmed details are saved or published. The draft remains available. |
| **Rules** | R1: The Asset Coordinator must check the asset facts before saving AI suggestions.<br>R2: Categories and subcategories must come from the approved list.<br>R3: Corrections are recorded for review; they do not automatically retrain the AI. |

## UC-11: Ask the AI Assistant

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| **Use case ID and name** | UC-11: Ask the AI Assistant |
| **Description** | A Department Representative asks a question or prepares a resource request in everyday language. The assistant answers from permitted information and asks for confirmation before submitting anything. |
| **Primary actor (initiator)** | Department Representative |
| **Secondary actors** | None required. A request prepared here is submitted through UC-3 and reviewed in UC-4. |
| **Related requirements** | FR-3.14; FR-7.A4; FR-11.1 to FR-11.10; AI-1 to AI-10; PER-2, PER-4 to PER-6; SAF-2; SEC-11; QUA-1, QUA-4, QUA-14, QUA-15; BR-9; LEG-1, LEG-3. |
| **Preconditions** | The Department Representative is signed in with a known role and department. Any factual answer must have a permitted supporting record or approved document. |
| **Trigger** | The Department Representative opens the assistant and enters a question or resource need. |
| **Main flow** | 1. The Department Representative enters a question or describes the resource they need.<br>2. The system identifies the task, checks permissions and asks a clarifying question if needed.<br>3. The assistant answers from permitted records or approved documents, with sources, confidence and limitations, or prepares a request draft. For a question, the use case ends here.<br>4. The Department Representative completes and checks the draft.<br>5. The system shows the final action and values and asks for confirmation.<br>6. The Department Representative confirms. The system rechecks access and the normal rules, submits through UC-3 and shows the actual result and request ID. |
| **Alternative flows** | A1 at step 2: The question is unclear. The assistant asks a specific follow-up question.<br>A2 at step 3: No supporting record or approved policy exists. The assistant says so and points to the responsible role or a permitted page.<br>A3 at step 4: The Department Representative prefers the standard form. The system copies the confirmed draft values into it.<br>A4 at step 5: The Department Representative cancels or does not confirm. Nothing is submitted or changed. |
| **Exception flows** | E1 at any step: The request asks for restricted information or an action outside the user's role. The system refuses that part without revealing protected details.<br>E2 at step 3: AI fails or takes too long. The system keeps the draft and offers standard search or forms.<br>E3 at step 6: Details are invalid, access has expired or availability has changed. The action is blocked and the system explains what needs correcting.<br>E4 at step 6: Submission fails or the result is uncertain. The assistant reports that status and avoids a duplicate submission on retry. |
| **Postconditions** | Success: The user receives a supported answer, or the confirmed request is |

| | |
|---|---|
| **Use case ID and name** | UC-11: Ask the AI Assistant |
| | saved with its actual ID and status.<br>Failure or cancellation: No unauthorized or unconfirmed change is made; the draft can be continued in the standard form. |
| **Rules** | R1: Confirmation covers only the action and values shown; important changes need confirmation again.<br>R2: Instructions in retrieved documents or asset details cannot change permissions or approval rules.<br>R3: Answers about policy and sustainability clearly separate facts, estimates and missing information. |

## UC-12: Generate a Savings and Sustainability Report

| | |
|---|---|
| **Use case ID and name** | UC-12: Generate a Savings and Sustainability Report |
| **Description** | A Finance Officer generates the savings and sustainability report (FR-12.2 type b) for a selected period. The system calculates the indicators with fixed formulas, obtains an AI-written summary of those figures, checks that the summary's figures match, and publishes the report only after the Finance Officer reviews and approves it. |
| **Primary actor (initiator)** | Finance Officer |
| **Secondary actors** | Sustainability Officer (optional reviewer); University Management (receives the published report). |
| **Related requirements** | FR-7.N1 to FR-7.N5; FR-7.R1 to FR-7.R8; FR-7.A4; FR-12.1 to FR-12.12; FR-13.1 to FR-13.13; AI-1 to AI-10; PER-4, PER-5; SAF-2; SEC-5, SEC-11; QUA-14; BR-9 to BR-11; LEG-3, LEG-5, LEG-6, LEG-8. |
| **Preconditions** | The user is signed in with the Finance Officer role. |
| **Trigger** | The Finance Officer selects 'Generate savings and sustainability report'. |
| **Main flow** | 1. The Finance Officer selects the report period (for example a month) and the scope (campus or a department).<br>2. The system calculates the indicators for that period and scope (FR-13) and shows them with their data coverage, assumptions and missing data.<br>3. The Finance Officer requests the AI summary; the system sends only aggregated, non-personal figures to the AI service.<br>4. The system receives the draft, checks every figure against the calculated values, and shows it labeled 'AI-generated' with the model version, period, data sources and assumptions.<br>5. The Finance Officer edits the draft if needed and approves it; the system checks the figures in the edited version again.<br>6. The system stores the draft and the approved version, records the approval, publishes the report to authorized users (hiding financial figures from those without financial access) and notifies them. |
| **Alternative flows** | A1 (step 3) The Finance Officer needs only the figures: the system produces the standard report and indicators (FR-7, FR-13) without an AI summary, and the use case ends. |

| | |
|---|---|
| **Use case ID and name** | UC-12: Generate a Savings and Sustainability Report |
| | A2 (step 5) The Finance Officer is not satisfied: they regenerate the draft or discard it; a discarded draft is kept as 'Discarded' and nothing is published.<br>A3 (step 5) The Finance Officer sends the draft to a Sustainability Officer for review before approving it. |
| **Exception flows** | E1 The AI service is unavailable or does not respond within the time limit in PER-5: the system informs the Finance Officer and offers the standard report and indicators instead.<br>E2 There is no recorded activity in the selected period: the system says so and does not generate a summary.<br>E3 A figure in the draft or edited version does not match the calculated value: the system highlights the statement and blocks approval until it is corrected or the draft is regenerated.<br>E4 Some records lack a value an indicator needs and have no category default: the system excludes them and lists them under 'Missing data' (FR-13.8). |
| **Postconditions** | Success: the approved report is stored with its draft, model version, period and approver, and is available to authorized users; the audit log records the generation and approval.<br>Failure: nothing is published; the standard report and indicators remain available. |
| **Rules** | R1 Only approved AI reports are shared with others.<br>R2 Every figure comes from the fixed-formula calculations (FR-7, FR-13); the AI does not produce its own numbers.<br>R3 Sustainability and savings figures are labeled as estimates, with their assumptions.<br>R4 Financial values are in AED and visible only to roles allowed to see financial information (SEC-5).<br>R5 Recommended actions are suggestions only and are never carried out automatically. |

## UC-13: Register an Account and Give Consent

| | |
|---|---|
| **Use case ID and name** | UC-13: Register an Account and Give Consent |
| **Description** | An Applicant applies for a ReSource account, reads the privacy notice and terms, and gives explicit consent so that the application can be reviewed and activated (UC-8). |
| **Primary actor (initiator)** | Applicant |
| **Secondary actors** | System Administrator (reviews and activates the application in UC-8). |
| **Related requirements** | FR-6.1 to FR-6.4, FR-6.6; FR-7.N1 to FR-7.N5; PER-3; SEC-2, SEC-6 to SEC-8; LEG-1; ORG-3. |
| **Preconditions** | The Applicant has no active account, or holds a valid invitation (FR-6.4). The current privacy notice and terms are published. |
| **Trigger** | The Applicant opens the registration page or an invitation link. |
| **Main flow** | 1. The system shows the privacy notice, the terms, the intended data uses and |

| | |
|---|---|
| **Use case ID and name** | UC-13: Register an Account and Give Consent |
| | how to withdraw consent.<br>2. The Applicant enters name, university ID, university email, department and a password.<br>3. The Applicant ticks the consent choice, which is initially unticked, and submits.<br>4. The system checks the required fields, the password rules and duplicates, and records consent with the time and notice version.<br>5. The system saves the application as Pending approval and notifies the System Administrator.<br>6. The system tells the Applicant that the account can be used after activation (UC-8). |
| **Alternative flows** | A1 at step 3: The Applicant does not consent. Nothing is saved, and the system explains that an account cannot be created without consent.<br>A2 at step 1: The Applicant arrived through an invitation. At step 5 the application goes to the System Administrator for activation as in UC-8. |
| **Exception flows** | E1 at step 4: A required field is missing, the password breaks SEC-2 or the email is not a permitted university address. The system names the problem and keeps the other entries except the password.<br>E2 at step 4: The university ID or email already has an account. The system refuses the application and offers sign-in or credential recovery (FR-6.6).<br>E3 at step 5: Saving fails. No application or consent record is kept, and the system reports the failure. |
| **Postconditions** | Success: A Pending approval application with a consent record exists; the Applicant cannot use asset functions until activation.<br>Failure or cancellation: No personal data is retained. |
| **Rules** | R1: Consent is explicit and recorded before any registration data is kept.<br>R2: Applicants cannot choose privileged roles.<br>R3: Only the personal data listed in SEC-7 is collected. |

# Appendix A: Glossary

| Term | Definition |
|---|---|
| Administrator / System Administrator | Administrator controls delegated business approvals. System Administrator controls technical access and configuration. These are separate roles. |
| AED | United Arab Emirates dirham, the currency used for all money values. |
| AI / semantic matching | Artificial intelligence / finding related resource descriptions by meaning rather than only identical words. |
| Applicant | A person who has applied for an account that a System Administrator has not yet activated. |
| Approved vendor list | The Administrator-maintained list of recycling and disposal vendors with an unexpired approval (FR-5.25). |
| Asset | An individually identified resource or a recorded group of equivalent units managed by the system. |
| Asset Coordinator | The role responsible for the records, availability and custody of the assets held by their department. |
| Audit trail | Attributable history of actions and decisions. Corrections are linked records; users cannot overwrite past entries. |
| Available quantity | Units that remain eligible for requests after reservations and other restrictions are considered. |
| Avoided purchase / net savings | Estimated new-purchase value avoided through reuse / that estimate adjusted for relevant transfer, repair and other recorded costs. |
| Category default value | A versioned value per asset category (replacement price, storage and handling cost, weight, additional years of use or emission factor) used when no actual value is recorded. |
| Compatibility score / confidence | A compatibility score (0 to 100) expresses estimated fit to a request. The confidence level (High, Medium or Low, AI-10) expresses how reliable the result is; neither authorizes an action. |
| Condition / life-cycle state | Condition describes physical quality, such as Good or Poor. Life-cycle state describes operational status, such as Available, In transfer or Under maintenance. |
| Consent | An explicit recorded choice to the stated data uses, linked to the notice version and time. It is separate from administrator approval of access. |
| CSV | Comma-separated values, a plain spreadsheet-compatible file format. |
| Department Representative | An authorized departmental user who searches for resources, submits and tracks requests for their department and confirms receipt. |
| Emission factor | Kilograms of $CO_2e$ avoided for each kilogram of an asset that is reused instead of bought new. Defaults come from EPA WARM version 16 [15] (Table 2.13-2). |

| Term | Definition |
|---|---|
| End-of-life action | An approved donation, recycling, disassembly, retirement or disposal action, with completion evidence. |
| End-of-life review flag | A system flag that asks for an end-of-life decision after an asset is closed as not repairable or reaches the maintenance-cost threshold (FR-5.11, FR-5.12). |
| Handling party | Facilities Management or an approved vendor assigned to handle hazardous or electronic-waste items and issue a certificate. |
| Handover / receipt | Handover records departure from the originating department. Receipt confirms arrival and condition and permits custody and location to change. |
| Hazard flag | A field recorded at registration that marks an asset as hazardous or electronic waste. |
| Human override | An authorized person’s recorded rejection or correction of an AI suggestion, with a reason where required. |
| KU | Khalifa University. |
| Model version / rule version | The identifier of the AI model or service and of the ranking or threshold rules that produced a result, recorded for traceability (AI-5). |
| OWASP | Open Worldwide Application Security Project, publisher of the Top 10 web security risks [13]. |
| PDF | Portable Document Format, used for uploaded documents and downloaded reports. |
| PDPL | The UAE Personal Data Protection Law, Federal Decree-Law No. 45 of 2021 [6]. |
| Procurement Team | The role that reviews reuse options for unmatched needs and gives the second approval for high-value requests; purchasing decisions stay under university procedures. |
| Project description role names | The project description [1] uses Asset Custodian, Requester and Procurement Officer. This document uses the Phase 2 roles [5]: Asset Coordinator, Department Representative (who submits requests for their department) and Procurement Team. |
| Publication | Making a validated, eligible asset visible in the authorized shared catalogue. |
| RBAC | Role-based access control: permissions depend on assigned role, department and record scope. |
| Release / approval | Release is the owning Asset Coordinator’s agreement to make the asset available for the request. Approval is the following authorized business decision, which reserves the units. |
| Requirement and item prefixes | FR functional requirement; AI general AI requirement; PER performance; SAF safety; SEC security; QUA quality; BR business rule; LEG legal; ORG organizational; UC use case; TBD decision to be determined. |

| Term | Definition |
| --- | --- |
| Reservation | Units set aside for an approved request. Submitting a request does not reserve units; the reservation is made at final approval and ends at receipt, cancellation or expiry. It does not change ownership or location. |
| ReSource | The proposed campus resource exchange and asset life-cycle management system. |
| Safety review | A recorded check by a Sustainability Officer that confirms or corrects the hazard classification before a hazardous asset is reused or an end-of-life action proceeds. |
| SRS | Software Requirements Specification: the documented behavior and constraints used for design and testing. |
| Waste diversion / $CO_2e$ | Material retained in use or routed away from disposal under the stated method / carbon dioxide equivalent, a unit for estimated climate impact. |
| WCAG | Web Content Accessibility Guidelines, the W3C accessibility standard used in QUA-2 and QUA-3 [14]. |
| Working day | A day counted by the approved university calendar; the prototype calendar and deadlines require confirmation in TBD-5. |

# Appendix B: Analysis Models (State Lists)

Each section has two tables: the states of one record type with their meaning, and the moves between states. For every move, the transition table gives the event that causes it, who causes it and the requirement that allows it. ‘System (automatic)’ means the system makes the move when the stated condition is met. Users cannot type a state; they cause a move only through the listed action and only if their role and department permit it (SEC-5). Final states cannot change, and corrections are recorded as linked records (SEC-12). For a multi-unit asset, a reservation or transfer may cover only some units; transferred units get a linked record (FR-4.15).

## B.1 Asset States

**Table B-1: Asset states**

| State | Meaning |
|---|---|
| **Draft** | Registered but not published; also used while a hazardous asset waits for its safety review |
| **Available** | Published; some or all units can be requested |
| **Reserved** | All available units are reserved for approved requests |
| **In transfer** | Handed over and not yet received |
| **In use** | Used by its owning department and not listed |
| **Withdrawn** | Removed from listing by the Asset Coordinator, or because it was marked not safe to use |
| **Under maintenance** | Has a maintenance request that is Under Inspection or Repair in Progress |
| **Pending Safety Review** | End-of-life action proposed; waiting for the safety review |
| **Pending End-of-Life Approval** | Safety review recorded; waiting for approval and the completion record |
| **Donated** | Given to an organization outside the university (final) |
| **Recycled** | Sent for recycling (final) |
| **Disassembled** | Taken apart; usable components may be registered as new linked assets (final) |
| **Retired** | Taken out of use and kept, for example in central storage (final) |
| **Disposed** | Disposed of as the last option (final) |
| **Archived** | Draft removed without being published; its ID is never reused (final) |

**Table B-2: Asset state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Asset saved as a draft | Asset Coordinator of the owning department | Draft | FR-1.1, FR-1.9 |
| **Draft** | Publication confirmed with all required fields valid; for a hazardous or electronic-waste asset, after the safety review | Asset Coordinator (safety review by a Sustainability Officer) | Available | FR-2.1, FR-1.15, SAF-1 |
| **Draft** | Draft removed without being published | Asset Coordinator | Archived | FR-1.7, FR-1.9 |
| **Available** | Final required approval saved for the last available units; the units are reserved in the same step | Administrator, or a Procurement Team member when a second approval is required | Reserved | FR-4.5, FR-3.8 |
| **Available** | Listing withdrawn while the asset has no open request | Asset Coordinator | Withdrawn | FR-1.11 |
| **Available, In use or Withdrawn** | Assigned inspection or repair started on a maintenance request | Assigned Maintenance Staff member | Under maintenance | FR-5.4, FR-5.9 |
| **Available, In use or Withdrawn** | End-of-life action proposed; no open reservation, transfer or maintenance request | Asset Coordinator or Sustainability Officer | Pending Safety Review | FR-5.20, FR-5.21 |
| **Reserved** | Approved request cancelled, or its transfer cancelled before handover | Requesting Department Representative or Asset Coordinator | Available | FR-3.11, FR-4.9, FR-3.8 |
| **Reserved** | Approved request not handed over within 10 working days (Desirable rule) | System (automatic) | Available | FR-3.9 |
| **Reserved** | Asset marked as handed over; no safety or maintenance restriction | Asset Coordinator of the owning department | In transfer | FR-4.9, FR-4.16 |
| **In transfer** | Receipt confirmed with the condition on arrival; owner, Asset Coordinator and location change | Department Representative or Asset Coordinator of the receiving department | In use | FR-4.10, FR-4.11 |

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **In transfer** | Dispute decided by accepting the asset | Administrator | In use | FR-4.12 |
| **In transfer** | Dispute decided by returning the asset to the original department | Administrator | Available | FR-4.12 |
| **In use or Withdrawn** | Asset published for reuse; no safety restriction | Asset Coordinator | Available | FR-2.1, SAF-5 |
| **Withdrawn** | Asset taken back into use by its department | Asset Coordinator | In use | FR-5.16 |
| **Under maintenance** | Maintenance completed with the updated condition and the asset safe to use | Assigned Maintenance Staff member (or an Administrator, BR-7) | State before maintenance (Available or In use) | FR-5.8, BR-7 |
| **Under maintenance** | Maintenance request cancelled with a reason | Reporting user, Asset Coordinator or Administrator | State before maintenance | FR-5.13 |
| **Under maintenance** | Asset recorded as not safe to use, or request closed as not repairable | Assigned Maintenance Staff member | Withdrawn | SAF-5, FR-5.12 |
| **Pending Safety Review** | Safety review recorded (handling party assigned for hazardous items) | Sustainability Officer other than the proposer | Pending End-of-Life Approval | FR-5.23, FR-5.24, SAF-3 |
| **Pending Safety Review** | Proposal rejected at safety review, with a reason | Sustainability Officer other than the proposer | State before the proposal | FR-5.28 |
| **Pending End-of-Life Approval** | Proposal rejected, with a reason | Administrator who is not the proposer | State before the proposal | FR-5.28, BR-8 |
| **Pending End-of-Life Approval** | Completion record saved after approval (with the certificate for hazardous items) | Proposer (Asset Coordinator or Sustainability Officer), after the Administrator’s approval | Donated, Recycled, Disassembled, Retired or Disposed | FR-5.26, FR-5.29 to FR-5.31 |

# B.2 Request States

**Table B-3: Request states**

| State | Meaning |
|---|---|
| **Open** | Need request with no asset chosen; waiting for a match (UC-9) |

| State | Meaning |
|---|---|
| **Submitted** | Asset chosen; waiting for the owning Asset Coordinator to release or decline it. No units are reserved |
| **Released** | Released by the Asset Coordinator; waiting for the required approvals |
| **Approved** | All required approvals recorded; the requested quantity is reserved |
| **In transfer** | Asset handed over; waiting for receipt |
| **Disputed** | The receiver reported a problem; the Administrator decides |
| **Completed** | Receipt confirmed (final) |
| **Rejected** | Declined by the Asset Coordinator or rejected by an approver, with a reason (final) |
| **Cancelled** | Cancelled before handover by the requesting Department Representative, or with its transfer by the Asset Coordinator, or ended after a dispute (final) |
| **Expired** | Not decided within 10 working days of submission, or not handed over within 10 working days of approval (final) |

**Table B-4: Request state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Need submitted without choosing an asset | Department Representative (own department only) | Open | FR-3.2, FR-3.5 |
| **New record** | Request for a published asset submitted; available quantity checked again | Department Representative (own department only) | Submitted | FR-3.1, FR-3.6, FR-3.7 |
| **Open** | Matched asset chosen and confirmed (UC-9) | Procurement Team member | Submitted | FR-8.7, FR-8.8 |
| **Open, Submitted, Released or Approved** | Request cancelled before handover | Requesting Department Representative | Cancelled | FR-3.11 |
| **Submitted** | Request released for approval | Asset Coordinator of the owning department | Released | FR-4.1 |
| **Submitted** | Request declined, with a reason | Asset Coordinator of the owning department | Rejected | FR-4.1, BR-3 |
| **Submitted or Released** | Not decided within 10 working days of submission (Desirable rule) | System (automatic) | Expired | FR-3.9 |

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **Released** | Final required approval saved; availability rechecked and the quantity reserved | Administrator, and a Procurement Team member when the value is above the limit or unknown | Approved | FR-4.2, FR-4.3, FR-4.5 |
| **Released** | Request rejected, with a reason | Administrator or Procurement Team member | Rejected | FR-4.2, FR-4.3 |
| **Approved** | Asset marked as handed over | Asset Coordinator of the owning department | In transfer | FR-4.9 |
| **Approved** | Transfer cancelled before handover | Asset Coordinator of the owning department | Cancelled | FR-4.9 |
| **Approved** | Not handed over within 10 working days of approval; reservation released (Desirable rule) | System (automatic) | Expired | FR-3.9 |
| **In transfer** | Receipt confirmed | Department Representative or Asset Coordinator of the receiving department | Completed | FR-4.10, FR-4.11 |
| **In transfer** | Problem reported on arrival | Department Representative or Asset Coordinator of the receiving department | Disputed | FR-4.12 |
| **Disputed** | Dispute decided by accepting the asset | Administrator | Completed | FR-4.12 |
| **Disputed** | Dispute decided by returning the asset | Administrator | Cancelled | FR-4.12 |

# B.3 Transfer States

**Table B-5: Transfer states**

| State | Meaning |
|---|---|
| **Scheduled** | Method, date and locations agreed |
| **In transfer** | Asset handed over |
| **Disputed** | Problem reported on arrival; the Administrator decides |

| State | Meaning |
|---|---|
| **Received** | Receipt confirmed; custody moved (final) |
| **Returned** | Asset sent back to the original department (final) |
| **Cancelled** | Stopped before handover; reserved units return to Available (final) |

**Table B-6: Transfer state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Transfer created for an approved request with method, date, locations and contact | Asset Coordinator of the owning department | Scheduled | FR-4.7 |
| **Scheduled** | Asset handed over; safety and handling notes shown to the receiver | Asset Coordinator of the owning department | In transfer | FR-4.9, FR-4.16, SAF-7, SAF-9 |
| **Scheduled** | Transfer cancelled before handover, or its request cancelled | Asset Coordinator or Requesting Department Representative | Cancelled | FR-4.9, FR-3.11 |
| **Scheduled** | Request expired before handover (Desirable rule) | System (automatic) | Cancelled | FR-3.9 |
| **In transfer** | Receipt confirmed with the condition on arrival | Department Representative or Asset Coordinator of the receiving department | Received | FR-4.10 |
| **In transfer** | Problem reported on arrival | Department Representative or Asset Coordinator of the receiving department | Disputed | FR-4.12 |
| **Disputed** | Dispute decided by accepting the asset | Administrator | Received | FR-4.12 |
| **Disputed** | Dispute decided by returning the asset | Administrator | Returned | FR-4.12 |

# B.4 Maintenance States

**Table B-7: Maintenance request states**

| State | Meaning |
|---|---|
| **Reported** | Defect logged (FR-5.1) |

| State | Meaning |
|---|---|
| **Inspection Requested** | Inspection requested (FR-5.2) |
| **Under Inspection** | The assigned Maintenance Staff member has started the inspection |
| **Repair in Progress** | Repair or refurbishment under way |
| **Completed** | Work finished; updated condition recorded and asset returned to service (final) |
| **Closed – Not Repairable** | Cannot be repaired; asset flagged for end-of-life review (final) |
| **Cancelled** | Cancelled with a reason before completion (final) |

**Table B-8: Maintenance request state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Defect reported | Asset Coordinator, Department Representative or Maintenance Staff | Reported | FR-5.1 |
| **New record** | Inspection requested with a reason and date | Asset Coordinator or Maintenance Staff | Inspection Requested | FR-5.2 |
| **Reported or Inspection Requested** | Inspection started (after assignment under FR-5.3) | Assigned Maintenance Staff member | Under Inspection | FR-5.4 |
| **Under Inspection** | Repair or refurbishment started | Assigned Maintenance Staff member | Repair in Progress | FR-5.4 |
| **Under Inspection or Repair in Progress** | Inspection or repair result and updated condition recorded | Assigned Maintenance Staff member (or an Administrator, BR-7) | Completed | FR-5.5, FR-5.6, FR-5.8 |
| **Under Inspection or Repair in Progress** | Asset found not repairable | Assigned Maintenance Staff member | Closed – Not Repairable | FR-5.12 |
| **Any state before Completed** | Request cancelled, with a reason | Reporting user, Asset Coordinator or Administrator | Cancelled | FR-5.13 |

# B.5 End-of-Life Proposal States

**Table B-9: End-of-life proposal states**

| State | Meaning |
|---|---|
| **Pending Safety Review** | Proposal submitted; waiting for a Sustainability Officer other than the proposer (FR-5.23) |
| **Pending End-of-Life Approval** | Safety review recorded; waiting for Administrator approval (BR-8) |
| **Approved** | All approvals recorded; waiting for the completion record |
| **Completed** | Completion record saved; asset in its final state (final) |
| **Rejected** | Rejected at safety review or approval, with a reason; asset returns to its previous state (final) |

**Table B-10: End-of-life proposal state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Action proposed with a justification and confirmed hazard flag; no blocking items | Asset Coordinator or Sustainability Officer | Pending Safety Review | FR-5.20, FR-5.21, FR-5.23, SAF-4 |
| **Pending Safety Review** | Safety review recorded (handling party assigned for hazardous items) | Sustainability Officer other than the proposer | Pending End-of-Life Approval | FR-5.23, FR-5.24, SAF-3 |
| **Pending Safety Review** | Rejected at safety review, with a reason | Sustainability Officer other than the proposer | Rejected | FR-5.28 |
| **Pending End-of-Life Approval** | Approved (own reason recorded if a Low-confidence AI recommendation is followed) | Administrator who is not the proposer | Approved | FR-5.26, BR-8, SAF-6 |
| **Pending End-of-Life Approval** | Rejected, with a reason | Administrator who is not the proposer | Rejected | FR-5.28, BR-8 |
| **Approved** | Completion record saved (certificate required for hazardous items) | Proposer (Asset Coordinator or Sustainability Officer) | Completed | FR-5.29 to FR-5.32, FR-5.24 |

## B.6 Account States

**Table B-11: Account states**

| State | Meaning |
|---|---|
| **Pending approval** | Application with recorded consent; waiting for activation (UC-13) |
| **Active** | Account can sign in and use its permitted functions |
| **Suspended** | Access stopped temporarily by a System Administrator |
| **Closed** | Access ended, for example after consent is withdrawn; history kept (final) |

**Table B-12: Account state transitions**

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | Registration with consent submitted, or invitation completed | Applicant | Pending approval | FR-6.1, FR-6.2, FR-6.4 |
| **Pending approval** | Account activated with an approved role and department | System Administrator | Active | FR-6.3 |
| **Pending approval, Active or Suspended** | Account closed with a reason | System Administrator | Closed | FR-6.10 |
| **Pending approval, Active or Suspended** | Consent withdrawn | Applicant or account holder | Closed | FR-6.11 |
| **Active** | Access stopped temporarily with a reason; sessions ended | System Administrator | Suspended | FR-6.10 |
| **Suspended** | Access restored with a reason (UC-8) | System Administrator | Active | FR-6.8, FR-6.10 |

## B.7 AI Report States

**Table B-13: AI report states**

| State | Meaning |
|---|---|
| **Draft** | AI-generated draft waiting for review (FR-12.9) |
| **Published** | Approved and shared with permitted recipients (final) |
| **Discarded** | Rejected by the reviewer and kept for reference (final) |

### Table B-14: AI report state transitions

| From state | Event | Who causes it | To state | Requirement |
|---|---|---|---|---|
| **New record** | AI report generated for a period and scope | Finance Officer, Sustainability Officer, Procurement Team member or Administrator | Draft | FR-12.1 |
| **Draft** | Report reviewed, edited if needed and approved | Finance Officer or Administrator; also a Sustainability Officer if the report has no financial figures | Published | FR-12.9, BR-11 |
| **Draft** | Report rejected by the reviewer | The reviewer (as for Published) | Discarded | FR-12.10 |

# Appendix C: To Be Determined List

Each item has a named team coordinator who will obtain confirmation from the stated stakeholder. Ownership identifies the follow-up responsibility; it does not imply that the stakeholder has been contacted or has approved the proposal. Open items must be resolved or explicitly accepted as prototype assumptions before their stated milestone.

| TBD ID | Item to confirm | Where used | Owner | Status |
|---|---|---|---|---|
| **TBD-1** | Confirm the named business owner, delegated approvers, role assignments and reuse-first policy through real stakeholder review. Appendix E simulations do not establish approval. | 1.3; 3.5; 5.1 | Mohamed Almarzooqi with Facilities Management and IT | Open (before requirements sign-off) |
| **TBD-2** | Confirm the high-value approval limit, treatment of unknown value, distinct approvers, repair-spending authority and end-of-life delegation. No threshold is claimed as university policy. | FR-4.3; BR-3, BR-7, BR-8 | Mohammad Alsuwaidi with Procurement, Finance and Facilities Management | Open (before approval-rule configuration) |
| **TBD-3** | Confirm account identity checks, permitted email domains, recovery method, role-authorizing contacts and stricter IT authentication rules. Prototype local accounts remain the baseline; university single sign-on is future work. | FR-6; SEC-2 to SEC-4; ORG-3; UC-13 | Mohamed Almarzooqi with IT Department | Open (before access testing) |
| **TBD-4** | Approve privacy-notice wording, applicable lawful bases, consent evidence, withdrawal response, retention periods (including audit entries), anonymization and erasure. Reconcile retained audit history with data-removal requests. The 24-hour erasure period in SEC-8 is a proposed prototype default, not a statutory rule. | FR-6.2, FR-6.11; SEC-8; FR-7.A5; 4.1; LEG-1 | Rashid Alzaabi with IT and the university privacy/legal contact | Open (before any real personal data) |
| **TBD-5** | Confirm working days, holidays, reservation expiry and reminder rules. Ten working days for expiry and five for receipt reminders are proposed prototype defaults, not verified campus policy. | FR-3.9; FR-4.13; FR-7.N6 | Mohammad Alsuwaidi with Facilities Management | Open (before deadline tests) |
| **TBD-6** | Confirm email delivery service and addresses, failure escalation and channels for inactive users. In-application notifications are essential; email copies are desirable. | FR-7.N; SEC-3 | Mohamed Almarzooqi with IT Department | Open (before notification testing) |

| TBD ID | Item to confirm | Where used | Owner | Status |
|---|---|---|---|---|
| **TBD-7** | Confirm maintenance and end-of-life terminology, return-to-service authority, hazardous-item reviewer qualifications, approved handlers and required evidence. Confirm the 50% repair-cost review default (FR-5.11) and the 90-day inspection window for risky transfers (SAF-9). | FR-5; SAF-1 to SAF-9; Appendix B | Rashid Alzaabi with Maintenance Staff and Facilities Management | Open (before workflow sign-off) |
| **TBD-8** | Approve financial and environmental formulas, baseline definitions, replacement prices, weights, emission factors and sources. Confirm treatment of repeated reuse, double counting and zero denominators; validate estimates with reviewed examples. Confirm whether the EPA WARM version 16 default emission factors (Table 2.13-2) are acceptable or should be replaced with UAE-specific factors. | FR-13; Appendix A | Rashid Alzaabi with Finance and Sustainability Officers | Open (before indicator acceptance tests) |
| **TBD-9** | Select AI services, model versions, data locations, permitted input fields and fallback settings; obtain IT approval before sending any real university data. | AI-5 to AI-10; SEC-11; LEG-3 | Ahmed AlAbdouli with IT Department | Open (before AI integration) |
| **TBD-10** | Confirm role-level visibility of department, financial and maintenance data, including who may enter repair costs. Default access is limited by SEC-5, and BR-11 sets who may approve reports with financial figures. | FR-2.5; FR-7.R3; FR-12; FR-13.11 | Mohamed Almarzooqi with Finance and IT | Open (before permission-matrix approval) |
| **TBD-11** | Confirm university IT governance, real-pilot approval, need and scope of a privacy impact assessment, code and report ownership, licensing and approved terms of use. | LEG-1 to LEG-5; ORG-1, ORG-2 | Rashid Alzaabi with university IT and legal contacts | Open (before any real pilot) |
| **TBD-12** | Confirm prototype workload, reviewed evaluation datasets, measurable acceptance targets and the AI confidence thresholds in AI-10. The document uses the 60-minute recovery target in QUA-6, and SAF-8 sets the 24-hour backup interval. | PER-1 to PER-6; SAF-8; QUA-1 to QUA-15; AI-10 | Ahmed AlAbdouli with all team members and lab instructors | Open (before test-plan baseline) |

| TBD ID | Item to confirm | Where used | Owner | Status |
|---|---|---|---|---|
| **TBD-13** | Confirm category lists, permitted file formats and limits, AI-report change alert (proposed 25%) and upload restrictions. Image/PDF uploads of at most 10 MB remain the draft baseline. | FR-1.6; FR-12.8; SEC-13 | Mohammad Alsuwaidi with Rashid Alzaabi | Open (before detailed design) |
| **TBD-14** | Confirm the sections required for the Phase 3 deliverable. Resolved: the Phase 3 lab slides [3] require the introduction, the template's system features and nonfunctional requirements (template Sections 4 and 5.1 to 5.4), one use case diagram and the use case descriptions. The template's Sections 2 and 3 are omitted and the remaining sections are numbered in order (1.2). | 1.2; 1.5 | Mohamed Almarzooqi with lab instructors | Closed |
| **TBD-15** | Confirm with real stakeholders the role names, state lists, use cases and requirement priorities agreed by the team. The Appendix E interviews were AI simulations, and real stakeholder interviews remain outstanding. | Sections 2 to 5; Appendices B, D and E | Mohamed Almarzooqi coordinates all four members | Open (before the design baseline) |
| **TBD-16** | Confirm with the lab instructors that sustainable-action recommendations (FR-10) and the AI assistant (FR-11) may be Desirable rather than part of the first prototype, as the Phase 2 scope note requires for any reduction in AI scope. | 2.10; 2.11; Appendix F | Ahmed AlAbdouli with lab instructors | Open (before the design baseline) |

# Appendix D: Requirements Traceability

Table D-1 traces every requirement ID in this document to its Phase 2 source [5], the main stakeholder role, the linked use cases and its priority. Consecutive requirements that share all four values are grouped on one row. Table D-2 shows how the requirements respond to each risk in the Phase 2 risk register.

**Table D-1: Requirements traceability matrix**

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **FR-1.1 to FR-1.8** | FR-1 | Asset Coordinator | UC-1 | Essential |
| **FR-1.9** | FR-1 | Asset Coordinator | UC-1, UC-2 | Essential |
| **FR-1.10** | FR-1 | Asset Coordinator | UC-1, UC-5 | Essential |
| **FR-1.11** | FR-1 | Asset Coordinator | UC-1 | Desirable |
| **FR-1.12** | FR-1 | Asset Coordinator | UC-1, UC-10 | Essential |
| **FR-1.13** | FR-1 | Asset Coordinator | UC-1 | Essential |
| **FR-1.14** | FR-1 | Asset Coordinator | UC-1 | Future |
| **FR-1.15** | FR-1 | Asset Coordinator | UC-1 | Essential |
| **FR-2.1** | FR-2 | Department Representative, Asset Coordinator | UC-1, UC-2 | Essential |
| **FR-2.2 to FR-2.6** | FR-2 | Department Representative, Asset Coordinator | UC-2 | Essential |
| **FR-2.7** | FR-2 | Department Representative, Asset Coordinator | UC-2, UC-3 | Essential |
| **FR-2.8** | FR-2 | Department Representative, Asset Coordinator | UC-2 | Desirable |
| **FR-3.1** | FR-3 | Department Representative | UC-3 | Essential |
| **FR-3.2** | FR-3 | Department Representative | UC-3, UC-9 | Essential |
| **FR-3.3 to FR-3.7** | FR-3 | Department Representative | UC-3 | Essential |
| **FR-3.8** | FR-3 | Department Representative | UC-3, UC-4 | Essential |
| **FR-3.9** | FR-3 | Department Representative | UC-3, UC-4, UC-5 | Desirable |
| **FR-3.10 to FR-3.11** | FR-3 | Department Representative | UC-3 | Essential |
| **FR-3.12** | FR-3 | Department Representative | UC-3, UC-4 | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **FR-3.13** | FR-3 | Department Representative | UC-3 | Desirable |
| **FR-3.14** | FR-3 | Department Representative | UC-3, UC-11 | Desirable |
| **FR-3.15** | FR-3 | Department Representative | UC-3 | Essential |
| **FR-4.1 to FR-4.2** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-4 | Essential |
| **FR-4.3** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-4 | Desirable |
| **FR-4.4 to FR-4.5** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-4 | Essential |
| **FR-4.6** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-4, UC-5 | Essential |
| **FR-4.7** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-5 | Essential |
| **FR-4.8** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-5 | Desirable |
| **FR-4.9 to FR-4.12** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-5 | Essential |
| **FR-4.13** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-5 | Desirable |
| **FR-4.14** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-4, UC-5 | Essential |
| **FR-4.15 to FR-4.16** | FR-4 | Asset Coordinator, Administrator, Procurement Team | UC-5 | Essential |
| **FR-5.1 to FR-5.8** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6 | Essential |
| **FR-5.9** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-2, UC-3, UC-5, UC-6, UC-9 | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **FR-5.10** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6 | Essential |
| **FR-5.11** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6, UC-7 | Desirable |
| **FR-5.12** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6, UC-7 | Essential |
| **FR-5.13 to FR-5.15** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6 | Essential |
| **FR-5.16** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-1, UC-4, UC-5, UC-6, UC-7 | Essential |
| **FR-5.17 to FR-5.18** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-6, UC-7 | Essential |
| **FR-5.19** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-2, UC-7 | Desirable |
| **FR-5.20 to FR-5.21** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-7 | Essential |
| **FR-5.22** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-7 | Desirable |
| **FR-5.23 to FR-5.26** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-7 | Essential |
| **FR-5.27** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-2, UC-3, UC-5, UC-7, UC-9 | Essential |
| **FR-5.28 to FR-5.32** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-7 | Essential |
| **FR-5.33** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-1, UC-7 | Future |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **FR-5.34** | FR-5 | Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator | UC-7 | Essential |
| **FR-6.1 to FR-6.2** | FR-6 | System Administrator, Applicant | UC-13 | Essential |
| **FR-6.3 to FR-6.4** | FR-6 | System Administrator, Applicant | UC-8, UC-13 | Essential |
| **FR-6.5** | FR-6 | System Administrator, Applicant | UC-1 to UC-13 | Essential |
| **FR-6.6** | FR-6 | System Administrator, Applicant | UC-13 | Essential |
| **FR-6.7** | FR-6 | System Administrator, Applicant | UC-1 to UC-13 | Essential |
| **FR-6.8 to FR-6.12** | FR-6 | System Administrator, Applicant | UC-8 | Essential |
| **FR-7.N1 to FR-7.N5** | FR-7 | All roles | UC-3 to UC-9, UC-12, UC-13 | Essential |
| **FR-7.N6** | FR-7 | All roles | UC-3, UC-4, UC-5 | Desirable |
| **FR-7.R1 to FR-7.R6** | FR-7 | Report users (Section 5.1) | UC-12 | Essential |
| **FR-7.R7** | FR-7 | Report users (Section 5.1) | UC-12 | Desirable |
| **FR-7.R8** | FR-7 | Report users (Section 5.1) | UC-12 | Future |
| **FR-7.A1 to FR-7.A2** | FR-7 | System Administrator, Administrator | UC-1 to UC-13 | Essential |
| **FR-7.A3** | FR-7 | System Administrator, Administrator | UC-8 | Essential |
| **FR-7.A4** | FR-7 | System Administrator, Administrator | UC-7, UC-9, UC-10, UC-11, UC-12 | Essential |
| **FR-7.A5 to FR-7.A6** | FR-7 | System Administrator, Administrator | UC-1 to UC-13 | Essential |
| **FR-8.1 to FR-8.9** | FR-8 | Procurement Team | UC-9 | Essential |
| **FR-9.1 to FR-9.9** | FR-9 | Asset Coordinator | UC-10 | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **FR-10.1 to FR-10.8** | FR-10 | Sustainability Officer, Maintenance Staff | UC-7 | Desirable |
| **FR-11.1** | FR-11 | Department Representative | UC-11 | Desirable |
| **FR-11.2 to FR-11.3** | FR-11 | Department Representative | UC-3, UC-11 | Desirable |
| **FR-11.4 to FR-11.10** | FR-11 | Department Representative | UC-11 | Desirable |
| **FR-12.1** | FR-12 | Finance Officer, Sustainability Officer | UC-12 | Essential |
| **FR-12.2** | FR-12 | Finance Officer, Sustainability Officer | UC-12 | Essential (a, b); Desirable (c–g) |
| **FR-12.3 to FR-12.7** | FR-12 | Finance Officer, Sustainability Officer | UC-12 | Essential |
| **FR-12.8** | FR-12 | Finance Officer, Sustainability Officer | UC-12 | Desirable |
| **FR-12.9 to FR-12.12** | FR-12 | Finance Officer, Sustainability Officer | UC-12 | Essential |
| **FR-13.1 to FR-13.2** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-12 | Essential |
| **FR-13.3** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-5, UC-6, UC-7, UC-12 | Essential |
| **FR-13.4 to FR-13.5** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-12 | Essential |
| **FR-13.6** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-1, UC-7, UC-12 | Essential |
| **FR-13.7 to FR-13.12** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-12 | Essential |
| **FR-13.13** | New in Phase 3 | Finance Officer, Sustainability Officer, University Management | UC-12 | Desirable |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **AI-1 to AI-7** | FR-8 to FR-12 | All users of AI features | UC-7, UC-9 to UC-12 | Essential |
| **AI-8** | FR-8 to FR-12 | All users of AI features | UC-7, UC-9, UC-11, UC-12 | Essential |
| **AI-9 to AI-10** | FR-8 to FR-12 | All users of AI features | UC-7, UC-9 to UC-12 | Essential |
| **PER-1** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-13 | Essential |
| **PER-2** | Phase 2 technical feasibility | All users; IT Department | UC-2, UC-3, UC-11 | Essential |
| **PER-3** | Phase 2 technical feasibility | All users; IT Department | UC-1, UC-3 to UC-8, UC-13 | Essential |
| **PER-4 to PER-5** | Phase 2 technical feasibility | All users; IT Department | UC-7, UC-9 to UC-12 | Essential |
| **PER-6** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-7, UC-9 to UC-11 | Essential |
| **SAF-1** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-1, UC-2, UC-3, UC-5, UC-7, UC-9 | Essential |
| **SAF-2** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-4 to UC-7, UC-9 to UC-12 | Essential |
| **SAF-3 to SAF-4** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-7 | Essential |
| **SAF-5** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-1, UC-2, UC-3, UC-5, UC-6 | Essential |
| **SAF-6** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-7 | Essential |
| **SAF-7** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-5, UC-7 | Desirable |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **SAF-8** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-1 to UC-13 | Essential |
| **SAF-9** | Phase 2 risks and ethics | Sustainability Officer, Maintenance Staff, Facilities Management | UC-5, UC-6 | Essential |
| **SEC-1** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-1 to UC-13 | Essential |
| **SEC-2** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-13 | Essential |
| **SEC-3 to SEC-4** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-1 to UC-13 | Essential |
| **SEC-5** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-2, UC-6, UC-7, UC-8, UC-9, UC-12 | Essential |
| **SEC-6 to SEC-8** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-8, UC-13 | Essential |
| **SEC-9** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-8 | Essential |
| **SEC-10** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-7, UC-9 | Essential |
| **SEC-11** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-9, UC-10, UC-11, UC-12 | Essential |
| **SEC-12** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-4, UC-5, UC-6, UC-7 | Essential |
| **SEC-13** | FR-6; Phase 2 legal analysis | System Administrator; IT Department | UC-1, UC-5, UC-6, UC-10 | Essential |
| **QUA-1** | Phase 2 technical feasibility | All users; IT Department | UC-1, UC-3, UC-9, UC-10, UC-11 | Essential |
| **QUA-2 to QUA-3** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-13 | Essential |
| **QUA-4** | Phase 2 technical feasibility | All users; IT Department | UC-1, UC-3 to UC-7, UC-9 to UC-11 | Essential |
| **QUA-5 to QUA-6** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-13 | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **QUA-7** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-7, UC-9, UC-10 | Essential |
| **QUA-8** | Phase 2 technical feasibility | All users; IT Department | UC-1 to UC-13 | Essential |
| **QUA-9** | Phase 2 technical feasibility | All users; IT Department | UC-10 | Essential |
| **QUA-10** | Phase 2 technical feasibility | All users; IT Department | UC-9 | Essential |
| **QUA-11** | Phase 2 technical feasibility | All users; IT Department | UC-7 | Desirable |
| **QUA-12 to QUA-13** | Phase 2 technical feasibility | All users; IT Department | UC-7, UC-9 | Essential |
| **QUA-14** | Phase 2 technical feasibility | All users; IT Department | UC-7, UC-9 to UC-12 | Essential |
| **QUA-15** | Phase 2 technical feasibility | All users; IT Department | UC-11 | Desirable |
| **BR-1** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | None (policy rule) | Essential |
| **BR-2** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-1 to UC-4 | Essential |
| **BR-3** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-4 | Essential |
| **BR-4** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-4, UC-7, UC-8 | Essential |
| **BR-5** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-3 to UC-5 | Essential |
| **BR-6** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-5 | Essential |
| **BR-7** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-6 | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **BR-8** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-7 | Essential |
| **BR-9** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-7, UC-9 to UC-12 | Essential |
| **BR-10** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-9, UC-12 | Essential |
| **BR-11** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-12 | Essential |
| **BR-12** | Phase 2 stakeholder roles | Head of Facilities Management; approving roles | UC-1 to UC-13 | Essential |
| **LEG-1** | Phase 2 legal analysis | University legal contact; IT Department | UC-8, UC-9, UC-11, UC-13 | Essential |
| **LEG-2** | Phase 2 legal analysis | University legal contact; IT Department | None (process requirement) | Essential |
| **LEG-3** | Phase 2 legal analysis | University legal contact; IT Department | UC-9, UC-10, UC-11, UC-12 | Essential |
| **LEG-4** | Phase 2 legal analysis | University legal contact; IT Department | UC-1 to UC-13 | Essential |
| **LEG-5** | Phase 2 legal analysis | University legal contact; IT Department | UC-12 | Essential |
| **LEG-6** | Phase 2 legal analysis | University legal contact; IT Department | UC-9, UC-12 | Essential |
| **LEG-7** | Phase 2 legal analysis | University legal contact; IT Department | UC-7 | Essential |
| **LEG-8** | Phase 2 legal analysis | University legal contact; IT Department | UC-12 | Desirable |
| **LEG-9** | Phase 2 legal analysis | University legal contact; IT Department | UC-9 | Essential |
| **ORG-1** | Phase 2 operational feasibility | IT Department; development team | None (process requirement) | Future (required before a real pilot) |
| **ORG-2** | Phase 2 operational feasibility | IT Department; development team | None (process requirement) | Essential |

| Requirement ID | Phase 2 source | Main stakeholder | Use case | Priority |
|---|---|---|---|---|
| **ORG-3** | Phase 2 operational feasibility | IT Department; development team | UC-8, UC-13 | Essential |
| **ORG-4** | Phase 2 operational feasibility | IT Department; development team | None (process requirement) | Essential |

**Table D-2: Phase 2 risks and the requirements that address them**

| Phase 2 risk [5] | Requirements that address it |
|---|---|
| R1: Insufficient or poor-quality data for AI functions | FR-1.2, FR-1.5 (complete, validated records); FR-9.4 (missing details flagged); FR-13.8 (missing data excluded and listed); ORG-2 (synthetic or reviewed test data); QUA-9, QUA-10 (separate labeled test sets). |
| R2: AI classification or matching gives inaccurate results | AI-1, AI-10 (confidence shown); AI-3, FR-8.7, FR-9.6 (human review and override); AI-6, FR-8.9 (standard search fallback); QUA-9, QUA-10 (accuracy targets). |
| R3: AI or LLM service fails or is unavailable | AI-6; PER-5, PER-6; FR-9.8, FR-11.9, FR-12.11 (a manual or standard option keeps the core prototype usable). |
| R4: Project scope is too large for the semester | Priorities in every requirement table (1.2): Essential requirements form the first prototype; FR-10, FR-11 and the Desirable requirements of other features can be deferred; FR-1.14, FR-5.33, FR-7.R8 and ORG-1 are Future; Appendix F.2 checks the Essential set against the Phase 2 schedule. |
| R5: Integration problems between modules | ORG-4 (shared repository and regular commits); QUA-4 (records stay correct under retries and simultaneous updates); QUA-7 (category and rule changes without workflow changes). |
| R6: Unauthorized access or exposure of information | FR-6.5, FR-6.7; SEC-1 to SEC-9, SEC-11; FR-7.A1 to FR-7.A6; LEG-1, LEG-4. |
| R7: Bias or unfair prioritization | FR-8.5, LEG-9, QUA-13 (department name and size are not ranking factors); AI-3, FR-8.7 (authorized override); BR-9. |
| R8: Recommendations cannot be explained | AI-2, FR-8.4 (reasons for every match); AI-1 (confidence); QUA-14 (usefulness of explanations); SEC-10 (human review on request). |
| R9: Web or application security vulnerabilities | SEC-2 to SEC-4, SEC-6; SEC-13 (upload limits, input handling and OWASP test); FR-7.A6. |
| R10: Prompt injection or disclosure through the assistant | AI-7, AI-9; FR-11.3, FR-11.8 (confirmation and permission checks before any action); SEC-11. |
| R11: Team availability or uneven workload | A project-management risk handled in the Phase 1 plan [4]; ORG-4 makes each member’s contribution visible. |

| Phase 2 risk [5] | Requirements that address it |
|---|---|
| R12: Unexpected infrastructure or AI-service cost | A financial risk handled in the Phase 2 feasibility study [5]; TBD-9 keeps the choice of AI services under IT approval. |
| R13: Misleading sustainability or carbon estimates | FR-13.2, FR-13.7, FR-13.8, FR-13.10, FR-13.12; FR-12.4, FR-12.5; AI-8; TBD-8. |
| R14: Users find the workflows difficult to adopt | QUA-1 to QUA-3 (usability and accessibility targets); FR-2.6, FR-3.4 (clear validation messages); TBD-15 (confirmation with real stakeholders). |

# Appendix E: Stakeholder Interviews

As planned in Phase 2, stakeholder interviews are needed to understand the requirements. Real interviews were not yet possible, so an AI tool (ChatGPT-6 Astra Extra-High) was used on 9 October 2026 to simulate the answers of six stakeholder groups. The questions and the AI-simulated answers are shown below, each followed by a short summary and the requirements it informed. No real university stakeholders participated, and the answers are not evidence of university practice (TBD-15). The prompts used earlier role names (Asset Custodian, Procurement Officer and Requester), which correspond to Asset Coordinator, Procurement Team and Department Representative in this document.

## E.1 Facilities Management and Asset Coordinator

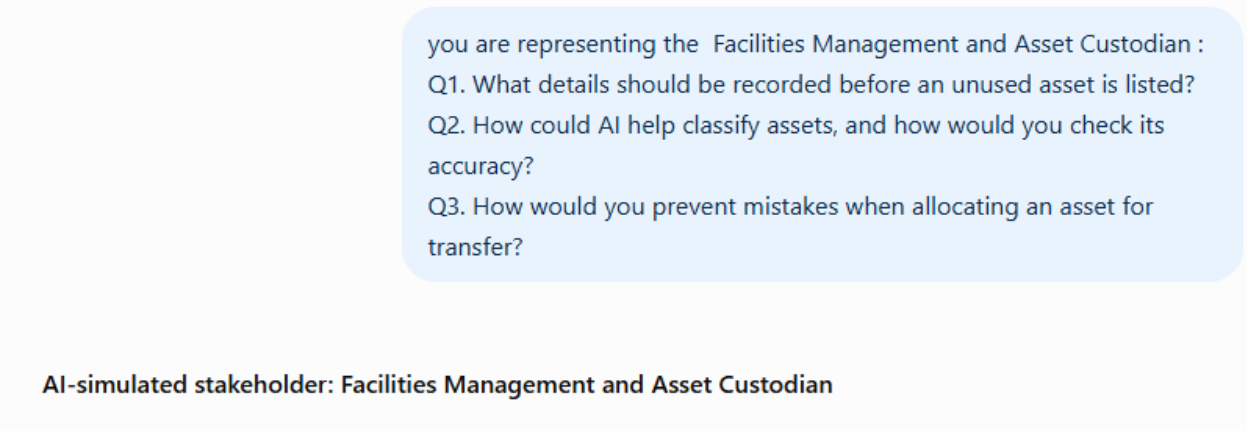
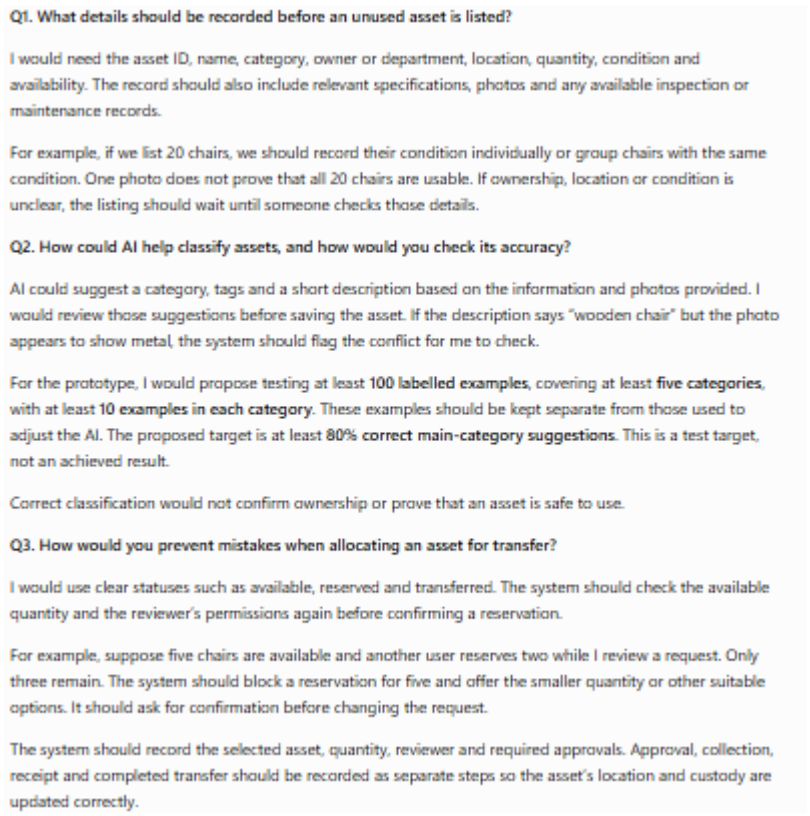



Figure E.1: AI-simulated interview, Facilities Management and Asset Coordinator questions



**Q1. What details should be recorded before an unused asset is listed?**

I would need the asset ID, name, category, owner or department, location, quantity, condition and availability. The record should also include relevant specifications, photos and any available inspection or maintenance records.

For example, if we list 20 chairs, we should record their condition individually or group chairs with the same condition. One photo does not prove that all 20 chairs are usable. If ownership, location or condition is unclear, the listing should wait until someone checks those details.

**Q2. How could AI help classify assets, and how would you check its accuracy?**

AI could suggest a category, tags and a short description based on the information and photos provided. I would review those suggestions before saving the asset. If the description says "wooden chair" but the photo appears to show metal, the system should flag the conflict for me to check.

For the prototype, I would propose testing at least **100 labelled examples**, covering at least **five categories**, with at least **10 examples in each category**. These examples should be kept separate from those used to adjust the AI. The proposed target is at least **80% correct main-category suggestions**. This is a test target, not an achieved result.

Correct classification would not confirm ownership or prove that an asset is safe to use.

**Q3. How would you prevent mistakes when allocating an asset for transfer?**

I would use clear statuses such as available, reserved and transferred. The system should check the available quantity and the reviewer's permissions again before confirming a reservation.

For example, suppose five chairs are available and another user reserves two while I review a request. Only three remain. The system should block a reservation for five and offer the smaller quantity or other suitable options. It should ask for confirmation before changing the request.

The system should record the selected asset, quantity, reviewer and required approvals. Approval, collection, receipt and completed transfer should be recorded as separate steps so the asset's location and custody are updated correctly.

Figure E.2: AI-simulated interview, Facilities Management and Asset Coordinator answers

Summary: listings need the asset ID, name, category, owning department, location, quantity, condition and availability, with specifications, photos and inspection records; uncertain items wait for checks. AI category and tag suggestions are reviewed before saving and conflicts are flagged, with a proposed test of at least 100 labeled examples and an 80% target. Availability and permissions are rechecked before a reservation, and approval, collection and receipt are recorded as separate steps.

Linked requirements: FR-1.2, FR-1.3; FR-9.1 to FR-9.9; QUA-9; FR-4.5; FR-8.6, FR-8.8; AI-3 to AI-5. Related features: FR-1, FR-4, FR-5.

# E.2 Procurement and Finance

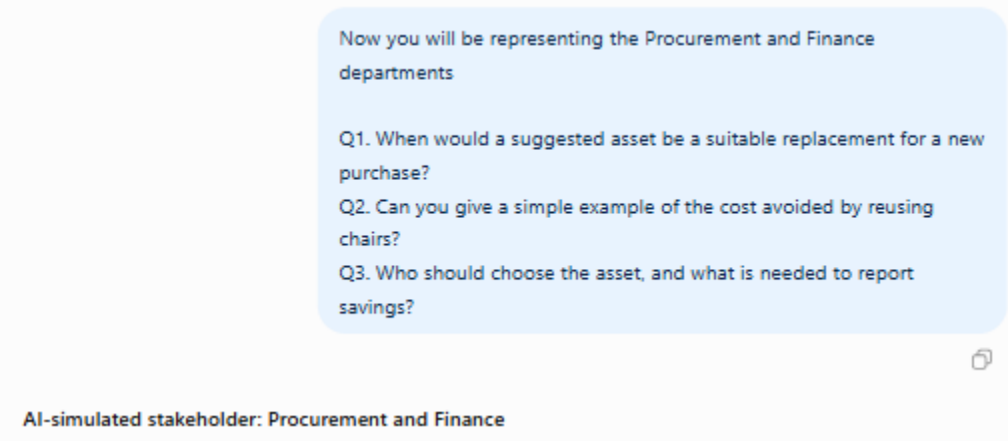
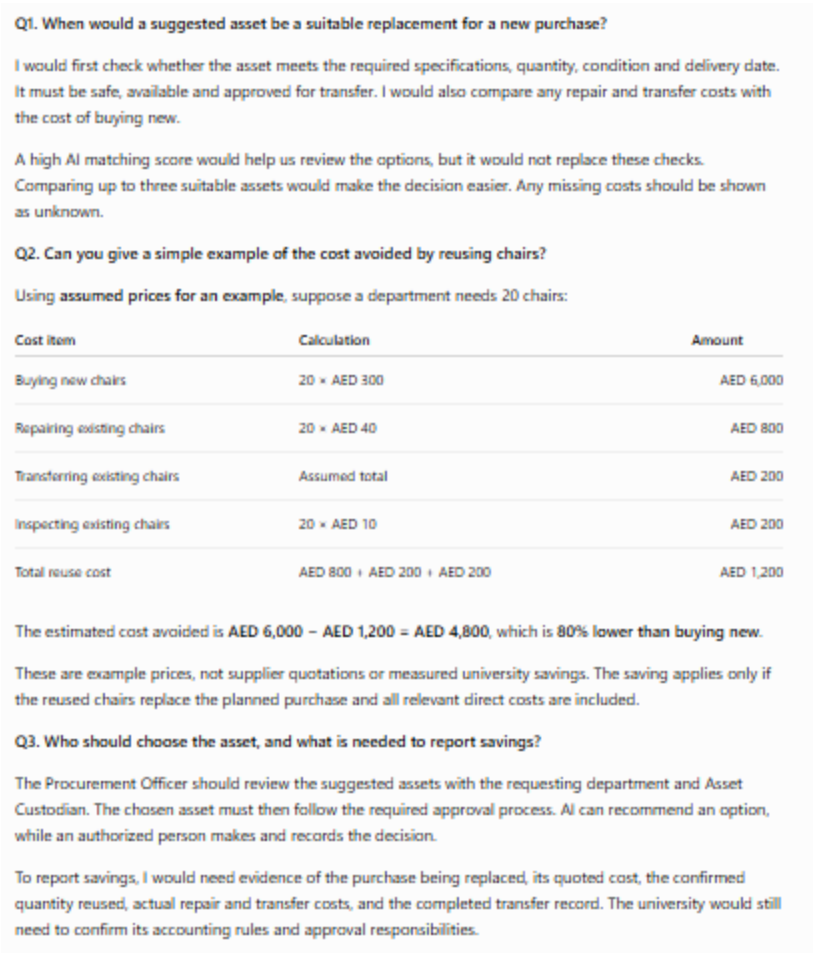



Figure E.3: AI-simulated interview, Procurement and Finance questions



**Q1. When would a suggested asset be a suitable replacement for a new purchase?**

I would first check whether the asset meets the required specifications, quantity, condition and delivery date. It must be safe, available and approved for transfer. I would also compare any repair and transfer costs with the cost of buying new.

A high AI matching score would help us review the options, but it would not replace these checks. Comparing up to three suitable assets would make the decision easier. Any missing costs should be shown as unknown.

**Q2. Can you give a simple example of the cost avoided by reusing chairs?**

Using **assumed prices for an example**, suppose a department needs 20 chairs:

| Cost item | Calculation | Amount |
|---|---|---|
| Buying new chairs | 20 × AED 300 | AED 6,000 |
| Repairing existing chairs | 20 × AED 40 | AED 800 |
| Transferring existing chairs | Assumed total | AED 200 |
| Inspecting existing chairs | 20 × AED 10 | AED 200 |
| Total reuse cost | AED 800 + AED 200 + AED 200 | AED 1,200 |

The estimated cost avoided is **AED 6,000 − AED 1,200 = AED 4,800**, which is **80% lower than buying new**.

These are example prices, not supplier quotations or measured university savings. The saving applies only if the reused chairs replace the planned purchase and all relevant direct costs are included.

**Q3. Who should choose the asset, and what is needed to report savings?**

The Procurement Officer should review the suggested assets with the requesting department and Asset Custodian. The chosen asset must then follow the required approval process. AI can recommend an option, while an authorized person makes and records the decision.

To report savings, I would need evidence of the purchase being replaced, its quoted cost, the confirmed quantity reused, actual repair and transfer costs, and the completed transfer record. The university would still need to confirm its accounting rules and approval responsibilities.

Figure E.4: AI-simulated interview, Procurement and Finance answers

Summary: a reused asset is suitable only if it meets the specifications, quantity, condition and date and is safe, available and approved; repair and transfer costs are compared with buying new, and missing costs are shown as unknown. An example with assumed prices compared 20

new chairs (AED 6,000) with reusing existing chairs (AED 1,200), an estimated AED 4,800 avoided; these are example figures, not measured savings. The Procurement Team reviews suggestions with the requesting department and an authorized person decides.

Linked requirements: FR-8.2 to FR-8.9; FR-4.7; FR-13.1 (indicators 3 and 4); AI-3 to AI-5, AI-8; LEG-6; BR-10. Related features: FR-4, FR-7, FR-12.

# E.3 Maintenance and Sustainability

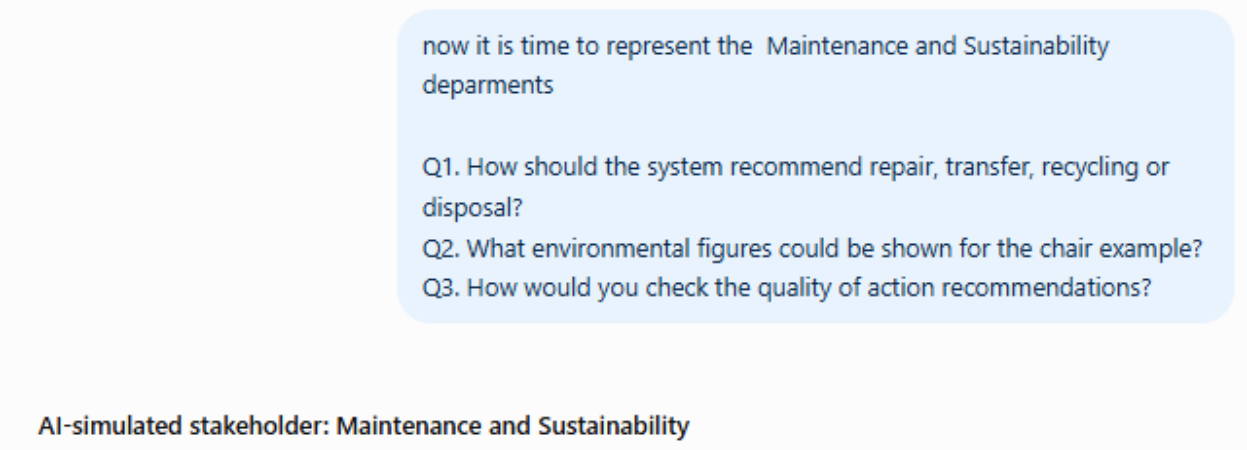



Figure E.5: AI-simulated interview, Maintenance and Sustainability questions

**Q1. How should the system recommend repair, transfer, recycling or disposal?**

I would expect the system to review the asset's condition, inspection records, maintenance history, remaining useful life, demand and available costs. It should compare suitable options and explain why it recommends one action over another.

For example, a chair with a damaged joint should wait for inspection before being offered for reuse. The system should consider keeping, repairing, refurbishing, transferring, donating, recovering usable parts or recycling before recommending disposal. Missing information should lead to manual review.

AI can suggest an action, but authorized staff must check safety and approve the next step.

**Q2. What environmental figures could be shown for the chair example?**

Using an **assumed weight of 8 kilograms per chair**, the 20 chairs would weigh:

**20 × 8 kg = 160 kg**

If the transfer is completed, we could describe this as **an estimated 160 kilograms of material kept in use**.

We could not automatically call it 160 kilograms of waste diverted from landfill, because we do not know what would otherwise have happened to the chairs. A carbon-reduction figure should remain unknown until we have a suitable calculation method and supporting emissions data.

**Q3. How would you check the quality of action recommendations?**

I would prepare at least **30 test scenarios** with the necessary asset information. Two reviewers should agree on the acceptable actions before testing, since more than one action may be reasonable.

The proposed target is for at least **24 out of 30 first recommendations—80%—to match an acceptable action**. This is a test target, not a completed result.

I would also test unsafe assets and missing information. The system must not recommend immediate use when an unresolved safety issue requires inspection. We should check that explanations match the records and track how often reviewers accept, reject or change suggestions. Acceptance alone does not prove that a recommendation is correct.

Figure E.6: AI-simulated interview, Maintenance and Sustainability answers

Summary: action advice should consider condition, inspections, maintenance history, remaining life, demand and costs; damaged items wait for inspection and disposal is the last option. An example of 20 chairs at an assumed 8 kg each gives an estimated 160 kg kept in use, which should not be reported as measured waste diversion; this informed the estimate labels in FR-13.7 and FR-13.12. Recommendation quality is tested on 30 scenarios agreed by two reviewers, with a target of 24 of 30.

Linked requirements: FR-10.1 to FR-10.8; FR-13.7, FR-13.12; QUA-11, QUA-12; AI-1 to AI-4, AI-8; SAF-4, SAF-5. Related features: FR-5, FR-13.

# E.4 IT and System Administrator

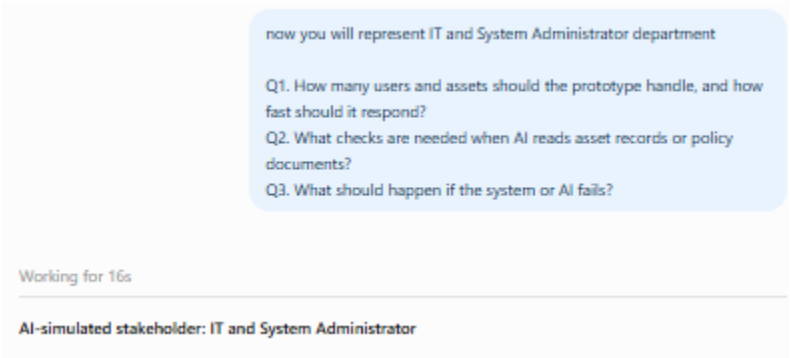
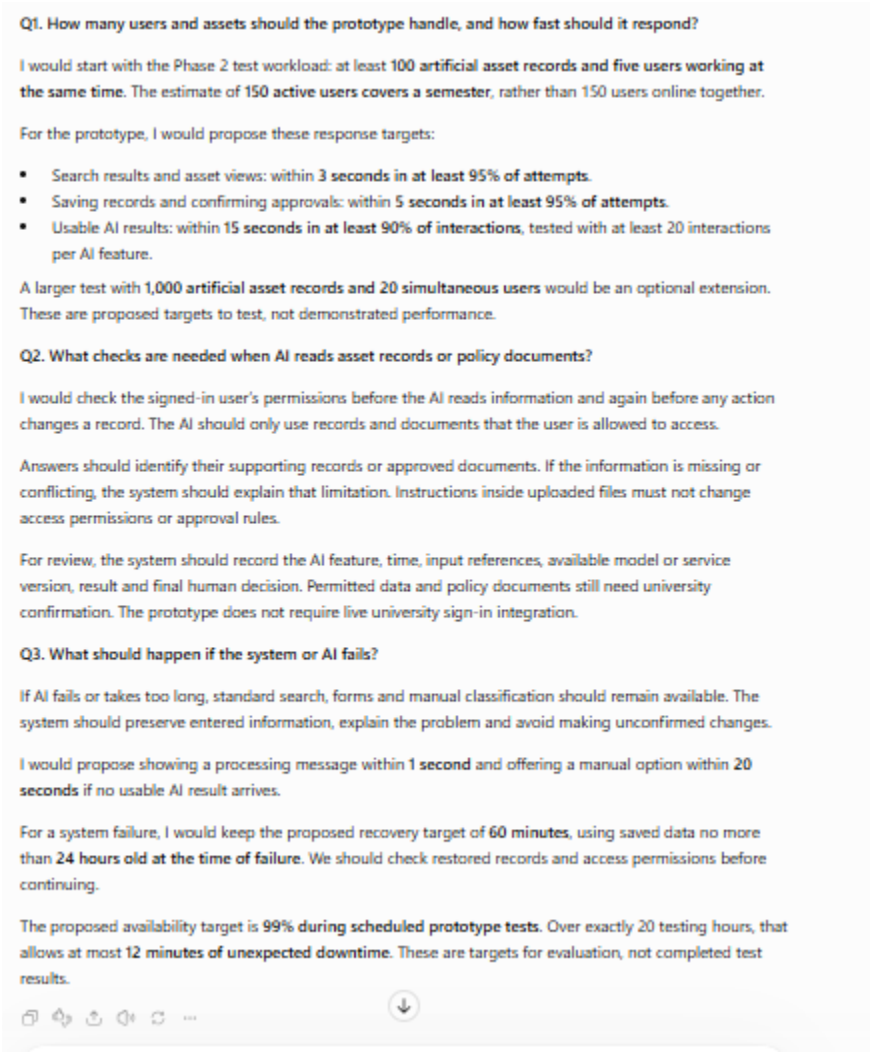



Figure E.7: AI-simulated interview, IT and System Administrator questions



**Q1. How many users and assets should the prototype handle, and how fast should it respond?**

I would start with the Phase 2 test workload: at least **100 artificial asset records and five users working at the same time**. The estimate of **150 active users covers a semester**, rather than 150 users online together.

For the prototype, I would propose these response targets:

- Search results and asset views: within **3 seconds in at least 95% of attempts**.
- Saving records and confirming approvals: within **5 seconds in at least 95% of attempts**.
- Usable AI results: within **15 seconds in at least 90% of interactions**, tested with at least 20 interactions per AI feature.

A larger test with **1,000 artificial asset records and 20 simultaneous users** would be an optional extension. These are proposed targets to test, not demonstrated performance.

**Q2. What checks are needed when AI reads asset records or policy documents?**

I would check the signed-in user's permissions before the AI reads information and again before any action changes a record. The AI should only use records and documents that the user is allowed to access.

Answers should identify their supporting records or approved documents. If the information is missing or conflicting, the system should explain that limitation. Instructions inside uploaded files must not change access permissions or approval rules.

For review, the system should record the AI feature, time, input references, available model or service version, result and final human decision. Permitted data and policy documents still need university confirmation. The prototype does not require live university sign-in integration.

**Q3. What should happen if the system or AI fails?**

If AI fails or takes too long, standard search, forms and manual classification should remain available. The system should preserve entered information, explain the problem and avoid making unconfirmed changes.

I would propose showing a processing message within **1 second** and offering a manual option within **20 seconds** if no usable AI result arrives.

For a system failure, I would keep the proposed recovery target of **60 minutes**, using saved data no more than **24 hours old at the time of failure**. We should check restored records and access permissions before continuing.

The proposed availability target is **99% during scheduled prototype tests**. Over exactly 20 testing hours, that allows at most **12 minutes of unexpected downtime**. These are targets for evaluation, not completed test results.

Figure E.8: AI-simulated interview, IT and System Administrator answers

Summary: the prototype should handle the Phase 2 workload of 100 artificial asset records and five simultaneous users, with search within 3 seconds and saves within 5 seconds in 95% of attempts and AI results within 15 seconds in 90% of interactions. Permissions are checked before AI reads data and before any action, and instructions inside files are ignored. A processing message appears within 1 second and a manual option within 20 seconds; recovery within 60 minutes with at most 24 hours of data loss, and 99% availability during scheduled tests.

Linked requirements: PER-1 to PER-6; QUA-4 to QUA-6; AI-5 to AI-7, AI-9; SAF-8; FR-11.5.

# E.5 Department Representatives

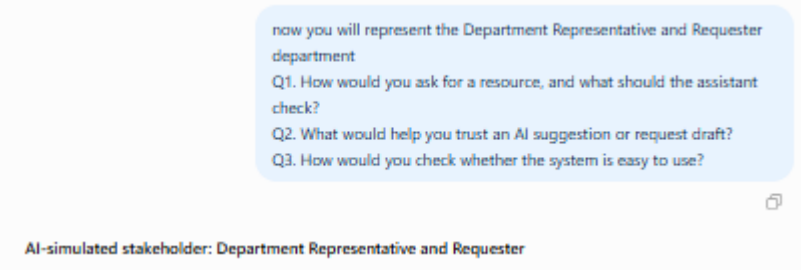



Figure E.9: AI-simulated interview, Department Representative questions

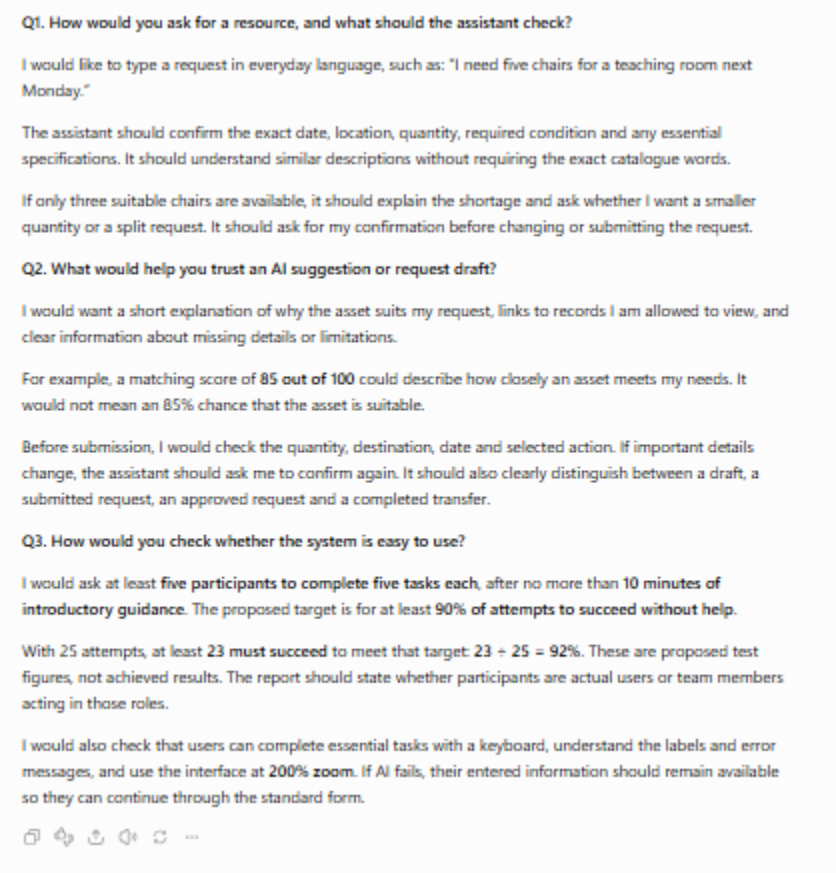



Figure E.10: AI-simulated interview, Department Representative answers

Summary: representatives want to type needs in everyday language; the assistant should confirm the date, location, quantity, condition and specifications, explain shortages and ask for confirmation before submitting. Trust depends on short explanations, links to permitted records and clear missing-information notes. Usability is tested with five participants on five tasks after at most 10 minutes of guidance, with a 90% success target, keyboard-only use and 200% zoom.

Linked requirements: FR-11.1 to FR-11.4; FR-8.1, FR-8.4, FR-8.6; AI-1, AI-2, AI-4, AI-6; PER-2 to PER-5; QUA-1 to QUA-3, QUA-15.

# E.6 University Management

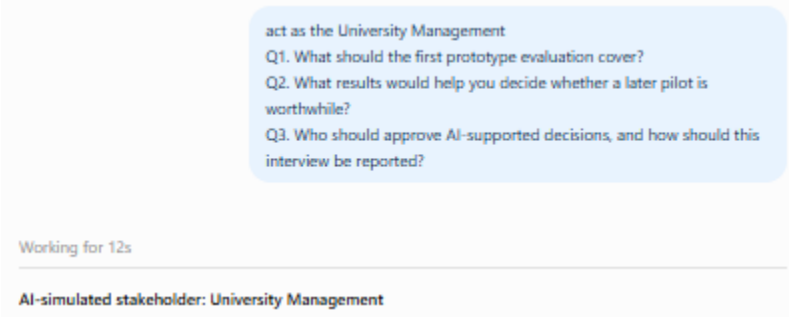
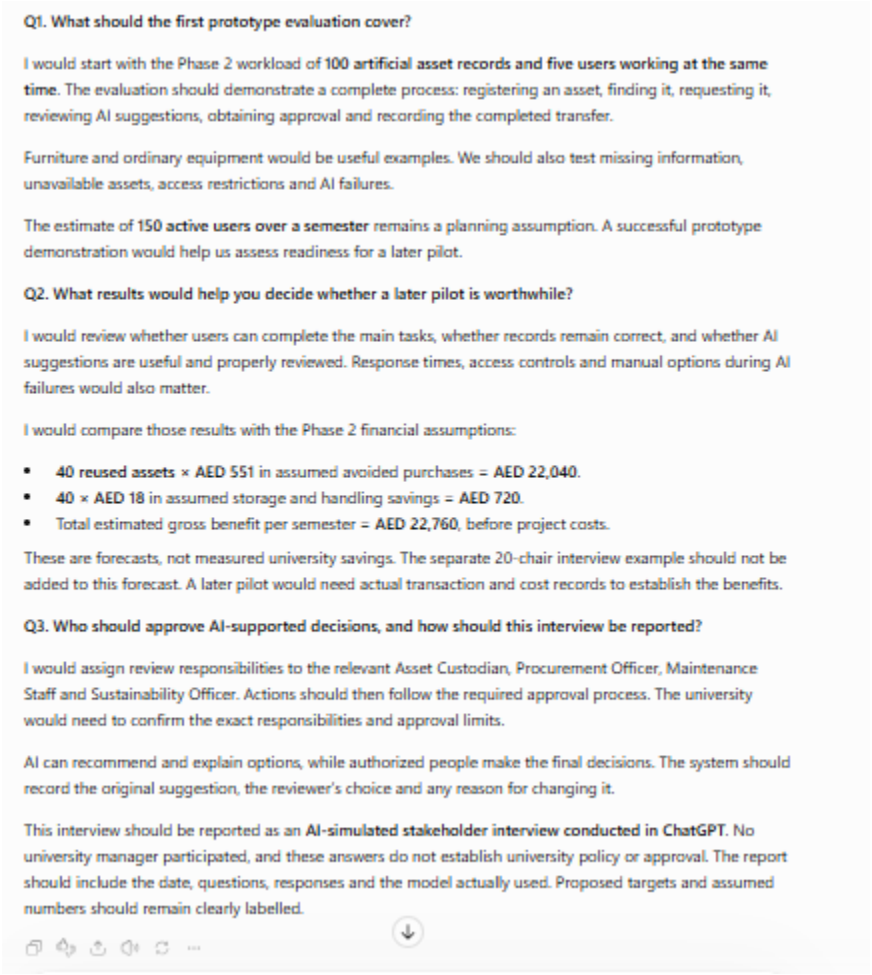



Figure E.11: AI-simulated interview, University Management questions



**Q1. What should the first prototype evaluation cover?**

I would start with the Phase 2 workload of **100 artificial asset records and five users working at the same time**. The evaluation should demonstrate a complete process: registering an asset, finding it, requesting it, reviewing AI suggestions, obtaining approval and recording the completed transfer.

Furniture and ordinary equipment would be useful examples. We should also test missing information, unavailable assets, access restrictions and AI failures.

The estimate of **150 active users over a semester** remains a planning assumption. A successful prototype demonstration would help us assess readiness for a later pilot.

**Q2. What results would help you decide whether a later pilot is worthwhile?**

I would review whether users can complete the main tasks, whether records remain correct, and whether AI suggestions are useful and properly reviewed. Response times, access controls and manual options during AI failures would also matter.

I would compare those results with the Phase 2 financial assumptions:

- **40 reused assets** × **AED 551** in assumed avoided purchases = **AED 22,040**.
- **40** × **AED 18** in assumed storage and handling savings = **AED 720**.
- Total estimated gross benefit per semester = **AED 22,760**, before project costs.

These are forecasts, not measured university savings. The separate 20-chair interview example should not be added to this forecast. A later pilot would need actual transaction and cost records to establish the benefits.

**Q3. Who should approve AI-supported decisions, and how should this interview be reported?**

I would assign review responsibilities to the relevant Asset Custodian, Procurement Officer, Maintenance Staff and Sustainability Officer. Actions should then follow the required approval process. The university would need to confirm the exact responsibilities and approval limits.

AI can recommend and explain options, while authorized people make the final decisions. The system should record the original suggestion, the reviewer's choice and any reason for changing it.

This interview should be reported as an **AI-simulated stakeholder interview conducted in ChatGPT**. No university manager participated, and these answers do not establish university policy or approval. The report should include the date, questions, responses and the model actually used. Proposed targets and assumed numbers should remain clearly labelled.

Figure E.12: AI-simulated interview, University Management answers

Summary: the first evaluation should demonstrate the complete process (register, find, request, review AI suggestions, approve and record the transfer) with the Phase 2 workload, and test missing information, unavailable assets, access limits and AI failures. A later pilot decision would compare results with the Phase 2 financial forecast, which is an estimate and not measured savings. Review responsibilities are assigned to the relevant roles, AI only recommends, and the interview is reported as an AI simulation.

Linked requirements: AI-3 to AI-5, AI-8; FR-13.12; PER-1; QUA-8 to QUA-14; BR-9. Related features: FR-4, FR-7, FR-12.

# Appendix F: Alignment with Phase 2

This appendix records what changed from the Phase 2 feasibility study [5] and checks that the Essential requirements fit the Phase 2 schedule.

## F.1 Changes from Phase 2

**Table F-1: Changes from Phase 2**

| Change | Section | Reason |
|---|---|---|
| The twelve high-level Phase 2 requirements (FR-1 to FR-12) were refined into 259 numbered requirements, each with a priority, linked use cases and an acceptance criterion. | Sections 2 to 4 | Phase 2 stated that detailed requirements and acceptance criteria would be developed in Phase 3 [5], and the SRS template requires them [1]. |
| New feature FR-13, Sustainability Indicators and Dashboard: nine indicators with fixed formulas, versioned default values and a named emission-factor source. | 2.13 | Phase 2 placed sustainability indicators in scope and in the development phase but had no requirement for them; the project description requires them [1]. |
| FR-7, Operational Reporting, was split into notifications (FR-7.N), standard reports (FR-7.R) and the audit log (FR-7.A). | 2.7 | The three parts have different users, rules and tests. |
| Reservation moved from request submission to final approval: submitting a request no longer reserves units. | FR-3.7, FR-3.8, FR-4.5; BR-5 | Phase 2 FR-3 let a request reserve assets. Reserving at final approval stops unapproved requests from blocking assets and prevents two requests from holding the same units. |
| A release step was added before approval: the owning Asset Coordinator releases or declines a request, the Administrator approves it, a Procurement Team member gives a second approval above a configured value, and one person cannot fill two approval steps. | FR-4.1 to FR-4.4; BR-3, BR-4 | Phase 2 named the approving roles but not the order of decisions or the separation of duties. |
| Receipt is confirmed by a Department Representative or the Asset Coordinator of the receiving department. | FR-4.10; BR-6 | Phase 2 gave receipt confirmation to Asset Coordinators in general; only the receiving department can check what arrived. |
| Sustainable-action recommendations (FR-10) and the AI assistant (FR-11) are Desirable, Medium-priority features. | 2.10, 2.11; F.2 | They are not in the minimum prototype in the project description [1], and Phase 2 rated scope and time (R4) as one of its two highest risks. Phase 2 requires lab instructor confirmation for any reduction in AI scope (TBD-16). |

| Change | Section | Reason |
|---|---|---|
| Every requirement has an Essential, Desirable or Future priority, and every feature a High or Medium priority, checked against the Phase 2 schedule. | 1.2; F.2 | Phase 2 Recommendation 6 asked for this classification so that essential functions are protected if time runs short. |
| Phase 2 stakeholder responsibilities became system actors and a permission table, and an Applicant actor was added for registration and consent. | 5.1, Table 5.1-1; UC-13 | Phase 2 required an explicit consent step at registration under PDPL Article 6 [5]; an applicant has no role until a System Administrator activates the account. |
| Phase 2 planned real stakeholder interviews before Phase 3 was completed; Phase 3 uses AI-simulated interviews, labeled as such, and lists the decisions that still need real stakeholders. | Appendix E; Appendix C (TBD-15) | Real interviews could not be arranged in the Phase 3 period, so the simulated answers are not treated as evidence. |
| University single sign-on, raised in Phase 2 as an open dependency, is not used by the prototype; it is added only after KU IT approval. | ORG-3; TBD-3 | Follows the Phase 2 working assumption of standalone accounts and keeps the prototype independent of an external approval. |
| The Phase 2 AI response target (90% of interactions within 15 seconds) became PER-4, with a progress indicator, a 20-second cut-off and a standard fallback. | PER-4 to PER-6 | Phase 2 required a clear fallback when AI is slow or unavailable. |
| The Phase 2 pilot size (about 150 active users) became the performance test workload. | PER-1 | Keeps performance tests consistent with the Phase 2 cost and operational assumptions. |
| The Phase 2 legal analysis became testable requirements: consent, data minimization, a data protection impact assessment, no personal data sent to AI services or outside the UAE, and cybercrime and copyright obligations. | 3.3; 4.2 | Phase 2 identified the obligations; Phase 3 states how the system meets them. |
| Each Phase 2 risk (R1 to R14) is linked to the requirements that address it. | Appendix D (Table D-2) | Shows that the Phase 2 risk responses are carried into the requirements. |

# F.2 Priority and Schedule Check

Phase 2 recommended building a working baseline before AI enhancements, and classifying features as essential, desirable or future so that essential end-to-end functions are protected if time runs short (Recommendation 6 and the schedule risk controls) [5]. Scope and time (risk R4) was one of the two highest Phase 2 risks.

The Phase 3 priorities follow that plan. The baseline workflow that Phase 2 listed first (authentication, assets, requests, search, approval, transfer and maintenance) is Essential, and so are the functions Phase 2 placed next in development: AI classification, matching with

explanations, sustainability indicators and reports. Sustainable-action recommendations (FR-10) and the assistant (FR-11) are Desirable because they are not in the minimum prototype in the project description [1]; Phase 2 listed them in scope at prototype scale, so this change is to be confirmed with the lab instructors (TBD-16). Overall, 218 requirements are Essential, 37 Desirable and 4 Future.

Capacity uses the Phase 2 assumption of four members working 8 hours a week each, or 32 person-hours a week [5]. The check uses the Phase 2 deadlines because they are earlier than the tentative schedule in the project description [1]; meeting them also meets the official deadlines.

**Table F-2: Schedule and available effort**

| Work period | Phase 2 deadline [5] | Project description deadline [1] | Available person-hours |
|---|---|---|---|
| **Phases 4–5, design (weeks of 19 and 26 Oct)** | Week of 26 Oct | Week of 1 Nov | 64 |
| **Phase 6, development (weeks of 2, 9 and 16 Nov)** | Week of 16 Nov | Week of 22 Nov | 96 |
| **Phases 7–8, testing and final submission (week of 23 Nov)** | Week of 23 Nov | Week of 6 Dec | 32 |
| **Total** | | | 192 |

**Table F-3: Planned work and estimated effort**

| Work package | Essential requirements | Phase 2 order | Estimate (person-hours) | Week of |
|---|---|---|---|---|
| Design document: architecture, detailed design and interface designs | All (design basis) | Phases 4–5 | 36 | 19–26 Oct |
| Project set-up, data model and sample data | 4.1; ORG-2, ORG-4 | Set-up | 12 | 19–26 Oct |
| AI proof of concept on sample data | FR-8, FR-9 (trial) | Prototype AI early | 12 | 19–26 Oct |
| Accounts, roles, departments and consent | FR-6 (12); SEC-1 to SEC-9 | 1. Baseline | 10 | 2 Nov |
| Asset registration, publication and search | FR-1 (13), FR-2 (7) | 1. Baseline | 10 | 2 Nov |
| Requests, approvals and transfers | FR-3 (12), FR-4 (13); BR-2 to BR-6 | 1. Baseline | 13 | 2–9 Nov |
| Maintenance, life-cycle history and end-of-life actions | FR-5 (30); SAF-1 to SAF-7, SAF-9; BR-7, BR-8 | 1. Baseline | 13 | 9 Nov |

| Work package | Essential requirements | Phase 2 order | Estimate (person-hours) | Week of |
|---|---|---|---|---|
| Notifications, standard reports and audit log | FR-7 (17); BR-12 | 1. Baseline (audit from the start) | 9 | 9 Nov |
| AI classification and matching in the workflow, with fallback | FR-8 (9), FR-9 (9); AI-1 to AI-10 | 2. After baseline | 12 | 16 Nov |
| AI reports (types a and b), indicators and dashboard | FR-12 (11), FR-13 (12) | 2. After baseline | 13 | 16 Nov |
| Integration and defect fixing | All | Continuous integration | 10 | 2–16 Nov |
| Performance, security, quality and AI accuracy tests; legal checklist; final documents and demonstration | PER, SEC, QUA, AI-10, LEG | Phases 7–8 | 30 | 23 Nov |

Result: the plan needs an estimated 60 of 64 person-hours for Phases 4–5, 90 of 96 for Phase 6 and 30 of 32 for Phases 7–8, or 180 of 192 in total, leaving a reserve of 12 hours. The later deadlines in the project description add about two weeks, or roughly 64 more person-hours. The estimates are the team’s judgment, based on the stack proposed in the Phase 2 technical analysis (Python and Django, whose built-in accounts, permissions, administration pages and forms cover much of the baseline, with MySQL, Sentence Transformers and a hosted language model) [5]; they are not measured values.

Desirable and Future requirements are not in the estimate; they are built only if the reserve is still unused after the Phase 6 milestone. Progress is reviewed weekly against Table F-3, as the Phase 2 schedule risk controls require. If the build falls more than 6 hours behind the plan, the team will use the framework’s standard administration pages for back-office functions (accounts, the vendor list and category values) instead of custom pages, so that every Essential requirement is still met on time.

# AI Contributions

## Mohamed Almarzooqi

An AI assistant was used to help draft and organize the introduction, FR-2, FR-6, the notification requirements (FR-7.N), the business rules (3.5), the actors list, the use case diagram, UC-2, UC-4, UC-8 and UC-13, the glossary and the TBD list, and to check them against the project description and the Phase 2 report. All content was reviewed and edited by the author.

## Mohammad Alsuwaidi

An AI assistant was used to help draft and organize FR-1, FR-3, FR-4, the data requirements (4.1), the state lists (Appendix B), the traceability matrix (Appendix D) and UC-1, UC-3 and UC-5, and to check them against the project description and the Phase 2 report. All content was reviewed and edited by the author.

## Ahmed AlAbdouli

An AI tool (ChatGPT-6 Astra Extra-High) was used to simulate six stakeholder roles and answer interview questions about the ReSource project (Appendix E). The interviews helped explore user needs and clarify the proposed requirements; no real university stakeholders participated. Numerical examples and proposed test targets are labeled as such. The author verified the content, checked that it stayed within the project scope and made the final decisions about what to include.

## Rashid Alzaabi

An AI assistant (Claude, Anthropic) was used to help organize the requirements in sections 2.5, 2.7 (reports and audit log), 2.12, 2.13, 3.2, 3.3 and 4.2 and the use cases UC-6, UC-7 and UC-12. It was also used to check them against the project description and the commitments made in the Phase 2 feasibility study. All requirements, measurements and default values were reviewed and edited by the author, and values that still need confirmation from Khalifa University are listed as TBD in Appendix C.

## Integration Review

Integration review: an AI assistant (Claude, Anthropic) was used to check the combined document for consistency across sections (role names, the reservation rule, state lists, cross-references, priorities and traceability) and to propose corrections. Each member checked the corrected text of their own sections.